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**Report prepared for the WHO annual consultation on
the composition of influenza vaccine for the Northern
Hemisphere 2019-2020**

18th – 20th February 2019

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WHO Vaccine Recommendation Meeting for the Northern Hemisphere 2019-2020 influenza season.

WHO CC, London - Final Report for WHO meeting: 18 - 20 February 2019.

Summary of Influenza Activity in the WHO European Region: October 1 2018 – January 27 2019

Summary

- The influenza season started the week of December 3 (week 49/2018) and as of January 27 (week 4/2019), influenza activity continued to increase.
- From January 14, 2019 (week 3/2019), influenza-positivity rates exceeded 50% among ILI and ARI specimens (for both sentinel and non-sentinel sources).
- Influenza A(H1N1)pdm09 virus detections dominated, but A(H3N2) viruses co-circulated widely. Very few influenza B viruses were detected.
- In hospitalised influenza-infected patients admitted to ICU wards, 39% of influenza A virus infections were subtyped; of these 80% were A(H1N1)pdm09 positive, but 40% of the patients over 75 years of age were infected with A(H3N2) viruses.
- In hospitalised patients with SARI, 24% were positive for an influenza virus; 99% were influenza A. Almost all influenza A viruses (95%) were subtyped and 79% were A(H1N1)pdm09.

Timing and geographic spread

Based on data from 50 countries and Kosovo¹ in the WHO Europe Region, the course of the 2018-2019 season (weeks 40/2018 to 4/2019) has been characterised by a later start to the influenza season compared to the last two seasons. Week 49/2018 was the first in which the proportion of respiratory specimens, from patients presenting to sentinel primary care sites with influenza-like illness (ILI) or acute respiratory infection (ARI) symptoms, that tested positive for an influenza virus was greater than 10% (one week later than the previous season and three weeks later than the 2016/2017 season). Since week 49/2018, influenza activity has continued to increase through to week 4/2019 (**Figure E1-A**). More than 40% of specimens have been positive for influenza since week 1/2019, and this proportion has continued to increase, exceeding 50% in week 3/2019. This percent positivity for ILI and ARI specimens is nearing the peak percentage of the past four influenza seasons (**Figure E1-A**). When examining country-reported influenza intensity, countries in the eastern part of the region experienced increased intensity first, with Armenia, Georgia, Turkey and Ukraine being the first to report medium influenza intensity. An exception to this is Malta, in the south, which reported medium intensity in week

¹ in accordance with. United Nations Security Council. Resolution 1244 (1999).

48/2018, but reported low/baseline levels after this week until week 1/2019 (**Figure E2**). In addition, no country located west of Malta has reported a week of high influenza intensity so far in the season, while Malta, Romania, Georgia and Turkey have all experienced consecutive weeks with reported high influenza intensity (**Figure E2**).

Distribution of viruses

In the WHO European Region between weeks 40/2018 and 4/2019, 57,443 influenza detections from sentinel and non-sentinel primary care sources were reported to The European Surveillance System (TESSy) (**Table E1-A**). For combined sentinel and non-sentinel data, almost all influenza viruses detected were type A (99%), and this was consistent when the data were stratified by sub-region (**Table E1-A**). Of the 21,835 (38%) influenza A specimens that were subtyped (sentinel and non-sentinel combined), 65% were A(H1N1)pdm09 and 35% were A(H3N2). However, the distribution of viruses varied by sub-region. The distribution of viruses in both southern Europe and eastern Europe reflected the results from the overall region. In northern Europe and central Asia, A(H1N1)pdm09 virus detections represented over 80% of subtyped influenza A detections. In contrast, nearly equal circulation of A(H1N1)pdm09 (41%) and A(H3N2) (59%) viruses was observed in western Europe and influenza A(H3N2) viruses dominated in West Asia, representing 64% of subtyped influenza A detections (**Table E1-A**). Of note, a very low percentage of influenza A positive specimens were subtyped in Northern (25%) and Western (22%) Europe, and so the results shown may not reflect the true distribution of influenza A subtypes in these regions. There were few influenza B viruses detected, and of those ascribed to a lineage (n=55), the majority (78%) were B/Yamagata.

When virologic data was restricted to sentinel specimens only, the regional distribution of influenza A subtypes remained similar to the distribution described above. However, there was an increased percentage of A(H1N1)pdm09 viruses (75%) in sentinel specimens, although 25% of type A viruses were not subtyped (**Table E1-B**). Influenza B virus detections remained low (1%) among viruses from both sources. From sentinel sources, among the few viruses (n=32) ascribed to a lineage, B/Yamagata represented the great majority of detections (91%) (**Figure E1-B/C**).

Of the 42 countries in the region for which a dominant type (comprising at least 60% of influenza virus detections) could be calculated using sentinel virologic data, all countries reported a predominance of influenza A type through week 4/2019 (**Figure E3**). Twenty-six countries reported a predominately influenza A(H1N1)pdm09 virus season. Belarus, Belgium, Israel, Luxembourg and Turkey reported a predominately A(H3N2) virus season, while eight countries (Bulgaria, France, Germany, Hungary, Lithuania, Portugal, Romania and Spain) reported co-dominance of influenza A(H1N1)pdm09 and A(H3N2) viruses. The remaining three countries had not subtyped enough influenza A specimens to determine subtype dominance.

Influenza virus characterisation performed by National Influenza Centres

Genetic characterisation

National influenza Centres from 19 countries reported genetic characterisations, to TESSy, of 1,314 influenza viruses collected between weeks 40/2018 to 4/2019.

All 800 A(H1N1)pdm09 viruses characterised belonged to genetic subgroup 6B.1, represented by A/Michigan/45/2015, the recommended vaccine component for this northern hemisphere season.

Among 485 A(H3N2) viruses, 459 were attributed to a genetic clade. The majority (n=394, 81%) belonged to clade 3C.2a represented by A/Hong Kong/4801/2014 and 62 (13%) belonged to clade 3C.3a represented by A/England/538/2018. Of the viruses in clade 3C.2a, 308 belonged to the 3C.2a1b subgroup represented by A/Alsace/1746/2018 and 37 belonged to the 3C.2a1 subclade represented by the recommended vaccine virus A/Singapore/INFIMH-16-0019/2016. Thirty-four viruses belonged to the 3C.2a2 subclade, 14 to the 3C.2a3 subclade, and the one remaining virus belonged to the 3C.2a4 subclade.

Twenty-nine influenza B specimens were characterised genetically, 12 B/Victoria-lineage and 17 B/Yamagata-lineage viruses. Of the 12 B/Victoria-lineage viruses one virus was not attributed to a clade. All others belonged to the B/Brisbane/60/2008 clade (clade 1A), but four belonged to a subgroup represented by B/Colorado/06/2017, the recommended influenza B component of current trivalent vaccines. Viruses in this subgroup carry an HA1 double amino acid deletion, Δ 162-163. Five viruses belonged to a subgroup represented by B/Hong Kong/269/2017, which carry an HA1 triple amino acid deletion, Δ 162-164. All 17 influenza B/Yamagata-lineage viruses belonged to genetic clade 3, represented by B/Phuket/3073/2013, the recommended B/Yamagata-lineage component of quadrivalent vaccines for the northern hemisphere 2018-2019 season.

Antiviral susceptibility

Neuraminidase inhibitor susceptibility was assessed for 726 viruses with collection dates between week 40/2018 and 4/2019: 454 A(H1N1)pdm09, 260 A(H3N2) and 12 type B. Of the 12 type B viruses, none showed evidence of reduced inhibition (RI) by oseltamivir, but one virus displayed RI by zanamivir. One A(H1N1)pdm09 virus carried H275Y amino acid substitution in neuraminidase and showed evidence of highly reduced inhibition (HRI) by oseltamivir; all A(H1N1)pdm09 viruses showed normal inhibition (NI) by zanamivir. One A(H3N2) virus showed evidence of HRI by oseltamivir; all A(H3N2) viruses showed NI by zanamivir.

Hospitalised cases

Between weeks 40/2018 and 4/2019, 12 countries reported a total of 5,850 laboratory-confirmed hospitalised influenza cases. Eleven of these countries reported 2,610 cases from patients in intensive care units (ICUs) and eight countries reported 3,240 cases from other hospital wards. Among cases admitted to ICUs, 98% were infected with influenza

type A viruses. Of the 1,010 (39%) influenza A viruses from ICU-admitted patients that were subtyped, 80% were influenza A(H1N1)pdm09 virus and 20% were influenza A(H3N2) virus. Approximately 42% of the influenza-associated ICU admissions with a recorded age (n=1,062) were in patients aged 65 years or older, an additional 22% were in individuals aged 55-64 (**Figure E4-A**). The proportion of A(H3N2) viruses among subtyped influenza A viruses was higher in individuals above 55 years of age than younger individuals. Among ICU-admitted patients aged 75 or older, over 40% of subtyped influenza A viruses were influenza A(H3N2). The countries conducting this surveillance were the Czech Republic, Denmark, Estonia, France, Ireland, Romania, the Russian Federation, Slovakia, Spain, Sweden, the United Kingdom and Ukraine.

This season, 18 countries, territories or areas with surveillance systems for Severe Acute Respiratory Infection (SARI) reported a total of 21,916 cases, the majority of which were in children. Of 5,244 SARI cases tested for influenza viruses since October 1 2018, 1,242 (24%) were positive and almost all detections were type A viruses (99%). The number of SARI cases, specimens tested for influenza viruses, and percentage positive varied widely by country (**Figure E4-B**). Of 1,186 (95%) influenza A viruses subtyped, 79% were A(H1N1)pdm09 and 21% were A(H3N2). The two influenza B viruses ascribed to a lineage were both B/Yamagata. The countries, territories, or areas conducting SARI surveillance were Albania, Armenia, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Georgia, Kazakhstan, Kosovo², Kyrgyzstan, Montenegro, Republic of Moldova, Russian Federation, Serbia, Tajikistan, Turkmenistan, Ukraine and Uzbekistan.

Mortality data

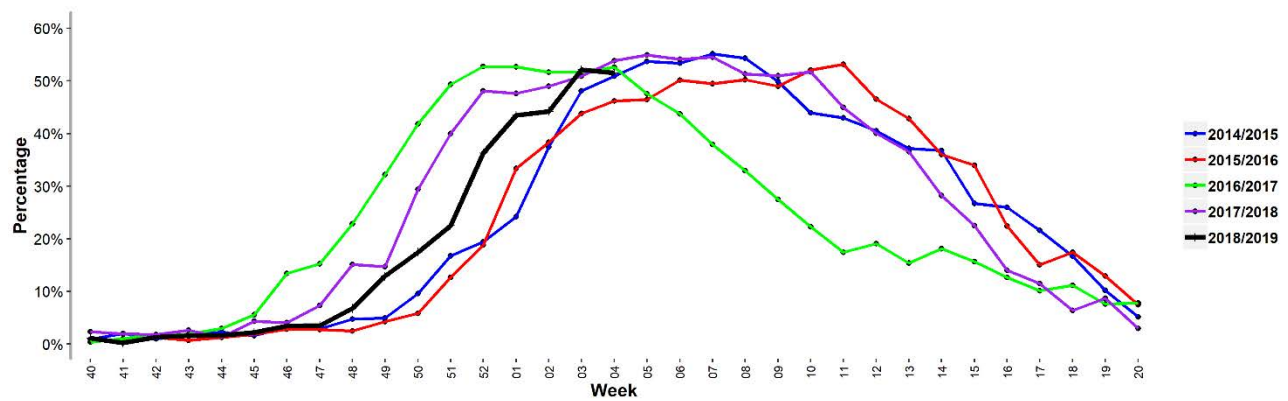
Analyses of all-cause and influenza-attributable mortality from 24 European countries or subnational regions participating in the European monitoring of excess mortality for public health action project (EuroMOMO – <http://www.euromomo.eu> network) indicated that mortality levels were within expected levels throughout much of the season so far. Pooled all-cause excess mortality began increasing in persons aged 65 years and older in certain countries in week 3/2019, and as of week 4/2019, excess mortality was observed in adults and the elderly in the pooled data. This observation appeared to be driven by excess mortality reported in certain countries, particularly Austria and Portugal.

Participating countries are Austria, Belgium, Berlin (Germany), Denmark, England (United Kingdom), Estonia, Finland, France, Greece, Hesse (Germany), Hungary, Ireland, Italy, Luxemburg, Malta, the Netherlands, Northern Ireland (United Kingdom), Norway, Portugal, Scotland (United Kingdom), Spain, Sweden, Switzerland and Wales (United Kingdom).

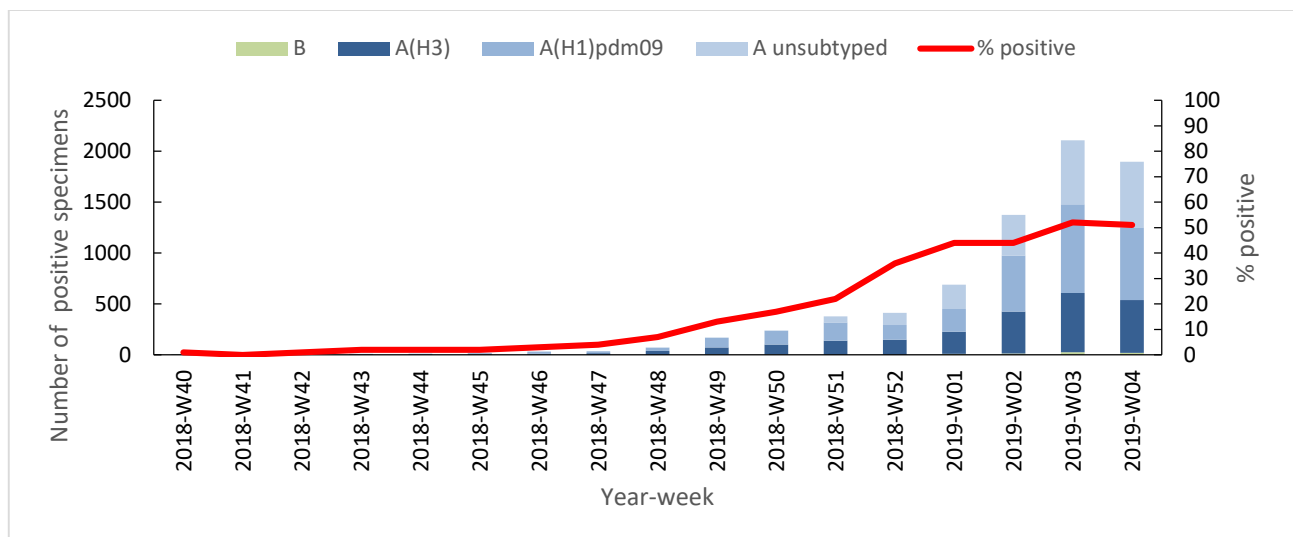
²in accordance with. United Nations Security Council. Resolution 1244 (1999).

Figure E1. Influenza Activity in the WHO European Region 2018-2019 (weeks 40/2018-4/2019)

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel source influenza detections 2018-2019 season, weeks 40 through 4



C. Non-sentinel source influenza detections 2018-2019 season, weeks 40 through 4

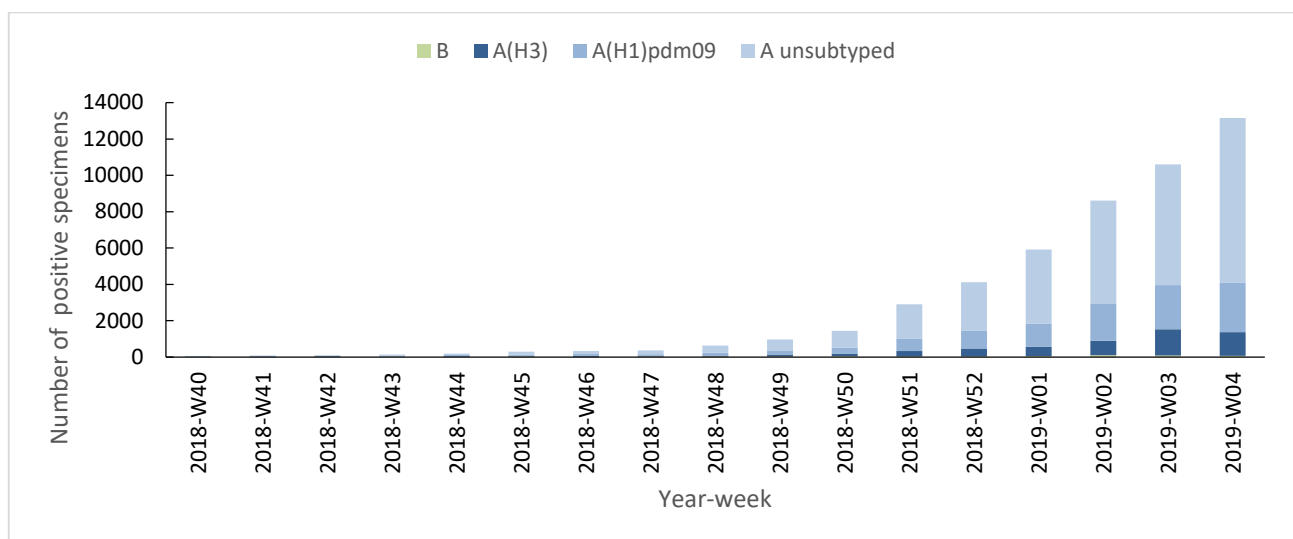


Figure E2. Development of the 2018-2019 Influenza Season in the WHO European Region

Country-Reported Intensity of Influenza Activity and Calculated Dominant Virus (sub)type (based on at least 60% of all cumulative [sentinel and non-sentinel] detections) by Week of the Influenza Season

		Week																	
		40	41	42	43	44	45	46	47	48	49	50	51	52	01	02	03	04	
Very High = exceptionally high level of activity																			
High = level of activity higher than usual																			
Medium = usual level of activity																			
Low = no activity or activity at baseline ^a level																			
Iceland	-	-	-	-	H3	H3	H3	H3	H3	H3	H3	H3	H3	H3	H3	H1/H3	H1/H3	-	
Ireland	-	-	-	-	-	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
Portugal	-	-	-	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1/H3	H1/H3	H1/H3	H1/H3	
Spain	-	-	-	-	-	-	H1	H1	H1	H1	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	
United Kingdom (Northern Ireland)	-	-	-	-	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
United Kingdom (Wales)	-	-	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	-	
United Kingdom (Scotland)	-	A/B	A/B	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
United Kingdom (England)	A	A	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
France	-	H1/H3	H3	H3	H3	H3	H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H3	H3	H3	H3	H3	
Belgium	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A	H3	H3	H3	
Netherlands	-	-	-	-	A	A	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
Luxembourg	-	-	-	-	-	-	-	-	-	-	-	-	H3	H3	H3	H3	H3	H3	
Switzerland	-	-	-	-	A	A	A	A	A	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1	H1	H1	
Germany	-	-	-	-	-	-	-	-	H1/H3	H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	
Denmark	-	-	A	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
Norway	-	H3	H3	H3	H1/H3	H1/H3	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	H1	
Italy	-	-	-	-	-	-	-	-	H3	H3	H3	H1/H3	H1/H3	H1/H3	H1	H1	H1	H1	
Austria	-	-	-	-	-	-	-	-	-	-	-	A	A	H1/H3	H1	H1	H1	H1	
Malta	-	-	-	-	H3	H3	H3	H3	H3	H3	H3	H1/H3	H1/H3	H1/H3	H1/H3	H1	-	-	
Slovenia	-	-	-	-	-	-	-	-	-	-	-	A	H1	H1	H1	H1	H1	H1	
Sweden	-	A	A	A	A	A	A	A	A	H1	H1	H1	H1	H1	H1	H1	H1	H1	
Croatia	-	-	-	-	-	-	-	-	-	-	H1	-	H1	H1	H1	H1	H1	H1	
Czech Republic	-	-	-	-	-	-	-	-	-	-	A/B	A/B	A/B	A/B	A/B	A	H1	H1	
Bosnia and Herzegovina	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Slovakia	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A	A	H1/H3	
Albania	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	-	
Hungary	-	-	-	-	-	-	-	-	-	-	-	-	A	H3	H3	H3	H1/H3	H1/H3	
Poland	-	-	-	-	-	-	-	-	-	-	-	-	-	A	H1	H1	H1	H1	
Montenegro	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	
Serbia	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	
Kosovo*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	A	A	
The Former Yugoslav Republic of Macedonia	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	
Greece	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	
Lithuania	-	-	-	-	-	-	-	-	-	-	H3	H3	H3	H3	H3	H3	H3	H3	
Bulgaria	-	-	-	-	-	-	-	-	-	-	H3	H1/H3	H1/H3	H1/H3	H1	H1	H1	H1	
Latvia	-	-	-	-	-	-	-	-	-	-	H1/H3	H1	H1	H1	H1	H1	H1	H1	
Romania	-	-	-	-	-	-	-	-	-	-	A	H1	H3	H3	H1/H3	H1	H1	H1	
Estonia	-	-	-	-	-	-	-	-	-	-	A	H1	H1	H1	H1	H1	H1	H1	
Finland	-	-	-	-	-	-	-	-	-	H1/H3	H1/H3	H1	H1	H1	H1	H1	H1	H1	
Belarus	-	-	-	-	-	-	-	-	-	-	-	-	-	A	H1/H3	H1/H3	H1/H3	H1	
Republic of Moldova	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	H1	
Ukraine	-	-	-	-	-	-	-	-	-	A	H3	H3	H3	H3	H3	H3	H3	H3	
Cyprus	-	-	-	-	-	-	-	-	-	-	-	-	-	A	A	A	A	A	
Israel	-	-	-	A	A	H1	H1	H1	H1	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	
Turkey	-	-	-	-	-	-	-	H3	H3	H3	H3	H3	H3	H3	H3	H3	H3	H3	
Georgia	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	H1	H1	
Armenia	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	H1	-	
Azerbaijan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	H1	H1	
Turkmenistan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Uzbekistan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Kazakhstan	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	H1	H1	
Russian Federation	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	H1/H3	
Tajikistan	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Kyrgyzstan	-	-	-	-	-	-	-	-	-	-	-	H1	H1	H1	H1	H1	H1	H1	

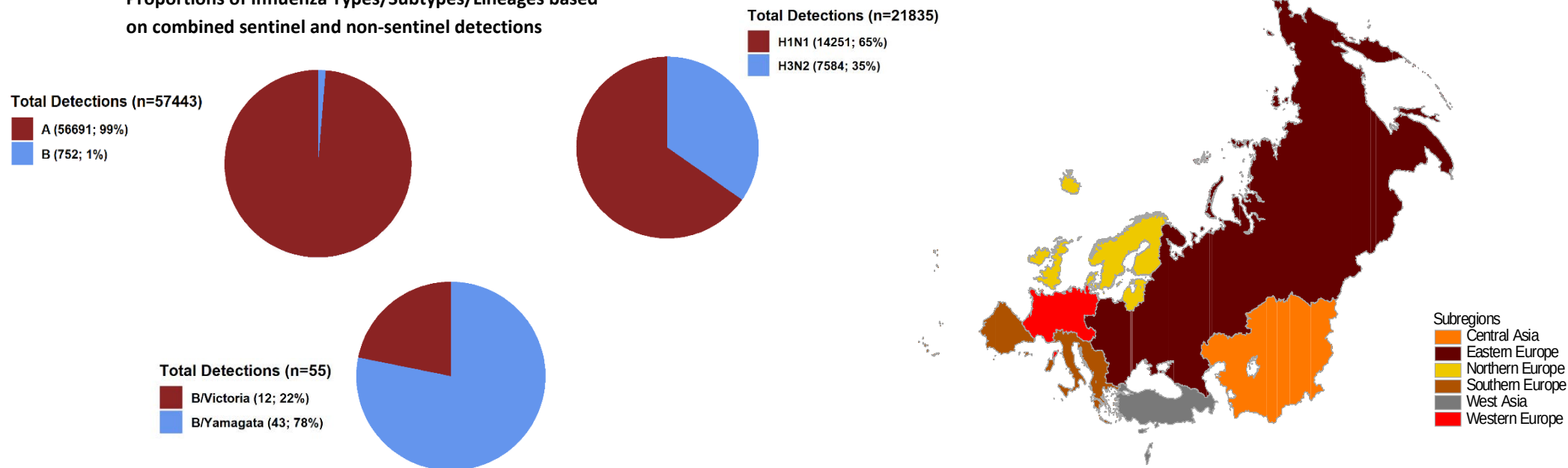
WEST

EAST

*in accordance with. United Nations Security Council. Resolution 1244 (1999).

Table E1. Distribution of Influenza Detections in the WHO European Region^a, 2018-2019 Season (weeks 40/2018-4/2019)

Proportions of Influenza Types/Subtypes/Lineages based on combined sentinel and non-sentinel detections



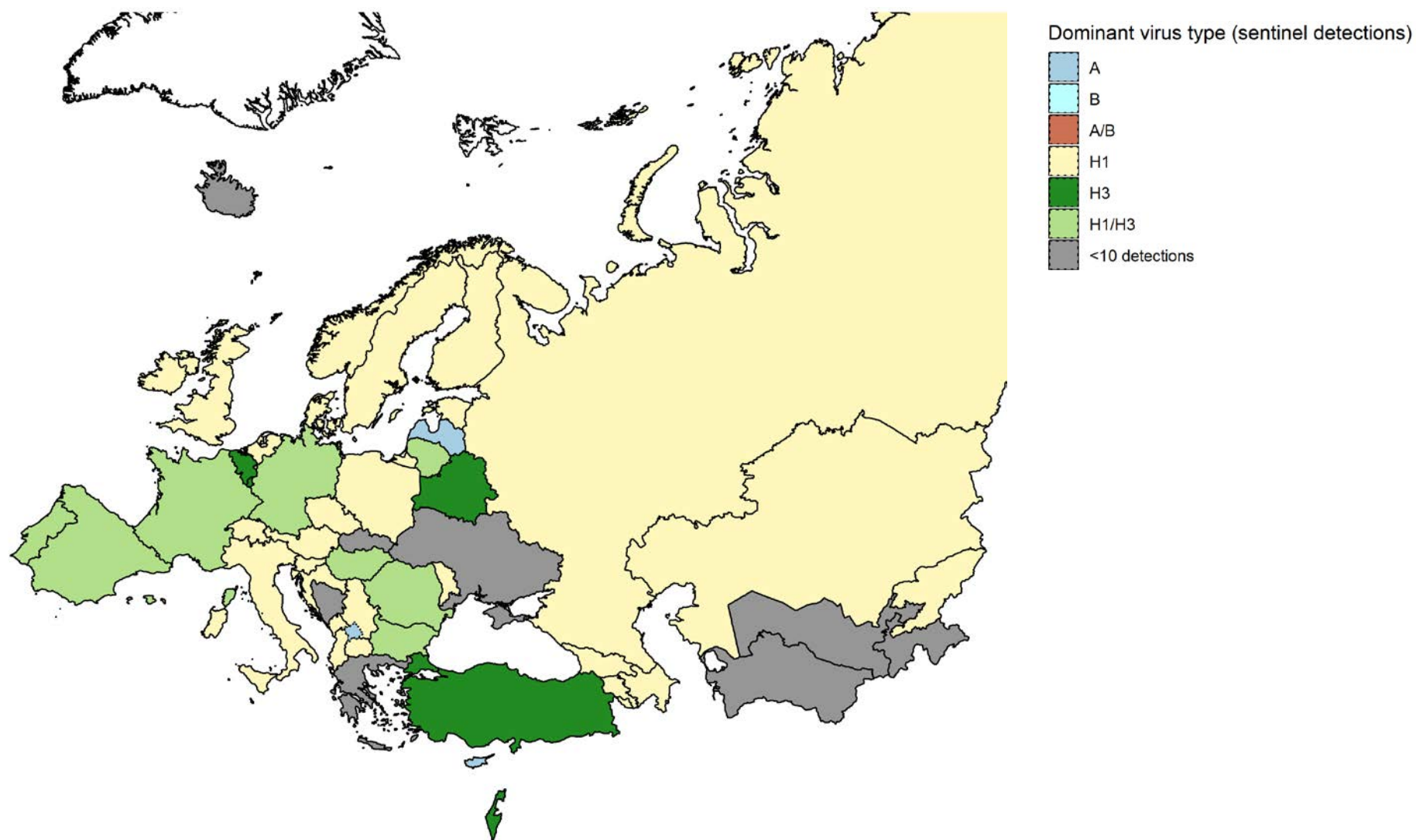
A. Influenza detections from sentinel and non-sentinel source specimens*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	9839	99%	9562	99%	5292	99%	29570	98%	2127	99%	301	99%	56691	99%
Influenza A subtyped	5725		2102		4578		7462		1667		301		21835	
<i>A (H1N1)pdm09</i>	3642	64%	866	41%	2800	61%	6106	82%	594	36%	243	81%	14251	65%
<i>A (H3N2)</i>	2083	36%	1236	59%	1778	39%	1356	18%	1073	64%	58	19%	7584	35%
Influenza B	101	1%	121	1%	61	1%	452	2%	15	1%	2	1%	752	1%
Influenza B lineage determined	35		1		1		18		0		0		55	
<i>B/Yamagata lineage</i>	28	80%	1	100%	1	100%	13	72%	0	-	0	-	43	78%
<i>B/Victoria lineage</i>	7	20%	0	0%	0	0%	5	28%	0	-	0	-	12	22%
Total	9940		9683		5353		30022		2142		303		57443	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

^aWHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: <http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>

B. Influenza detections from sentinel source ILI or ARI specimens	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	3628	99%	837	99.5%	557	99%	965	99%	1222	99%	168	99%	7377	99%
Influenza A subtyped	1759		766		533		881		1167		168		5274	
<i>A (H1N1)pdm09</i>	1128	64%	373	49%	377	71%	695	79%	300	26%	140	83%	3013	75%
<i>A (H3N2)</i>	631	26%	393	51%	156	29%	186	21%	867	74%	28	17%	2261	25%
Influenza B	63	1%	3	0.5%	7	1%	6	1%	11	1%	2	1%	92	1%
Influenza B lineage determined	29		1		1		1		0		0		32	
<i>B/Yamagata lineage</i>	27	93%	1	100%	1	100%	0	0%	0	-	0	-	29	91%
<i>B/Victoria lineage</i>	2	7%	0	0%	0	0%	1	100%	0	-	0	-	3	9%
Total	3691		840		564		971		1233		170		7469	

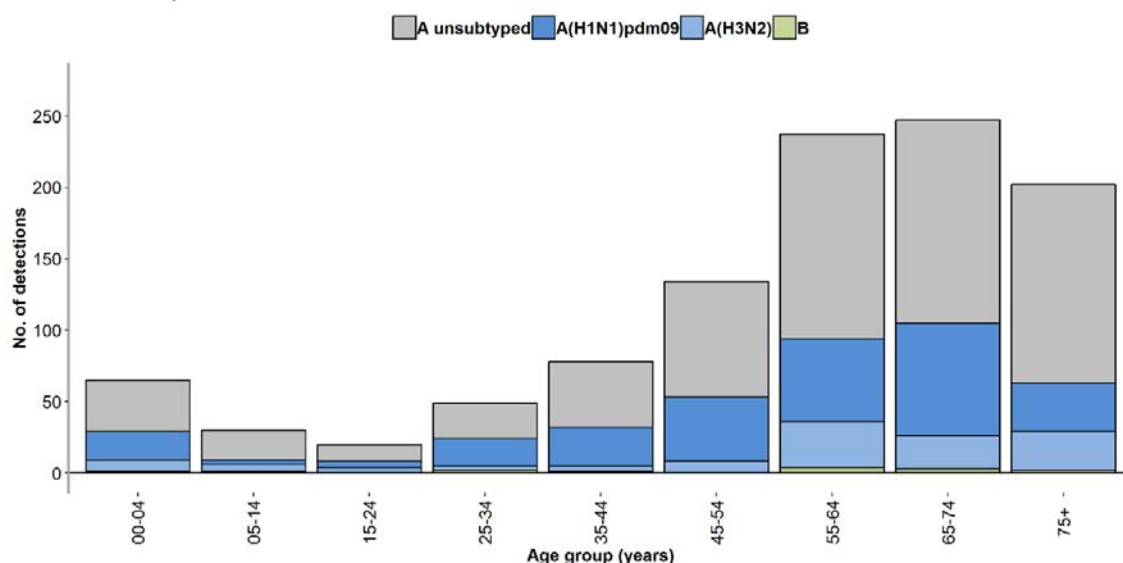
Figure E3. Patterns of influenza type circulation in the WHO European Region^a, weeks 40/2018-4/2019



^aDominance of (sub)type is based on at least 60% of sentinel detections and co-dominance is based on between 40% and 60% of detections. Calculations only considered valid if at least 10 viruses detected from sentinel sources.

Figure E4. Hospitalised influenza cases

A. Distribution of virus (sub)type by age group in laboratory-confirmed hospitalised cases in intensive care units^a, weeks 40/2018-4/2019



^aReporting countries: Czech Republic, Denmark, Estonia, Finland, France, Ireland, the Netherlands, Portugal, Romania, Russian Federation, Spain, Sweden, Ukraine and the United Kingdom

B. Number of Severe Acute Respiratory Infection (SARI) cases and influenza positives by country, territory or area, weeks 40/2018-4/2019

Country, territory or area	SARI cases	SARI Specimens Tested for Influenza	Influenza A	Influenza B	Percent Positive
Albania	291	291	128	0	44%
Armenia	2087	735	163	0	22%
Azerbaijan	53	5	5	0	100%
Belarus	268	263	4	0	2%
Belgium	704	704	77	2	11%
Bosnia and Herzegovina	186	191	131	0	69%
Georgia	2320	412	199	0	48%
Kazakhstan	9785	1118	189	1	17%
Kyrgyzstan	2022	186	73	1	40%
Moldova, Republic of	2360	171	30	0	18%
Montenegro	22	21	18	0	86%
Russian Federation	676	655	67	0	10%
Serbia	265	119	43	0	36%
Tajikistan	74	13	0	0	0%
Turkmenistan	103	57	1	0	2%
Ukraine	532	251	103	0	41%
Uzbekistan	145	29	3	0	10%
Kosovo ^b	23	23	4	0	17%
Total	21916	5244	1238	4	24%

^bin accordance with. United Nations Security Council. Resolution 1244 (1999)

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2018-08-31, totalling 1,855, have been received from 57 countries in Europe, Africa, the Middle East, the Indian Ocean, Central Asia and the Far East. Virus isolates (V) represented 28% of the samples received and 62% were sent solely as clinical specimens (CS), 165 samples were sent as CS/V pairs. Of the unique samples, approximately 92% were type A viruses, with influenza A(H1N1)pdm09 viruses and A(H3N2) viruses being received in similar numbers (**Table 1**). Only small numbers of influenza B viruses have been received with viruses of the B/Yamagata lineage and the B/Victoria lineage received in similar numbers.

Table 2 shows the recovery/isolation rates from samples received with collection dates after 2018-08-31. Viruses were recovered efficiently. For samples sent as CS, that were taken for culture, virus recoveries of between 70% and 78% were achieved, depending on virus subtype or lineage. Recoveries of samples sent as viruses were $\geq 95\%$. For A(H3N2) viruses, efficient recovery was obtained as assessed by detection of virus neuraminidase activity using the MUNANA sialidase assay, but only a minority (approximately 30%) of these could be assessed by HI assay as the majority of viruses had low HA titres (<4 HA units in the presence of 20nM oseltamivir).

Table 3 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and viruses of the B/Victoria and B/Yamagata lineages with post-infection ferret antisera raised against vaccine viruses or reference viruses in HI assays. For A(H3N2) viruses, the reactivity of test viruses in plaque reduction neutralisation assays (PRNA) is also summarised. Numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown. For A(H1N1)pdm09 test viruses the HI titres of an antiserum raised against the vaccine virus, egg-propagated A/Michigan/45/2015, were compared to the homologous titre (that with the vaccine virus). For A(H3N2) test viruses the HI titres of antisera raised against cell culture-propagated A/Stockholm//6/2014 (a 3C.3a virus), cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a virus), the vaccine virus egg-propagated A/Singapore/INFIMH-16-0019/2016 (a 3C.2a1 virus) and the vaccine virus egg-propagated A/Switzerland/8060/2017 (a 3C.2a2 virus) were compared to the titres of the antisera with their homologous viruses. For the A(H3N2) test viruses assessed by PRNA the 50% neutralising titres of antisera raised against the tissue culture-propagated and the egg-propagated cultivars of A/Singapore/INFIMH-16-0019/2016 and egg-propagated A/Switzerland/8060/2017 were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Victoria lineage the HI titres of test viruses with antisera raised against one of two tissue culture-propagated B/Brisbane/60/2008-like equivalents (B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016 – clade 1A viruses) and also tissue-culture propagated B/Colorado//06/2017 (a 1A- Δ 2 virus) were compared to the titres of the antisera with their homologous viruses. For Influenza B viruses of the B/Yamagata lineage the HI titres of test viruses with an antiserum raised against the egg-propagated vaccine virus, B/Phuket/3073/2013, were compared to the titre of the antiserum with the homologous virus.

Table 1a. Summary of clinical samples and virus isolates received, with collection dates after 2018-08-31, by country*

MONTH	TOTAL RECEIVED	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2018													
SEPTEMBER													
Cote d'Ivoire	7									4	in process	3	in process
Egypt	7			2	0	2	0	3	0				
France	7			6	3	2	3					1	1
Ghana	3	1	0							1	1		
Iran	4			1	1								
Madagascar	7			4	4							1	1
Mauritius	4			6	in process	2	0	1				2	2
Oman	4					2	in process						
Saudi Arabia	6	5	in process	2	in process							1	in process
Spain	1			1	in process								
Sweden	1			1	in process								
OCTOBER													
Cameroon	32					26	in process	6	in process				
Cote d'Ivoire	7					1	in process			3	0	3	0
Czech Republic	2			2	2								
Denmark	2					2	0	2					
Egypt	1					1	0	1					
Estonia	3	1	0	1	0	1	0	1					
Finland	2			1	1	1	0	1					
France	9			3	in process	6	4	2					
Germany	1					1	0	1					
Ghana	12	4	in process	1	1	1	0	1		5	2	1	1
Hong Kong	1					1	in process						
Iceland	2					1	0	1				1	1
Ireland	3			2	in process	1	0	1					
Israel	1			1	in process								
Jordan	1					1	in process						
Latvia	1			1	1								
Madagascar	11			11	in process								
Mauritius	4			1	1	3	3	0					
Netherlands	1					1	0	1					
Norway	29			12	7	14	0	8				3	1
Oman	187			49	in process	125	in process			3	3	2	0
Palestine	1			1	in process								
Portugal	2			2	in process								
Slovenia	1			1	in process								
United Kingdom	3			1	1	2	0	2					
NOVEMBER													
Austria	4			1	1	3	1	2					
Bosnia and Herzegovina	2					2	in process						
Belgium	5			3	in process	2	in process						
Bulgaria	1			1	in process								
Cameroon	46			1	in process	35	in process	10	in process	9	in process	2	0
Cote d'Ivoire	15			1	0	3	0						
Croatia	1			1	in process								
Czech Republic	1			1	1								
Denmark	12			8	in process	3	0	3		1	1		
Estonia	3			3	in process								
Finland	4			2	2	2	0	2					
France	16			10	in process	6	3	2					
Georgia	2			2	in process								
Germany	8			4	in process	4	0	4					
Ghana	22	1	0	8	in process	7	in process			5	3	1	1
Hong Kong	9			1	in process	2	in process			5	5	1	1
Iceland	15			4	3	11	7	3					
Iran	6					5	0	5				1	1
Ireland	17			12	in process	4	0	3				1	1
Israel	11			3	in process	8	in process						
Italy	10			2	in process	8	in process						
Jordan	4			2	in process	2	in process						
Kazakhstan	2			2	in process								
Kyrgyzstan	10			10	in process								
Latvia	2					2	1	1					
Lithuania	5					5	in process						
Madagascar	3			1	0							2	1
Mauritius	2			1	1	1	0	0					
Moldova	3			3	3								
Netherlands	3			2	2	1	0	1					
Norway	26			14	in process	12	1	10					
Oman	204			69	in process	120	in process			5	0	2	2
Palestine	5	1	in process	4	in process							8	4
Portugal	1											1	in process
Russia	7			5	5	2	1	1					
Spain	8			2	in process	6	in process						
Sweden	1			1	in process								
Switzerland	3			1	in process	2	1	1					
Turkey	9			1	in process	8	in process						
Turkmenistan	1			1	in process								
Ukraine	4					4	in process						
United Kingdom	11			6	in process	4	0	1				1	1

Table 1b. Summary of clinical samples and virus isolates received, with collection dates after 2018-08-31, by country*

MONTH	TOTAL RECEIVED		A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage				
Country			Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹			
DECEMBER																	
Armenia	7				7	in process											
Austria	1						1	0	1								
Bosnia and Herzegovina	9				1	1	8	0	1								
Belgium	6				2	in process	4	in process									
Bulgaria	9				5	in process	4	0	4								
Cameroon	4				1	in process	1	in process		1	in process						
Cote d'Ivoire	3				1	in process	2	in process				1	0				
Croatia	7				5	in process	2	0									
Cyprus	3				3	in process											
Denmark	7				5		2	0	2								
Estonia	18	1	in process		16	in process	1	0									
France	33				17	in process	14	in process			1	1	1	in process			
Georgia	9				5	in process	4	in process									
Germany	11				5	in process	6	in process									
Ghana	10				1	0	4	0	2	1	in process	4	in process				
Greece	11				8	in process	3	in process									
Hungary	6				4	4	2	1	1								
Hong Kong	13				6	in process	4	0	4			1	1	2	2		
Iceland	3				3	3											
Iran	0																
Ireland	3				3	3											
Israel	16				9	in process	7	in process									
Italy	1				1	1											
Jordan	10				3	in process	7	in process									
Kazakhstan	22				16	in process	6	in process									
Kyrgyzstan	20				17	in process	3	in process									
Latvia	6				5	5	1	1	0								
Lithuania	14	1	in process		5	3	8	in process									
Macedonia	1						1	in process									
Madagascar	9								3	0	5	in process	1	0			
Moldova	9				5	in process	4	in process									
Montenegro	2				1	in process					1	1					
Netherlands	5				4	in process	1	in process									
Norway	15				6	in process	7	in process			2	in process					
Palestine	4	1	in process		1	in process	2	in process									
Poland	1				1	0											
Portugal	18				8	in process	9	in process					1	1			
Romania	12				2	in process	10	in process									
Russia	18				7	in process	11	in process									
Serbia	4				4	in process											
Slovenia	3				1	in process					2	2					
Spain	28				15	in process	13	in process									
Sweden	11				7	in process	4	3	1								
Switzerland	13				9	in process	4	0	4								
Tunisia	4						3	in process		1	in process						
Turkey	49				7	in process	41	in process	1	in process							
Ukraine	32	3	in process		9	in process	20	in process									
United Kingdom	2						2	0									
2019 JANUARY																	
Albania	14				3	in process	11	in process									
Armenia	10				7	in process	3	1	2								
Bosnia and Herzegovina	3				2	in process	1	0	0								
Belgium	47				8	in process	39	in process									
Bulgaria	8				7	in process	1	0	1								
Croatia	2												2	1			
Cyprus	17				17	in process											
France	26				11	in process	15	in process									
Georgia	9				9	in process											
Germany	11				4	in process	7	in process									
Greece	24				17	in process	7	in process									
Hong Kong	7				5	5	2	0	2								
Hungary	2						2	2									
Italy	4				3	3	1	0	1								
Jordan	6				2	in process	4	in process									
Kazakhstan	6				6	in process											
Lithuania	1				1	1							1	1			
Luxembourg	25				10	in process	14	in process									
Macedonia	19				16	in process	3	in process									
Moldova	8				7	in process	1	0	1								
Montenegro	8	2	0		6	in process											
Netherlands	12				8	in process	4	2	2								
Norway	19				10	in process	7	in process					2	in process			
Poland	6				6	3											
Portugal	5				1	1	4	3	1								
Romania	13				11	in process	2	in process									
Russia	1				1	1											
Serbia	5				5	in process											
Slovenia	11				8	in process	3	0	3								
Spain	73				30	in process	43	in process									
Switzerland	4				3	in process	1	0	1								
Tunisia	2				1	in process	1	in process									
Turkey	13				3	in process	9	in process	1	in process							
Ukraine	3	1	in process				2	in process									
United Kingdom	35	3	0		29	0	3	0									
57 Countries	1855	25	0	798	76	43.0%	891	38	102	48.0%	40	0	55	22	46	22	2.5%
						92.4%								7.6%			

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

1 virus is H1N2

Table 2. Isolation rates for specimens collected after 2018-08-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated ¹	Percent recovered	Number processed	Number propagated ¹	Percent recovered
H1	82	61	74%	153	149	97%
H3	70	5²	7%	170	49²	29%
		44³	70% (10%)⁴ 63%		112³	95% (30%)⁴ 66%
B Victoria	8	6	75%	22	21	95%
B Yamagata	9	7	78%	15	15	100%

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 3. Summary of isolates with collection dates after 2018-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays (PRNA)

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ¹	210	209 (99.5%)	1 (0.5%)	0 (0%)
H3 ^{2a}	54	15 (27.8%)	18 (33.3%)	21 (38.9%)
H3 ^{2b}	54	23 (42.6%)	15 (27.8%)	16 (29.6%)
H3 ^{2c}	54	7 (13.0%)	21 (38.9%)	26 (48.1%)
H3 ^{2d}	54	0 (0%)	1 (1.9%)	53 (98.1%)
H3 ^{3a}	173	48 (27.8%)	84 (48.6%)	41 (23.7%)
H3 ^{3b}	173	10 (5.8%)	24 (13.9%)	139 (80.4%)
H3 ^{3c}	173	17 (9.8%)	2 (1.2%)	154 (89.0%)
B/Victoria ⁴	27	7 (25.9%)	0 (0.0%)	20 (74.1%)
B/Victoria ⁵	27	8 (29.6%)	7 (25.9%)	12 (44.4%)
B/Yamagata ⁶	22	18 (81.8%)	2 (9.1%)	2 (9.1%)

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Ferret antiserum raised against vaccine virus A/Michigan/45/2015

2a. Ferret antiserum raised against tissue culture-propagated A/Stockholm/6/2014 (genetic subgroup 3C.3a)

2b. Ferret antiserum raised against tissue culture-propagated A/Hong Kong/5738/2014 (genetic subgroup 3C.2a)

2c. Ferret antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (genetic subgroup 3C.2a1)

2d. Ferret antiserum raised against egg-propagated A/Switzerland/8060/2017 (genetic subgroup 3C.2a2)

3a. Ferret antiserum raised against tissue culture-propagated A/Singapore/INFIMH-16-0019/2016 (genetic subgroup 3C.2a1) in PRNA (2-fold)

3b. Ferret antiserum raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (genetic subgroup 3C.2a1) in PRNA (2-fold)

3c. Ferret antiserum raised against egg-propagated A/Switzerland/8060/2017 (genetic subgroup 3C.2a2) in PRNA (2-fold)

4. Ferret antisera raised against at least one of two tissue-culture propagated B/Brisbane/60/2008-like equivalents (B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016)

5. Ferret antisera raised against tissue-culture propagated B/Colorado/06/2017 (1AΔ2)

6. Ferret antiserum raised against egg-propagated vaccine virus B/Phuket/3073/2013

Influenza A(H1N1)pdm09 virus analyses.

Influenza A(H1N1)pdm09 viruses have been received from NICs in 43 countries or administrative regions in four WHO Regions: African (3: Ghana, Madagascar, and Mauritius), Eastern Mediterranean (2: Iran and Oman), European (37) and Western Pacific (1: Hong Kong SAR) (**Table 1**). **Table 4** summarises the antigenic (HI) results for viruses recovered from clinical specimens with collection dates after 2018-08-31.

Tables 5-1 to 5-11 show the results of HI assays of H1N1 viruses carried out since our September 2018 report. Test viruses in each table are sorted by genetic group and then by date of collection and viruses with collection dates after 2018-08-31 are shown below red lines in **Tables 5-5 to 5-11**. A summary of the HI results for viruses with collection dates after 2018-08-31 is shown in **Table 5-12** and a summary of viruses by genetic group is shown in **Table 5-13**.

The vast majority (99.5%) were antigenically indistinguishable from the egg-propagated vaccine virus, A/Michigan/45/2015, being recognised at titres within 2-fold of the titre of the antiserum with the homologous virus (**Table 4** and **Table 5-12**) with only a single virus, A/Hong Kong/110/2019, being recognised at a titre 4-fold lower than the homologous titre of the antiserum.

Antisera raised against seven reference viruses (A/Bayern/69/2009, A/Astrakhan/1/2011, A/Hong Kong/5659/2012, A/Slovenia/2903/2015, A/Paris/1447/2017, A/Switzerland/2656/2017 and A/Norway/3433/2018) recognised ≥85% of test viruses with collection dates after 2018-08-31 at titres within 2-fold of the titres of the antisera with their homologous viruses and over 98% at titres within 4-fold of the respective homologous titres (**Table 5-12**). The antiserum raised against A/Switzerland/3330/2017 recognised 93% of test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus. The antisera raised against egg-propagated A/California/7/2009 and cell culture-propagated A/Lviv/N6/2009 recognised 89% and 63%, respectively, of test viruses at titres within 4-fold of the homologous titres. The antiserum raised against A/Lviv/N6/2009 is an unusual virus/antiserum combination with A/Lviv/N6/2009 encoding **HA1** amino acid substitutions of **G155G/E**, with E predominating, and **D222G**. The seasonal A(H1N2) reassortant virus recovered in Sweden, A/Ystad/1/2018, reacted at titres equivalent to or better than the respective homologous titres with all antisera in the panel with the exception of that raised against A/Lviv/N6/2009.

All test viruses for which the genetic group or subgroup were known, with collection dates after 2018-08-31, fell into clade 6B.1 which is defined by the amino acid substitutions **S84N**, **S162N** (introducing a potential N-linked glycosylation site) and **I216T** in **HA1** with the viruses clustering into a genetic subclade designated as 6B.1A and defined by the **HA1** amino acid substitutions **S74R**, **S164T** (which alters the glycosylation motif at residues **162 to 164**) and **I295V**. **Figure 1** shows a phylogenetic tree for the HA genes of H1N1 viruses sequenced at the Francis Crick Institute and other representative viruses. The HA of most test viruses carry the substitution **S183P** in **HA1**, although this is not retained in all genetic groups. Within subclade 6B.1A

clusters of viruses encoding a range of **HA1** amino acid substitutions have emerged, e.g. **T120A**, or **N260D** in combination with **N129D** many with **T185I**, or **N260D** with **E235D** and **V193A** in **HA2**, or **N129D** with **A141E**, or **K302T** and **N169S** and **E179D** in **HA2**, or **L161I** and **I77M** in **HA2**. **Table 5-13** shows that viruses in each of six genetic subgroups show similar patterns of recognition by the panel of antisera.

An amino acid sequence alignment of the HAs for representative H1N1 viruses is shown in **Figure 2**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 3 illustrates the location of amino acid residues on an H1-HA structure defining genetic clade 6.B1 and the substitutions defining the major genetic subgroup within the 6B.1A subclade (**S74R**, **S164T** and **I295V** in **HA1**) and those that define two main clusters of viruses (**T120A**, **S183P/S** and **N129D** and **N260D** in **HA1**).

Figure 4 shows a phylogenetic tree for the NA genes of H1N1 viruses sequenced at the Francis Crick Institute and other representative viruses. The vast majority encoded the amino acid substitutions **G77R**, **V81A**, **I188T** and **N449D**, compared with the NA of A/Michigan/45/2015 and many also carry the substitution **D416N** in combination with **N222D** or **Q51K** and **T452I**.

Figure 5 shows a NA amino acid alignment for the same representative set of H1N1 viruses as used in **Figure 2**, annotated as **Figure 2** but lacking a proteolytic-processing site.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 218 viruses with collection dates after 2018-08-31. All viruses but one showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 14**). The exception, A/Antalya/5/2018 from Turkey, showed highly reduced inhibition (HRI) by oseltamivir and normal inhibition by zanamivir and its NA carried a H275Y amino acid substitution.

Conclusion.

The vast majority of A(H1N1)pdm09 viruses received fell into a predominant genetic subgroup within clade 6B.1 encoding the **HA1** amino acid substitutions **S74R**, **S164T** (which alters the glycosylation motif at residues **162 to 164**) and **I295V**, subclade 6B.1A, with a number of virus clusters, defined by additional amino acid substitutions, being present within the subgroup. All test viruses but one remained antigenically similar to the vaccine virus, A/Michigan/45/2015, and the vast majority (>99%) also retained susceptibility to oseltamivir and zanamivir.

Table 4. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2018-08-31

MONTH	H1N1pdm09				
	Number received ¹	Number propagated ²	Fold reduction in HI titre		
Country			≤2 fold ³	4 fold ³	≥8 fold ³
2018					
SEPTEMBER					
Ghana	1	1	1		
Iran	4	4	4		
Madagascar	6	3	3		
OCTOBER					
Czech Republic	2	2	2		
Finland	1	1	1		
France	3	2	2		
Ghana	1	1	1		
Latvia	1	1	1		
Mauritius	1	1	1		
Norway	12	7	7		
United Kingdom	1	1	1		
NOVEMBER					
Austria	1	1	1		
Czech Republic	1	1	1		
Denmark	8	1	1		
Finland	2	2	2		
France	10	8	8		
Germany	4	3	3		
Ghana	8	2	2		
Iceland	4	3	3		
Ireland	12	3	3		
Mauritius	1	1	1		
Moldova	3	3	3		
Netherlands	2	2	2		
Norway	14	2	2		
Oman	69	7	7		
Russia	5	5	5		
United Kingdom	5	5	5		
DECEMBER					
Bosnia and Herzegovina	1	1	1		
Croatia	6	3	3		
Denmark	5	5	5		
Estonia	16	10	10		
France	17	6	6		
Hong Kong	6	1	1		
Hungary	4	4	4		
Iceland	3	3	3		
Ireland	3	3	3		
Israel	9	4	4		
Italy	1	1	1		
Latvia	5	5	5		
Lithuania	5	3	3		
Moldova	5	1	1		
Netherlands	4	2	2		
Norway	6	2	2		
Portugal	8	3	3		
Russia	7	1	1		
Serbia	4	1	1		
Sweden	7	5	5		
Switzerland	9	6	6		
Turkey	7	4	4		
Ukraine	9	3	3		
2019					
JANUARY					
Bosnia and Herzegovina		1	1		
Bulgaria	2	5	5		
Cyprus	7	3	3		
France	11	2	2		
Germany	4	3	3		
Georgia	9	6	6		
Greece	17	2	2		
Hong Kong	5	5	4	1	
Italy	3	3	3		
Lithuania	1	1	1		
Luxembourg	10	4	4		
Montenegro	6	3	3		
Poland	6	3	3		
Portugal	1	1	1		
Romania	11	4	4		
Russia	1	1	1		
Serbia	5	3	3		
Slovenia	8	6	6		
Spain	30	4	4		
43 Countries	466	210	209 99.5%	1 0.5%	0 0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assays

3. Reduction in HI titre with post-infection ferret antiserum raised against A/Michigan/45/2015 (egg-propagated vaccine virus)

Includes the A(H1N2) virus - A/Ystad/1/2018

Table 5-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-09-19)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg NIB F42/16 ^{*1}	A/Cal 7/09 Egg F07/16 ^{*1}	A/Bayern 69/09 MDCK F09/15 ^{*1}	A/Lviv N6/09 MDCK F14/13 ^{*1}	A/Astrak 1/11 MDCK F22/13 ^{*1}	A/HK 5659/12 MDCK F17/15 ^{*1}	A/Slov 2903/2015 Egg F02/16 ^{*1}	A/Paris 1447/17 MDCK F03/18 ^{*2}	A/Swit 2656/17 Egg F20/18 ^{*1}	A/Swit 3330/17 Egg F23/18 ^{*1}
				Passage history									
				Ferret number									
Genetic group				6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5
REFERENCE VIRUSES													
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	40	40	320	320	40	40	80	320	160	80
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK2	80	160	1280	1280	80	80	160	640	640	320
A/Astrakhan/1/2011		2011-02-28	MDCK1/MDCK7	640	1280	640	640	1280	1280	1280	2560	1280	640
A/Hong Kong/5659/2012		2012-05-21	MDCK4/MDCK2	320	640	160	80	320	640	640	1280	640	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	640	1280	320	160	640	640	1280	1280	1280	640
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	640	1280	640	320	1280	640	1280	2560	2560	640
A/Switzerland/2656/2017		2017-12-21	E5/E1	640	2560	640	640	1280	1280	2560	2560	2560	640
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E1	640	1280	320	320	640	640	1280	2560	1280	640
TEST VIRUSES													
A/Armenia/1/2018		2018-01-08	MDCK1/MDCK1	640	640	160	160	320	320	640	640	1280	640
A/Armenia/2/2018		2018-01-15	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	1280	640
A/Cote d'Ivoire/110/2018		2018-01-29	MDCK1	320	640	160	160	320	320	640	640	640	640
A/Armenia/3/2018		2018-02-15	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	1280	640
A/Armenia/5/2018		2018-02-22	MDCK1/MDCK1	640	640	320	160	640	640	1280	1280	1280	640
A/Armenia/6/2018		2018-02-23	MDCK1/MDCK1	640	640	320	160	640	320	1280	640	1280	640
A/Armenia/9/2018		2018-02-27	MDCK1/MDCK1	640	640	160	160	320	320	640	640	1280	640
A/Armenia/10/2018		2018-03-02	MDCK1/MDCK1	640	640	320	160	640	320	640	640	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

Vaccine

1 <= <40; 2 <= <80

Table 5-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-09-25)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Mich 45/15 Egg NIB F42/16 ^{*1}	A/Cal 7/09 Egg F07/16 ^{*1}	A/Bayern 69/09 MDCK F09/15 ^{*1}	A/Lviv N6/09 MDCK F14/13 ^{*1}	A/Astrak 1/11 MDCK F22/13 ^{*1}	A/HK 5659/12 MDCK F17/15 ^{*1}	A/Slov 2903/2015 Egg F02/16 ^{*1}	A/Paris 1447/17 MDCK F03/18 ^{*2}	A/Swit 2656/17 Egg F20/18 ^{*1}	A/Swit 3330/17 Egg F23/18 ^{*1}
				Passage history	Ferret number	Genetic group							
REFERENCE VIRUSES													
A/California/7/2009	clone 38-32	2009-04-09	E3/E3	640	1280	320	320	640	640	1280	1280	1280	640
A/Bayern/69/2009	G155E	2009-07-01	MDCK5/MDCK1	<	40	320	160	40	40	40	160	160	80
A/Lviv/N6/2009	G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	80	80	160	640	320	160
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK7	640	1280	640	640	1280	640	1280	2560	1280	640
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	640	160	160	320	320	640	640	640	320
A/Slovenia/2903/2015	clone 37	2015-10-26	E4/E2	640	1280	320	320	640	640	1280	1280	1280	640
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	640	640	320	160	640	640	1280	2560	1280	640
A/Switzerland/2656/2017		2017-12-21	E5/E1	1280	2560	1280	640	1280	1280	2560	5120	>5120	1280
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E1	640	640	640	320	640	640	1280	2560	1280	1280
TEST VIRUSES													
A/Armenia/484/2018		2018-02-15	MDCK1	640	1280	640	320	640	640	1280	2560	1280	1280
A/Armenia/481/2018		2018-02-15	MDCK1	640	1280	640	320	640	640	1280	2560	1280	1280
A/Armenia/541/2018		2018-02-19	MDCK1	640	1280	320	320	640	640	1280	2560	1280	1280
A/Armenia/538/2018		2018-02-19	MDCK1	1280	1280	640	320	640	1280	1280	2560	1280	1280
A/Armenia/556/2018		2018-02-21	MDCK1	640	1280	640	320	640	640	1280	2560	2560	1280
A/Armenia/589/2018		2018-02-27	MDCK1	640	1280	640	320	640	640	1280	2560	2560	1280
A/Armenia/610/2018		2018-02-28	MDCK1	1280	2560	640	640	1280	1280	2560	5120	2560	1280
A/Armenia/605/2018		2018-02-28	MDCK1	640	1280	640	320	640	640	1280	2560	2560	1280
A/Armenia/653/2018		2018-03-02	MDCK1	640	1280	320	320	640	640	1280	2560	1280	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

Vaccine

1 <= <40; 2 <= <80

Table 5-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-10-10)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					NEW											
					A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/Astrak 1/11	A/HK 5659/12	A/Slov 2903/2015	A/Paris 1447/17	A/Swit 2656/17	A/Swit 3330/17	
					Egg	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	Egg	
Ferret number	F07/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F22/13 ⁻¹	F33/18 ⁻²	F17/15 ⁻¹	F02/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹						
Genetic group	6B.1				5	5	6A	6B.1	6B.1A	6B.1A	6B.1A5					
REFERENCE VIRUSES																
A/Michigan/45/2015	clone 38-32	6B.1	2015-09-07	E3/E3	640	1280	320	320	640	640	640	1280	2560	1280	640	
A/California/7/2009			2009-04-09	E3/E3	640	640	320	320	320	320	640	1280	1280	640		
A/Bayern/69/2009		G155E	2009-07-01	MDCK5/MDCK1	<	40	320	160	<	<	<	40	320	160	80	
A/Lviv/N6/2009		G155E, D222G	2009-10-27	MDCK4/SIAT1/MDCK3	<	80	640	640	40	80	80	160	640	640	160	
A/Astrakhan/1/2011			5	2011-02-28	MDCK1/MDCK7	640	1280	640	320	640	640	640	1280	2560	1280	640
A/Hong Kong/5659/2012	6A		2012-05-21	MDCK4/MDCK2	320	320	160	80	160	160	320	640	640	640	320	
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	1280	320	320	640	640	640	1280	1280	1280	1280	
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	640	320	160	320	640	640	1280	2560	2560	640	
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	640	640	320	640	640	640	1280	1280	640	320	
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E1	640	640	320	320	320	320	640	1280	2560	1280	1280	
TEST VIRUSES																
A/Qatar/16-VI-18-0023245/2018		6B.1	2018-04-09	MDCK1	320	640	640	320	640	640	1280	1280	5120	640	640	
A/Qatar/10-VI-18-0028800/2018		6B.1	2018-05-03	MDCK1	1280	1280	640	320	640	1280	1280	2560	5120	2560	1280	
A/Qatar/16-VI-18-0034073/2018		6B.1	2018-06-04	MDCK1	640	1280	320	320	640	640	640	1280	2560	2560	1280	
A/Central African Republic/804/2018		6B.1	2018-06-18	MDCK1/MDCK1	1280	2560	640	320	640	1280	1280	2560	5120	2560	1280	
A/Central African Republic/843/2018		6B.1	2018-06-29	MDCK1/MDCK1	640	1280	320	320	320	640	640	1280	2560	2560	1280	
A/Qatar/16-vi-18-0039724/2018		6B.1	2018-07-16	MDCK1	640	640	320	160	320	640	640	1280	2560	1280	640	
A/Qatar/10-vi-18-0039784/2018		6B.1	2018-07-16	MDCK1	320	80	160	80	160	160	160	320	640	640	320	
A/Qatar/16-vi-18-0040447/2018		6B.1	2018-07-21	MDCK1	640	1280	640	320	640	640	640	1280	2560	2560	1280	
A/Qatar/10-VI-18-0042958/2018		6B.1	2018-08-09	MDCK1	640	640	320	160	320	640	640	1280	2560	1280	640	
A/Qatar/10-VI-18-0043130/2018		6B.1	2018-08-11	MDCK1	640	640	320	160	320	640	640	1280	2560	1280	640	
A/Qatar/10-VI-18-0043633/2018		6B.1	2018-08-15	MDCK1	640	640	320	320	320	320	640	1280	2560	1280	640	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Vaccine

Table 5-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-10-24)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg NIB F42/16 ⁻¹	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/HK 5659/12 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg
					F07/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F22/13 ⁻¹	F17/15 ⁻¹	F02/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹	
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	320	320	320	640	1280	2560	1280	640
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	640	1280	320	320	320	640	640	1280	1280	320
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	80	320	320	40	40	80	320	160	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	160	1280	1280	40	80	160	640	640	320
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	1280	2560	1280	640	1280	1280	2560	5120	2560	1280
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	640	1280	320	160	320	640	1280	2560	1280	640
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	1280	320	320	320	640	1280	1280	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	640	320	160	320	640	1280	2560	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	1280	1280	640	640	640	640	1280	2560	2560	1280
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E1	640	640	320	320	320	640	1280	2560	1280	1280
TEST VIRUSES														
A/Central African Republic/568/2018		6B.1	2018-04-24	MDCK1/MDCK2	1280	2560	640	320	640	1280	5120	>5120	>5120	>5120
A/Central African Republic/576/2018			2018-04-26	MDCK1/MDCK2	640	1280	640	320	320	640	1280	2560	2560	1280
A/Saudi Arabia/103697202/2018		6B.1	2018-05-13	MDCK1	640	1280	320	160	320	640	1280	2560	2560	1280
A/Saudi Arabia/P4649405/2018		6B.1	2018-05-15	MDCK1	640	640	320	160	320	320	1280	2560	2560	1280
A/Central African Republic/700/2018			2018-05-24	MDCK1/MDCK2	640	640	320	160	320	320	640	1280	1280	640
A/Saudi Arabia/1014307787/2018		6B.1	2018-06-06	SIAT1	320	640	160	160	160	160	640	1280	1280	320
A/Saudi Arabia/1152587505/2018		6B.1	2018-06-24	MDCK1	640	320	160	160	160	320	640	1280	1280	640
A/Oman/3647/2018		6B.1	2018-07-16	MDCK1	1280	2560	640	320	640	640	2560	>5120	>5120	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine

Table 5-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-11-20)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg NIB F42/16 ^{*1}	A/Cal 7/09 Egg F07/16 ^{*1}	A/Bayern 69/09 MDCK F09/15 ^{*1}	A/Lviv N6/09 MDCK F14/13 ^{*1}	A/Astrak 1/11 MDCK F22/13 ^{*1}	A/HK 5659/12 MDCK F17/15 ^{*1}	A/Slov 2903/2015 Egg F02/16 ^{*1}	A/Paris 1447/17 MDCK F03/18 ^{*2}	A/Swit 2656/17 Egg F20/18 ^{*1}	A/Swit 3330/17 Egg F23/18 ^{*1}
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	320	320	320	640	1280	2560	1280	640
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	320	1280	320	320	320	320	640	1280	1280	640
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	<	40	320	320	<	40	40	320	160	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	160	640	1280	40	80	160	640	320	320
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	640	1280	640	640	640	640	1280	2560	1280	640
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	640	160	80	160	320	640	1280	640	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	320	640	160	160	320	640	640	1280	1280	640
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	640	160	80	160	320	640	1280	1280	640
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	1280	320	320	320	640	1280	1280	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E1	320	640	160	160	320	320	640	1280	1280	640
TEST VIRUSES														
A/Saudi Arabia/24/2018		6B.1	2018-05-02	MDCK1	320	640	160	40	160	320	640	1280	1280	320
A/Ghana/2325/2018		6B.1	2018-07-03	C1/MDCK1	320	640	160	80	160	320	640	1280	1280	640
A/Ghana/2345/2018		6B.1	2018-07-09	C1/MDCK1	640	640	320	160	320	640	1280	2560	1280	640
A/Ghana/512/2018		6B.1	2018-07-17	C1/MDCK1	320	640	160	80	160	160	640	1280	640	320
A/Ghana/2597/2018		6B.1	2018-07-30	C1/MDCK1	640	640	160	80	160	320	640	2560	1280	640
A/Ghana/594/2018		6B.1	2018-08-23	C1/MDCK1	640	640	160	40	160	320	640	1280	1280	640
A/Ghana/561/2018		6B.1A5	2018-09-08	C1/MDCK1	640	640	320	80	160	320	640	2560	1280	640
A/Ghana/2419/2018		6B.1A5	2018-11-07	C1/MDCK1	640	1280	320	160	320	640	1280	2560	2560	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine

Table 5-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-11-27 + 2018-12-04)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/HK 5659/12	A/Slov 2903/2015	A/Paris 1447/17	A/Swit 2656/17	A/Swit 3330/17
					Egg	Egg	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	Egg
					F42/16 ⁻¹	F07/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F22/13 ⁻¹	F17/15 ⁻¹	F02/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹
Genetic group	6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5				
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	1280	320	320	640	640	1280	2560	1280	640
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	640	1280	320	320	320	640	1280	1280	640	320
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	80	320	320	40	40	80	320	80	80
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	160	1280	1280	40	160	320	640	320	160
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	1280	320	320	320	640	1280	1280	640	320
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	1280	320	160	320	640	640	1280	320	320
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	1280	320	320	320	640	1280	2560	1280	320
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	640	320	160	320	320	1280	1280	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	1280	640	320	640	640	1280	2560	1280	640
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	640	640	320	160	320	640	1280	1280	640	640
TEST VIRUSES														
A/Mauritius/I-545/2018		6B.1	2018-07-08	MDCK1/MDCK1	640	320	320	320	320	640	1280	2560	640	640
A/Mauritius/SF55/2018		6B.1	2018-07-16	MDCK1/MDCK1	1280	2560	640	640	640	1280	2560	5120	1280	640
A/Norway/3282/2018			2018-10-08	MDCK1/MDCK1	640	1280	320	160	320	640	1280	2560	640	320
A/Norway/3283/2018	S190X(S/R)	6B.1A5	2018-10-12	MDCK1/MDCK1	320	320	160	80	80	160	640	640	320	160
A/Mauritius/I-975/2018		6B.1A5	2018-10-27	MDCK1/MDCK1	640	640	320	320	160	320	640	1280	640	320
A/Norway/3415/2018		6B.1A5	2018-10-30	MDCK1	640	640	320	320	160	320	640	1280	640	320
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK1	640	1280	320	320	320	640	1280	2560	640	640
A/Norway/3401/2018		6B.1A5	2018-10-31	MDCK1	640	640	320	320	320	640	1280	2560	640	320
A/Mauritius/I-1002/2018		6B.1A5	2018-11-06	MDCK1/MDCK1	640	640	320	320	160	320	640	1280	320	320
A/Norway/3316/2018		6B.1A7	2018-10-19	MDCK1	640	640	320	160	320	320	640	1280	640	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

Vaccine

1 <= <40; 2 <= <80

Table 5-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2018-12-12)

				Haemagglutination inhibition titre										
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				A/Mich	A/Cal	A/Bayern	A/Lviv	A/Astrak	A/HK	A/Slov	A/Paris	A/Swit	A/Swit	
				45/15	7/09	69/09	N6/09	1/11	5659/12	2903/2015	1447/17	2656/17	3330/17	
				Egg	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	
				NIB	F07/16 ^{*1}	F09/15 ^{*1}	F14/13 ^{*1}	F22/13 ^{*1}	F17/15 ^{*1}	F02/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F23/18 ^{*1}	
F42/16 ^{*1}														
Genetic group				6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	
REFERENCE VIRUSES														
A/Michigan/45/2015		6B.1	2015-09-07	E3/E3	640	640	320	320	320	640	1280	2560	640	320
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	320	640	320	320	320	640	640	1280	640	320
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	80	320	160	<	40	80	320	80	40
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	160	640	640	40	160	160	640	160	160
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK5	640	1280	320	160	320	640	1280	2560	640	320
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	160	320	80	80	160	320	1280	640	160	80
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	640	1280	320	160	320	640	640	2560	640	320
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	640	160	80	160	320	640	2560	320	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	1280	640	320	320	640	1280	2560	640	320
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	640	160	160	160	320	640	1280	640	320
TEST VIRUSES														
A/Palestine/1JN2018/2017		6B.1	2017-12-13	MDCK2	320	320	160	80	160	320	640	640	320	160
A/Palestine/7H2018/2018		6B.1	2018-03-27	MDCK1	640	640	320	160	320	320	1280	1280	640	320
A/Palestine/3B2018/2018		6B.1	2018-04-02	MDCK1	640	640	320	160	320	320	1280	1280	640	320
A/Palestine/5Y2018/2018		6B.1	2018-05-24	MDCK1	320	640	160	160	160	320	640	1280	640	160
A/Norway/3399/2018		6B.1A7	2018-10-29	MDCK3	320	640	160	80	320	320	640	1280	640	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80

Vaccine

Table 5-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-01-16)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antisera									
					A/Mich 45/15 Egg NIB F42/16 ¹	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/HK 5659/12 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5
REFERENCE VIRUSES														
A/Michigan/45/2015	clone 38-32 G155E G155E, D222G	6B.1	2015-09-07	E3/E3	640	640	320	320	640	640	1280	2560	640	320
A/California/7/2009			2009-04-09	E3/E5	640	1280	640	640	640	1280	1280	2560	1280	640
A/Bavaria/69/2009			2009-07-01	MDCK5/MDCK1	<	<	320	160	<	<	40	160	40	40
A/Lviv/N6/2009			2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	640	40	80	160	640	320	160
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	640	640	640	320	320	640	640	2560	320	320
A/Hong Kong/5659/2012	clone 37	6A	2012-05-21	MDCK4/MDCK2	320	160	160	80	160	640	640	640	160	160
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	640	640	320	320	640	640	1280	2560	640	320
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	640	320	320	160	320	640	1280	2560	640	320
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	640	320	320	640	640	1280	2560	640	320
A/Switzerland/3330/2017		clone 35	6B.1A5	2017-12-20	E6/E2	320	320	160	160	160	320	320	1280	320
TEST VIRUSES														
A/Iran/137043/2018		6B.1A	2018-09-02	SIAT2/MDCK1	640	640	640	160	320	640	1280	5120	1280	320
A/Iran/137511/2018		6B.1A	2018-09-10	SIAT2/MDCK1	640	320	320	160	320	320	1280	2560	640	320
A/Iran/137742/2018		6B.1A	2018-09-15	SIAT2/MDCK1	640	320	320	160	320	320	1280	2560	640	320
A/Iran/138156/2018		6B.1A	2018-09-25	SIAT2/MDCK1	1280	640	320	320	320	640	1280	2560	1280	640
A/England/595/2018		6B.1A	2018-10-20	SIAT1/MDCK1	320	320	320	80	320	320	320	640	320	160
A/Caen/2372/2018		6B.1A	2018-10-24	MDCK1/MDCK1	640	320	320	160	320	320	1280	2560	640	320
A/England/603/2018		6B.1A	2018-11-05	MDCK1/MDCK1	640	320	320	160	320	640	1280	1280	640	320
A/England/607/2018		6B.1A	2018-11-14	SIAT1/MDCK1	320	320	320	80	320	320	640	1280	640	160
A/England/606/2018		6B.1A	2018-11-14	MDCK1/MDCK1	640	320	160	160	320	640	1280	640	640	320
A/Iceland/24352/2018		6B.1A	2018-11-27	MDCK1	640	320	320	80	160	320	640	1280	640	160
A/Iceland/24861/2018		6B.1A	2018-12-03	MDCK1	320	320	320	80	160	320	640	1280	320	320
A/Iceland/25090/2018		6B.1A	2018-12-06	MDCK1	640	640	320	160	320	640	1280	2560	640	320
A/Estonia/118411/2018		6B.1A	2018-12-28	SIAT1	640	320	320	160	320	640	1280	2560	640	320
A/Estonia/118399/2018		6B.1A	2018-12-28	SIAT1	640	640	640	160	320	640	1280	2560	1280	320
A/Estonia/118387/2018		6B.1A	2018-12-28	SIAT1	640	320	320	160	160	320	1280	2560	640	320
A/Paris/2313/2018		6B.1A5	2018-10-04	MDCK1/MDCK1	640	320	160	160	160	320	640	1280	320	160
A/England/608/2018		6B.1A5	2018-11-13	SIAT1/MDCK1	320	320	160	160	160	320	640	1280	320	320
A/Czech Republic/2018/2018		6B.1A5	2018-11-14	MDCK3/MDCK1	640	160	320	160	160	320	640	1280	320	320
A/Baden-Wuerttemberg/162/2018		6B.1A5	2018-11-20	C1/MDCK1	640	320	320	160	160	640	640	1280	320	320
A/Thuringen/61/2018		6B.1A5	2018-11-21	C1/MDCK1	320	320	320	160	320	320	640	1280	320	320
A/Moldova/754.47/2018		6B.1A5	2018-11-21	MDCK1	640	320	320	320	320	640	1280	2560	640	320
A/Norway/3594-2/2018		6B.1A5	2018-11-23	Px/MDCK1	320	160	160	160	160	320	640	1280	160	160
A/Iceland/111/2018		6B.1A5	2018-11-27	MDCK1/MDCK1	320	80	160	80	80	320	640	640	160	160
A/Picardie/2488/2018		6B.1A5	2018-11-27	MDCK1/MDCK1	320	160	160	80	160	160	320	640	160	160
A/Moldova/785.48/2018		6B.1A5	2018-11-28	MDCK1	640	640	640	320	320	640	1280	2560	640	320
A/Moldova/774.48/2018		6B.1A5	2018-11-29	MDCK1	320	320	320	160	160	320	640	1280	320	160
A/Austria/1108922/2018		6B.1A5	2018-11-29	SIAT1/MDCK1	1280	320	640	320	320	640	1280	2560	640	320
A/Norway/3750/2018		6B.1A5	2018-11-30	MDCK1	640	320	320	320	320	640	1280	2560	640	320
A/Iceland/24775/2018		6B.1A5	2018-12-01	MDCK1	640	640	640	320	320	640	1280	2560	640	320
A/Moldova/817.49/2018		6B.1A5	2018-12-03	MDCK1	640	320	320	320	320	640	1280	2560	640	320
A/Norway/3681/2018		6B.1A5	2018-12-04	MDCK1	320	160	320	160	160	320	640	1280	320	320
A/Norway/3743/2018		6B.1A5	2018-12-05	MDCK1	640	160	160	160	160	320	640	1280	160	160
A/Estonia/118397/2018		6B.1A5	2018-12-28	SIAT1	320	160	160	160	160	160	640	1280	160	160
A/Latvia/10-064408/2018		6B.1A7	2018-10-23	MDCK2/MDCK1	320	160	160	80	160	320	640	1280	320	160
A/Mecklenburg-Vorpommern/4/2018		6B.1A7	2018-11-01	C1/MDCK1	640	320	320	160	320	640	1280	2560	640	320
A/Paris/2406/2018		6B.1A7	2018-11-09	MDCK1/MDCK1	320	160	160	80	160	320	640	1280	320	160
A/England/605/2018		6B.1A7	2018-11-12	SIAT1/MDCK1	640	320	320	320	320	640	1280	2560	640	320
A/Picardie/2549/2018		6B.1A7	2018-12-04	MDCK1/MDCK1	640	640	320	320	320	640	1280	2560	640	640
A/Latvia/12-013993/2018		6B.1A7	2018-12-04	MDCK1/MDCK1	640	640	320	160	320	640	1280	2560	640	320
A/Latvia/12-014030/2018		6B.1A7	2018-12-05	MDCK1/MDCK1	640	640	320	160	320	640	1280	2560	640	320
A/Latvia/12-013970/2018		6B.1A7	2018-12-05	MDCK1/MDCK1	640	640	320	160	320	640	1280	1280	640	320
A/Latvia/12-027178/2018		6B.1A7	2018-12-11	MDCKx/MDCK1	320	160	160	80	160	160	640	1280	320	160
A/Czech Republic/1927/2018	A261S	6B.1A1	2018-10-13	MDCK4/MDCK1	640	320	160	80	160	640	1280	2560	320	160
A/Czech Republic/1928/2018		6B.1A1	2018-10-15	MDCK4/MDCK1	640	160	160	80	160	320	640	1280	320	160
A/Iceland/23365/2018		6B.1A1	2018-11-13	MDCK1	320	80	80	40	80	160	320	640	160	80
A/Dijon/2407/2018		6B.1A2	2018-11-07	MDCK1/MDCK1	640	640	320	160	320	640	1280	2560	640	320
A/Latvia/12-024461/2018		6B.1A6	2018-12-10	MDCKx/MDCK1	320	160	160	80	80	160	320	640	160	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine

Table 5-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-01-22)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					A/Mich 45/15 Egg	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/HK 5659/12 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg		
					F42/16 ⁻¹	F07/16 ⁻¹	F09/15 ⁻¹	F14/13 ⁻¹	F22/13 ⁻¹	F17/15 ⁻¹	F02/16 ⁻¹	F03/18 ⁻²	F20/18 ⁻¹	F23/18 ⁻¹		
					NIB											
Genetic group					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5		
REFERENCE VIRUSES																
A/Michigan/45/2015	clone 38-32 G155E G155E, D222G	6B.1	2015-09-07	E3/E3	320	320	320	320	320	320	640	1280	320	320		
A/California/7/2009		2009-04-09	E3/E3	320	320	320	640	320	640	640	1280	640	320			
A/Bavarn/69/2009		2009-07-01	MDCK5/MDCK1	<	<	320	160	<	<	40	160	40	<			
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	40	40	640	320	<	40	80	320	160	80			
A/Astrakhan/1/2011		2011-02-28	MDCK1/MDCK7	320	320	640	320	160	320	640	1280	320	320			
A/Hong Kong/5659/2012	clone 37	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	160	320	640	1280	160	160		
A/Slovenia/2903/2015		6B.1	2015-10-26	E4/E2	320	320	160	160	320	320	640	1280	320	320		
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	160	160	160	80	160	160	640	1280	160	160		
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E2	640	640	640	320	640	640	1280	2560	1280	640		
A/Switzerland/3330/2017		clone 35	6B.1A5	2017-12-20	E6/E2	320	160	160	160	160	160	640	1280	320	640	
TEST VIRUSES																
A/Ghana/3763/2018	D187X(D/N), S190X(S/R)	6B.1A	2018-11-07	C2/MDCK1	1280	640	320	160	320	640	1280	2560	640	640		
A/Oman/6052/2018		6B.1A	2018-11-14	MDCK2	320	80	160	80	80	80	320	640	160	160		
A/Estonia/118321/2018		6B.1A	2018-12-20	MDCK1	640	320	320	160	160	320	1280	1280	640	320		
A/Estonia/118316/2018		6B.1A	2018-12-20	MDCK1	160	80	80	40	80	80	320	640	160	80		
A/Estonia/118371/2018		6B.1A	2018-12-27	MDCK2	320	<	160	80	80	80	320	640	160	160		
A/Bulgaria/028/2019		6B.1A	2019-01-07	MDCK1	320	320	160	80	160	320	640	1280	320	320		
A/Bulgaria/041/2019		6B.1A	2019-01-08	MDCK1	640	320	320	160	160	320	1280	2560	640	320		
A/Ghana/3865/2018		6B.1A5	2018-10-02	MDCK1	320	160	160	80	80	160	640	1280	320	160		
A/Oman/5965/2018		6B.1A5	2018-11-11	MDCK1	640	80	160	160	160	160	640	1280	320	160		
A/Lyon/CHU-R18.112.40/2018		6B.1A5	2018-11-15	MDCK2/MDCK1	320	160	160	80	80	160	640	1280	160	160		
A/Lyon/2119/2018		6B.1A5	2018-11-22	MDCK2/MDCK1	640	160	320	160	160	320	640	1280	320	320		
A/Oman/6319/2018		6B.1A5	2018-11-23	MDCK1	320	1280	160	160	160	320	640	1280	320	320		
A/Oman/6338/2018		6B.1A5	2018-11-24	MDCK1	320	160	160	80	160	320	1280	1280	320	320		
A/Ireland/84720/2018		6B.1A5	2018-11-29	MDCK2/MDCK1	160	40	80	40	40	160	320	640	80	80		
A/Clermont-Ferrand/2180/2018		6B.1A5	2018-12-03	MDCK2/MDCK1	160	80	80	80	80	160	320	640	80	80		
A/Ireland/85958/2018		6B.1A5	2018-12-04	MDCK2/MDCK1	640	160	320	320	320	320	1280	2560	640	320		
A/Ireland/85950/2018		6B.1A5	2018-12-04	MDCK2/MDCK1	640	160	320	160	320	320	1280	1280	320	640		
A/Switzerland/4806/2018		6B.1A5	2018-12-10	MDCK2/MDCK1	640	320	320	320	320	320	1280	2560	640	640		
A/Switzerland/6296/2018		6B.1A5	2018-12-12	MDCK1/MDCK1	320	40	160	160	80	160	640	1280	160	160		
A/Lithuania/MB36587/2018		6B.1A5	2018-12-17	MDCK1/MDCK1	640	320	320	320	320	640	1280	2560	640	640		
A/Switzerland/7147/2018		6B.1A5	2018-12-18	MDCK1/MDCK1	640	320	320	160	320	320	1280	1280	640	320		
A/Estonia/118306/2018		6B.1A5	2018-12-19	MDCK1	640	320	320	160	160	320	640	1280	320	320		
A/Estonia/118323/2018		6B.1A5	2018-12-20	MDCK1	640	160	160	160	160	160	640	1280	160	160		
A/Estonia/118355/2018		6B.1A5	2018-12-21	MDCK1	1280	320	640	320	320	640	2560	2560	640	640		
A/Switzerland/671/2018		6B.1A5	2018-12-27	MDCK1/MDCK1	320	160	160	160	160	320	640	1280	320	320		
A/Bulgaria/043/2019		6B.1A5	2019-01-08	MDCK1	320	160	160	160	80	160	640	640	320	160		
A/Oman/6133/2018		6B.1A7	2018-11-18	MDCK1	640	160	160	80	160	320	640	1280	320	320		
A/Ireland/84816/2018		6B.1A7	2018-11-29	MDCK1/MDCK1	640	160	320	160	160	320	640	1280	320	160		
A/Denmark/1848/2018		6B.1A7	2018-12-03	MDCK3/MDCK1	320	160	160	80	80	160	640	1280	320	320		
A/Denmark/1839/2018		6B.1A7	2018-12-03	MDCK3/MDCK1	320	80	160	160	80	160	640	1280	320	320		
A/Denmark/1849/2018		6B.1A7	2018-12-05	MDCK3/MDCK1	320	80	80	40	80	160	320	640	160	160		
A/Denmark/1850/2018		6B.1A7	2018-12-06	MDCK3/MDCK1	320	40	160	80	80	160	640	640	320	160		
A/Poitiers/2104/2018		6B.1A1	2018-11-13	MDCK3/MDCK1	320	80	160	40	80	160	320	1280	160	80		
A/Oman/5755/2018		6B.1A2	2018-11-05	MDCK1	320	160	160	80	160	160	640	1280	320	160		
A/Bulgaria/023/2019		6B.1A2	2019-01-07	MDCK1	320	160	160	80	160	160	640	1280	320	160		
A/Oman/5940/2018		6B.1A6	2018-11-11	MDCK1	160	80	80	80	80	160	320	640	160	160		
A/Macon/2132/2018		6B.1A6	2018-11-27	MDCK2/MDCK1	320	160	160	80	160	160	640	1280	320	160		
A/Ireland/84630/2018		6B.1A6	2018-11-28	MDCK2/MDCK1	640	320	320	160	320	320	1280	2560	320	320		
A/Saint-Etienne/2167/2018		6B.1A6	2018-11-28	MDCK2/MDCK1	320	160	160	80	160	160	640	1280	320	320		
A/Denmark/1832/2018		6B.1A6	2018-11-29	MDCK3/MDCK1	320	<	160	160	80	160	640	1280	320	160		
A/Denmark/1841/2018	6B.1A6	2018-12-03	MDCK3/MDCK1	160	<	80	80	80	80	320	640	160	80			
A/Switzerland/8061/2018	6B.1A6	2018-12-19	MDCK1/MDCK1	320	160	160	80	160	160	640	1280	320	320			
A/Switzerland/9036/2018	6B.1A6	2018-12-21	SIAT1/MDCK1	160	80	80	80	80	160	320	640	160	160			

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80

Vaccine

Table 5-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-01-29)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											NEW	
					Post-infection ferret antisera												
					A/Mich 45/15 Egg NIB F42/16 ¹	A/Cal 7/09 Egg F07/16 ¹	A/Bayern 69/09 MDCK F09/15 ¹	A/Lviv N6/09 MDCK F14/13 ¹	A/Astrak 1/11 MDCK F22/13 ¹	A/HK 5659/12 MDCK F17/15 ¹	A/Slov 2903/2015 Egg F02/16 ¹	A/Paris 1447/17 MDCK F03/18 ²	A/Swit 2656/17 Egg F20/18 ¹	A/Swit 3330/17 Egg F23/18 ¹	A/Norway 3433/18 MDCK F04/19 ¹		
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5		
REFERENCE VIRUSES																	
A/Michigan/45/2015			2015-09-07	E3/E3	320	320	160	160	160	320	640	640	640	160	640		
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	320	320	320	160	320	320	640	640	320	160	1280		
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	<	<	320	160	<	<	40	160	<	40	160		
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	40	80	640	640	40	80	160	640	160	160	640		
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	160	160	320	320	160	160	320	1280	320	160	1280		
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	160	80	160	320	640	640	320	160	1280		
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	160	320	160	160	160	160	640	640	640	160	1280		
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	160	160	80	80	160	640	1280	320	160	1280		
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E1	640	640	320	320	320	1280	2560	640	320	2560	2560		
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	160	160	160	160	160	160	320	640	320	320	1280		
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK2	320	160	160	80	160	160	640	1280	320	160	1280		
TEST VIRUSES																	
A/Finland/971/2018			2018-10-31	MDCK1/MDCK1	1280	640	640	320	320	640	1280	2560	1280	640	2560		
A/Finland/976/2018			2018-11-15	MDCK1/MDCK1	320	80	160	160	80	160	640	1280	320	160	1280		
A/Finland/977/2018			2018-11-28	MDCK1/MDCK1	320	160	160	160	160	320	640	640	320	160	2560		
A/Bayern/1/2019			2019-01-03	P1/MDCK1	320	160	80	80	80	160	320	1280	160	160	1280		
A/Chitla/1/2018		6B.1A	2018-11-29	MDCK1/MDCK1	160	80	80	40	80	80	320	640	160	80	640		
A/Poland/261/2019		6B.1A	2019-01-02	MDCK2	160	160	160	40	80	160	640	640	320	160	640		
A/Bulgaria/032/2019		6B.1A	2019-01-04	MDCK2	160	160	80	40	80	160	320	320	160	160	640		
A/Lipetsk/2/2018		6B.1A5	2018-11-16	MDCK1/MDCK1	320	80	160	80	80	160	320	640	160	160	1280		
A/Lipetsk/1/2018		6B.1A5	2018-11-16	MDCK1/MDCK1	160	40	80	40	40	80	160	320	80	80	640		
A/Lipetsk/5/2018		6B.1A5	2018-11-21	MDCK1/MDCK1	320	160	160	160	80	160	320	640	320	160	2560		
A/Novosibirsk/1/2018		6B.1A5	2018-11-26	MDCK1/MDCK1	320	160	160	160	80	160	320	640	320	160	1280		
A/Kyiv/509/2018		6B.1A5	2018-12-16	SIAT1/MDCK1	640	320	160	160	160	320	640	1280	320	320	2560		
A/Kyiv/520/2018		6B.1A5	2018-12-17	SIAT2/MDCK1	640	320	160	160	160	160	640	1280	320	320	2560		
A/Kyiv/512/2018		6B.1A5	2018-12-17	SIAT2/MDCK1	640	320	320	160	160	160	640	1280	320	320	2560		
A/Linköping/70/2018		6B.1A5	2018-12-20	MDCK0/MDCK1	160	160	160	80	80	160	320	640	160	160	1280		
A/Linköping/69/2018		6B.1A5	2018-12-21	MDCK0/MDCK1	160	80	80	40	40	80	320	640	160	160	1280		
A/Israel/553/2018		6B.1A5	2018-12-23	C1/MDCK1	320	160	160	80	160	160	640	640	320	160	2560		
A/Parma/182/2018		6B.1A5	2018-12-24	MDCK1/MDCK1	640	160	160	160	320	320	640	1280	640	160	2560		
A/Uppsala/5/2018		6B.1A5	2018-12-25	MDCK0/MDCK1	640	320	320	160	320	320	640	1280	640	320	2560		
A/Bretagne/2731/2018		6B.1A5	2018-12-26	MDCK1/MDCK1	640	640	640	320	320	320	1280	1280	640	320	5120		
A/Hong Kong/94/2019		6B.1A5	2018-12-31	MDCK1/MDCK1	160	<	80	40	40	40	160	320	80	80	640		
A/Hong Kong/98/2019		6B.1A5	2019-01-01	MDCK1/MDCK1	640	320	320	320	320	320	1280	2560	640	320	5120		
A/Hong Kong/106/2019		6B.1A5	2019-01-01	MDCK1/MDCK1	640	320	160	160	160	320	640	1280	640	320	2560		
A/Hong Kong/134/2019		6B.1A5	2019-01-02	MDCK1/MDCK1	320	80	160	80	160	160	640	1280	320	320	1280		
A/Dijon/2783/2018		6B.1A5	2019-01-03	MDCK1/MDCK1	320	80	80	80	80	80	320	640	160	160	1280		
A/Nordrhein-Westfalen/1/2019		6B.1A5	2019-01-04	P1/MDCK1	320	160	160	160	320	320	640	1280	320	320	2560		
A/Slovenia/69/2019		6B.1A5	2019-01-07	SIATx/MDCK1	320	160	160	80	80	160	640	1280	320	160	1280		
A/Bucuresti/239582/2019		6B.1A5	2019-01-07	SIAT1/MDCK1	320	160	160	80	80	160	320	640	160	160	1280		
A/Parma/6/2019		6B.1A5	2019-01-07	MDCK1/MDCK1	640	320	320	320	320	320	640	1280	640	320	2560		
A/Bremen/1/2019		6B.1A5	2019-01-07	P1/MDCK1	320	160	160	80	80	160	640	640	160	160	1280		
A/Arges/239662/2019		6B.1A5	2019-01-08	SIAT1/MDCK1	320	160	160	160	80	160	640	1280	320	160	2560		
A/Bucuresti/239743/2019		6B.1A5	2019-01-10	SIAT1/MDCK1	320	160	160	160	160	160	640	1280	320	320	2560		
A/Ireland/86125/2018		6B.1A7	2018-12-05	MDCK2/MDCK2	320	160	160	160	160	160	640	1280	320	160	1280		
A/Eskilstuna/6/2018		6B.1A7	2018-12-17	MDCK0/MDCK1	160	160	80	40	80	160	320	320	160	80	640		
A/Israel/10429/2018		6B.1A7	2018-12-20	C1/MDCK1	320	320	160	160	160	320	640	2560	320	160	2560		
A/Lithuania/MB37407/2018		6B.1A7	2018-12-22	MDCK1	320	320	160	80	80	160	640	1280	320	160	1280		
A/Israel/596/2018		6B.1A7	2018-12-25	C1/MDCK1	640	320	320	160	320	320	640	2560	320	320	2560		
A/Lorraine/2690/2018		6B.1A7	2018-12-26	MDCK1/MDCK1	640	320	160	80	160	320	1280	1280	640	320	2560		
A/Picardie/2725/2018		6B.1A7	2018-12-27	MDCK1/MDCK1	640	320	320	160	320	640	1280	2560	1280	640	2560		
A/Lorraine/2721/2018		6B.1A7	2018-12-27	MDCK1/MDCK1	320	160	160	40	80	160	640	640	320	160	1280		
A/Bosnia and Herzegovina/72/2018		6B.1A7	2018-12-31	MDCK1	320	320	160	160	160	320	640	1280	320	160	1280		
A/Franche-Comte/021/2019		6B.1A7	2019-01-02	MDCK1/MDCK1	160	160	80	40	80	80	320	640	160	80	640		
A/Lithuania/MB357/2019		6B.1A7	2019-01-03	MDCK1	320	160	160	80	160	160	640	1280	320	160	2560		
A/Parma/2/2019		6B.1A7	2019-01-04	MDCK1/MDCK1	320	160	160	80	80	160	320	640	160	160	1280		
A/Bucuresti/239595/2019		6B.1A7	2019-01-07	SIAT1/MDCK1	320	160	160	80	80	160	640	1280	160	160	1280		
A/Slovenia/107/2019		6B.1A7	2019-01-08	SIATx/MDCK1	640	320	320	160	160	320	1280	2560	640	320	2560		
A/Slovenia/92/2019		6B.1A7	2019-01-08	MDCKx/MDCK1	320	160	160	80	160	160	640	1280	320	160	1280		
A/Slovenia/131/2019		6B.1A7	2019-01-09	SIATx/MDCK1	1280	640	320	320	320	320	1280	2560	640	320	2560		
A/Slovenia/125/2019		6B.1A7	2019-01-09	SIATx/MDCK1	640	640	320	160	160	320	1280	2560	640	320	2560		
A/Parma/7/2019		6B.1A7	2019-01-09	MDCK1/MDCK1	160	80	80	40	40	80	320	640	160	80	640		
A/Hong Kong/95/2019		6B.1A2	2019-01-01	MDCK1/MDCK1	640	320	320	160	160	160	640	1280	640	320	2560		
A/Hong Kong/110/2019		6B.1A2	2019-01-01	MDCK1/MDCK1	80	40	160	160	40	40	80	160	40	40	320		
A/Georgia/197/2019		6B.1A2	2019-01-08	MDCK1	160	80	80	40	80	80	320	640	160	80	1280		
A/Georgia/158/2019		6B.1A2	2019-01-08	MDCK1	320	160	160	80	160	160	640	1280	320	160	1280		
A/Georgia/175/2019		6B.1A2	2019-01-09	MDCK1	160	80	80	40	80	80	320	320	160	80	640		
A/Israel/516/2018		6B.1A6	2018-12-16	C1/MDCK1	640	320	160	80	160	320	640	1280	320	320	2560		
A/Lithuania/MB7/2018		6B.1A6	2018-12-31	MDCK1	160	80	80	40	40	80	160	640	160	80	640		
A/Poland/50/2019		6B.1A6	2019-01-07	MDCK1	640	320	320	320	320	640	1280	2560	640	640	2560		
A/Slovenia/54/2019		6B.1A6	2019-01-07	MDCKx/MDCK1	160	80	80	40	80	80	320	320	160	80	640		
A/Poland/57/2019		6B.1A6	2019-01-08	MDCK1	1280	640	320	320	640	640	2560	2560	1280	640	2560		

* Superscripts

Table 5-11. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2019-02-05)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre										
					Post-infection ferret antisera										
					A/Mich 45/15 Egg NIB F42/16 ¹	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/HK 5659/12 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	A/Norway 3433/18 MDCK
					F22/13 ¹	F07/16 ¹	F09/15 ¹	F14/13 ¹	F22/13 ¹	F17/15 ¹	F02/16 ¹	F03/18 ²	F20/18 ¹	F23/18 ¹	F04/19 ¹
					6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5
REFERENCE VIRUSES															
A/Michigan/45/2015			2015-09-07	E3/E3	640	640	640	320	640	640	1280	2560	1280	320	1280
A/California/7/2009	clone 38-32		2009-04-09	E3/E3	320	320	320	320	320	320	640	1280	640	160	1280
A/Bayern/69/2009	G155E		2009-07-01	MDCK5/MDCK1	40	<	320	160	<	<	80	160	<	40	320
A/Lviv/N6/2009	G155E, D222G		2009-10-27	MDCK4/SIAT1/MDCK3	80	80	1280	1280	80	80	320	640	320	160	1280
A/Astrakhan/1/2011		5	2011-02-28	MDCK1/MDCK7	320	320	640	640	320	320	640	2560	640	320	2560
A/Hong Kong/5659/2012		6A	2012-05-21	MDCK4/MDCK2	320	320	320	160	320	320	640	1280	320	320	1280
A/Slovenia/2903/2015	clone 37	6B.1	2015-10-26	E4/E2	320	640	320	320	320	320	1280	1280	640	320	1280
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	320	320	160	80	160	160	640	1280	320	160	1280
A/Switzerland/2656/2017		6B.1A	2017-12-21	E5/E3	640	640	320	320	320	320	1280	2560	640	640	2560
A/Switzerland/3330/2017	clone 35	6B.1A5	2017-12-20	E6/E2	320	320	320	320	320	320	640	1280	640	640	2560
A/Norway/3433/2018		6B.1A5	2018-10-30	MDCK2	320	160	320	160	320	320	640	1280	320	320	2560
TEST VIRUSES															
A/Antsirabe/3880/2018			2018-09-03	MDCK1/MDCK1	640	320	320	160	160	160	1280	1280	320	320	1280
A/Antananarivo/3877/2018			2018-09-03	MDCK1/MDCK1	640	320	320	160	160	320	1280	1280	640	320	1280
A/Antsirabe/3941/2018			2018-09-10	MDCK1/MDCK1	1280	640	640	320	640	640	1280	2560	1280	640	2560
A/Netherlands/02307/2018			2018-11-06	SIAT1/MDCK1	1280	640	320	320	320	640	2560	2560	1280	640	2560
A/Netherlands/10605/2018			2018-11-30	MDCK-MIX2/MDCK1	640	160	320	160	320	320	1280	1280	320	320	2560
A/Croatia/5358/2018			2018-12-04	MDCKx/MDCK1	1280	1280	640	320	640	640	1280	5120	1280	640	5120
A/Croatia/5354/2018			2018-12-04	MDCKx/MDCK1	1280	1280	640	640	640	640	2560	5120	1280	640	5120
A/Bursa/1897/2018			2018-12-13	Cx/MDCK1	640	320	320	320	320	640	1280	2560	640	320	5120
A/Hungary/1/2019			2018-12-13	Cx/MDCK1	640	320	320	320	320	640	1280	2560	640	320	2560
A/Croatia/5562/2018			2018-12-14	MDCKx/MDCK2	640	320	320	160	320	320	640	1280	640	320	2560
A/Hungary/20/2019			2018-12-17	Cx/MDCK1	320	160	160	160	160	320	640	1280	320	160	2560
A/Hungary/7/2019			2018-12-17	Cx/SIAT1	1280	640	640	320	640	640	1280	2560	1280	640	2560
A/Antalya/5/2018			2018-12-18	Cx/MDCK1	640	640	320	160	320	640	1280	2560	640	320	2560
A/Netherlands/10611/2018			2018-12-19	MDCK-MIX2/MDCK1	640	320	320	320	320	640	1280	2560	640	320	2560
A/Portugal/84/2018			2018-12-19	SIAT1/MDCK1	1280	640	640	320	640	640	1280	2560	1280	640	5120
A/Hungary/6/2019			2018-12-19	Cx/MDCK1	320	320	320	160	160	320	1280	1280	320	320	2560
A/Antalya/10/2018			2018-12-20	Cx/MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Antalya/9/2018			2018-12-20	Cx/MDCK1	1280	640	640	320	640	640	1280	2560	1280	640	2560
A/Netherlands/10612/2018			2018-12-21	MDCK-MIX2/MDCK1	1280	640	320	320	640	640	2560	2560	640	640	5120
A/Portugal/125/2018			2018-12-24	SIAT1/MDCK1	1280	640	640	160	640	640	1280	2560	1280	640	5120
A/Moscow/227/2018			2018-12-25	MDCK1/MDCK1	640	640	320	320	640	640	1280	2560	640	320	2560
A/Ystad/1/2018			2018-12-26	MDCK0/MDCK1	640	640	640	160	640	640	1280	2560	1280	640	2560
A/Portugal/90/2018			2018-12-27	SIAT1/MDCK1	1280	640	320	320	320	640	2560	2560	640	640	2560
A/Serbia/17151/2018			2018-12-31	C2/MDCK1	1280	640	640	320	640	640	1280	2560	640	640	2560
A/Greece/2/2019			2019-01-02	MDCK1/MDCK1	1280	640	1280	640	1280	1280	2560	5120	1280	640	5120
A/Serbia/39/2019			2019-01-03	C2/MDCK1	640	320	320	160	160	320	640	1280	320	160	1280
A/St. Petersburg/Rii-01/2019			2019-01-03	MDCK1/MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Greece/5/2019			2019-01-03	MDCK1/MDCK1	640	160	320	160	320	320	640	1280	320	320	2560
A/Serbia/34/2019			2019-01-04	C2/MDCK1	640	320	320	320	640	640	1280	2560	640	320	2560
A/Serbia/25/2019			2019-01-04	C2/MDCK1	1280	160	640	320	320	320	1280	2560	640	320	2560
A/Montenegro/103/2019			2019-01-05	MDCKx/MDCK1	640	160	320	160	160	160	640	1280	320	320	2560
A/Montenegro/107/2019			2019-01-06	MDCKx/MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Portugal/150/2019			2019-01-08	SIAT1/MDCK1	640	640	640	320	320	640	1280	2560	640	320	2560
A/Georgia/184/2019			2019-01-09	MDCK2	640	320	320	160	320	320	1280	2560	640	320	2560
A/Montenegro/123/2019			2019-01-09	MDCKx/MDCK1	640	320	320	160	160	320	640	1280	320	320	2560
A/Georgia/189/2019			2019-01-10	MDCK2	640	320	320	160	320	320	1280	2560	640	320	2560
A/Georgia/187/2019			2019-01-10	MDCK2	640	320	320	160	320	320	1280	1280	640	320	2560
A/Bosnia and Herzegovina/2760/2019			2019-01-11	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Luxembourg/2482/2019			2019-01-14	MDCK1	640	160	320	320	320	320	1280	2560	640	320	2560
A/Luxembourg/2180/2019			2019-01-14	MDCK1	640	320	640	320	320	640	1280	2560	640	640	5120
A/Cyprus/F107/2019			2019-01-14	MDCK1	320	320	320	160	320	320	640	1280	640	320	2560
A/Cyprus/F105/2019			2019-01-14	MDCK1	320	320	320	160	320	320	640	2560	640	320	2560
A/Luxembourg/2489/2019			2019-01-15	MDCK1	640	320	320	320	320	320	1280	1280	640	320	2560
A/Valladolid/37/2019			2019-01-15	MDCK1/MDCK1	1280	640	640	320	640	640	1280	2560	640	640	2560
A/Cyprus/FA17/2019			2019-01-15	MDCK1	640	320	320	160	320	320	1280	2560	640	320	2560
A/Valladolid/40/2019			2019-01-16	MDCK1/MDCK1	1280	640	640	320	640	640	2560	2560	1280	320	5120
A/Valladolid/39/2019			2019-01-16	MDCK1/MDCK1	640	320	320	160	320	320	640	2560	640	320	2560
A/Valladolid/38/2019			2019-01-16	MDCK1/MDCK1	640	320	320	320	320	320	1280	2560	640	320	2560
A/Luxembourg/3119/2019			2019-01-17	MDCK1	1280	640	640	320	640	640	2560	2560	1280	640	2560

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
 1 < = <40; 2 < = <80

Vaccine

Table 5-12. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2018-08-31 Summary

			Haemagglutination inhibition titre										
			Post-infection ferret antisera										
Viruses	Other information	NEW											
		A/Mich 45/15	A/Cal 7/09	A/Bayern 69/09	A/Lviv N6/09	A/Astrak 1/11	A/HK 5659/12	A/Slov 2903/2015	A/Paris 1447/17	A/Swit 2656/17	A/Swit 3330/17	A/Norway 3433/18	
		Passage history	Egg	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	Egg	Egg	MDCK	
		Ferret number	NIB F42/16 ^{*1}	F07/16 ^{*1}	F09/15 ^{*1}	F14/13 ^{*1}	F22/13 ^{*1}	F17/15 ^{*1}	F02/16 ^{*1}	F03/18 ^{*2}	F20/18 ^{*1}	F23/18 ^{*1}	F04/19 ^{*1}
Genetic group		6B.1				5	6A	6B.1	6B.1A	6B.1A	6B.1A5	6B.1A5	
REFERENCE VIRUSES													
A/Michigan/45/2015		6B.1	320	320	160	160	160	320	640	640	640	160	640
A/California/7/2009	clone 38-32		320	320	320	160	320	320	640	640	320	160	1280
A/Bayern/69/2009	G155E		<	<	320	160	<	<	40	160	<	40	160
A/Lviv/N6/2009	G155E, D222G		40	80	640	640	40	80	160	640	160	160	640
A/Astrakhan/1/2011		5	160	160	320	320	160	320	1280	320	160	160	1280
A/Hong Kong/5659/2012		6A	320	320	160	80	160	320	640	640	320	160	1280
A/Slovenia/2903/2015	clone 37	6B.1	160	320	160	160	160	160	640	640	640	160	1280
A/Paris/1447/2017		6B.1A	320	160	160	80	80	160	640	1280	320	160	1280
A/Switzerland/2656/2017		6B.1A	640	640	320	320	320	320	1280	2560	640	320	2560
A/Switzerland/3330/2017	clone 35	6B.1A5	160	160	160	160	160	160	320	640	320	320	1280
A/Norway/3433/2018		6B.1A5	320	160	160	80	160	160	640	1280	320	160	1280
TEST VIRUSES													
Number of viruses tested*			210	210	210	210	210	210	210	210	210	210	109
No with titre reduction ≤2-fold			209	144	188	39	198	188	196	199	183	153	108
%			99.5	68.5	89.5	18.6	94.3	89.5	93.3	94.8	87.1	72.9	99.1
No with titre reduction =4-fold			1	44	22	93	12	20	13	10	24	43	1
%			0.5	21.0	10.5	44.3	5.7	9.5	6.2	4.7	11.4	20.5	0.9
No with titre reduction ≥8-fold				22		78		2	1	1	3	14	
%				10.5		37.2		1.0	0.5	0.5	1.5	6.6	

* Of those with available HA sequence, all were clade 6B.1

Vaccine

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

**Table 5-13. Antigenic analyses of influenza A(H1N1)pdm09 viruses with collection dates after 2018-08-31
Summary by genetic subgroup**

Viruses		Haemagglutination inhibition titre										
		Post-infection ferret antisera										
		A/Mich 45/15 Egg NIB F42/16 ⁻¹	A/Cal 7/09 Egg	A/Bayern 69/09 MDCK	A/Lviv N6/09 MDCK	A/Astrak 1/11 MDCK	A/HK 5659/12 MDCK	A/Slov 2903/2015 Egg	A/Paris 1447/17 MDCK	A/Swit 2656/17 Egg	A/Swit 3330/17 Egg	NEW A/Norway 3433/18 MDCK
Passage history	Ferret number	Genetic group										
TEST VIRUSES												
Total number tested		156	156	156	156	156	156	156	156	156	156	56
Number tested	6B.1A	25	25	25	25	25	25	25	25	25	25	3
No with titre reduction ≤2-fold		25	10	22	4	25	21	24	22	19	21	3
%		100	40.0	88.0	16.0	100	84.0	96.0	88.0	76.0	84.0	100
No with titre reduction =4-fold			14	3	13		4	1	3	3	3	
%			56.0	12.0	52.0		16.0	4.0	12.0	12.0	12.0	
No with titre reduction ≥8-fold			1		8				3	3	1	
%			4.0		32.0				12.0	12.0	4.0	
Number tested	6B.1A1	4	4	4	4	4	4	4	4	4	4	
No with titre reduction ≤2-fold		4		3		3	3	3	3	3	2	
%		100		75.0		75.0	75.0	75.0	75.0	75.0	50.0	
No with titre reduction =4-fold			2	1		1	1	1	1	1	1	
%			50.0	25.0		25.0	25.0	25.0	25.0	25.0	25.0	
No with titre reduction ≥8-fold			2		4						1	
%			50.0		100						25.0	
Number tested	6B.1A2	8	8	8	8	8	8	8	8	8	8	5
No with titre reduction ≤2-fold		7	5	6		7	5	7	6	3	3	4
%		87.5	62.5	75.0		87.5	62.5	87.5	75.0	37.5	37.5	80.0
No with titre reduction =4-fold		1	2	2	5	1	2		1	4	4	1
%		12.5	25.0	25.0	62.5	12.5	25.0		12.5	50.0	50.0	20.0
No with titre reduction ≥8-fold			1		3		1	1	1	1	1	
%			12.5		37.5		12.5	12.5	12.5	12.5	12.5	
Number tested	6B.1A5	70	70	70	70	70	70	70	70	70	70	25
No with titre reduction ≤2-fold		70	43	64	23	63	63	67	66	41	59	25
%		100	61.4	91.4	32.9	90.0	90.0	95.7	94.3	58.6	84.3	100
No with titre reduction =4-fold			16	6	30	7	6	3	4	22	9	
%			22.9	8.6	42.9	10.0	8.6	4.3	5.7	31.4	12.9	
No with titre reduction ≥8-fold			11		17		1			7	2	
%			15.7		24.2		1.4			10.0	2.8	
Number tested	6B.1A6	14	14	14	14	14	14	14	14	14	14	5
No with titre reduction ≤2-fold		14	7	9	4	12	10	12	12	3	7	5
%		100	50.0	64.3	28.6	85.7	71.4	85.7	85.7	21.4	50.0	100
No with titre reduction =4-fold			4	5	6	2	4	2	2	8	6	
%			28.6	35.7	42.8	14.3	28.6	14.3	14.3	57.2	42.8	
No with titre reduction ≥8-fold			3		4					3	1	
%			21.4		28.6					21.4	7.2	
Number tested	6B.1A7	35	35	35	35	35	35	35	35	35	35	18
No with titre reduction ≤2-fold		35	26	31	5	34	32	35	34	24	29	18
%		100	74.3	88.6	14.3	97.2	91.4	100	97.2	68.6	82.9	100
No with titre reduction =4-fold			5	4	14	1	3		1	10	6	
%			14.3	11.4	40.0	2.8	8.6		2.8	28.6	17.1	
No with titre reduction ≥8-fold			4		16					1		
%			11.4		45.7					2.8		

Vaccine

Figure 1. Phylogenetic comparison of A(H1N1)pdm09 HA genes

Vaccine virus
Reference viruses

Collection date

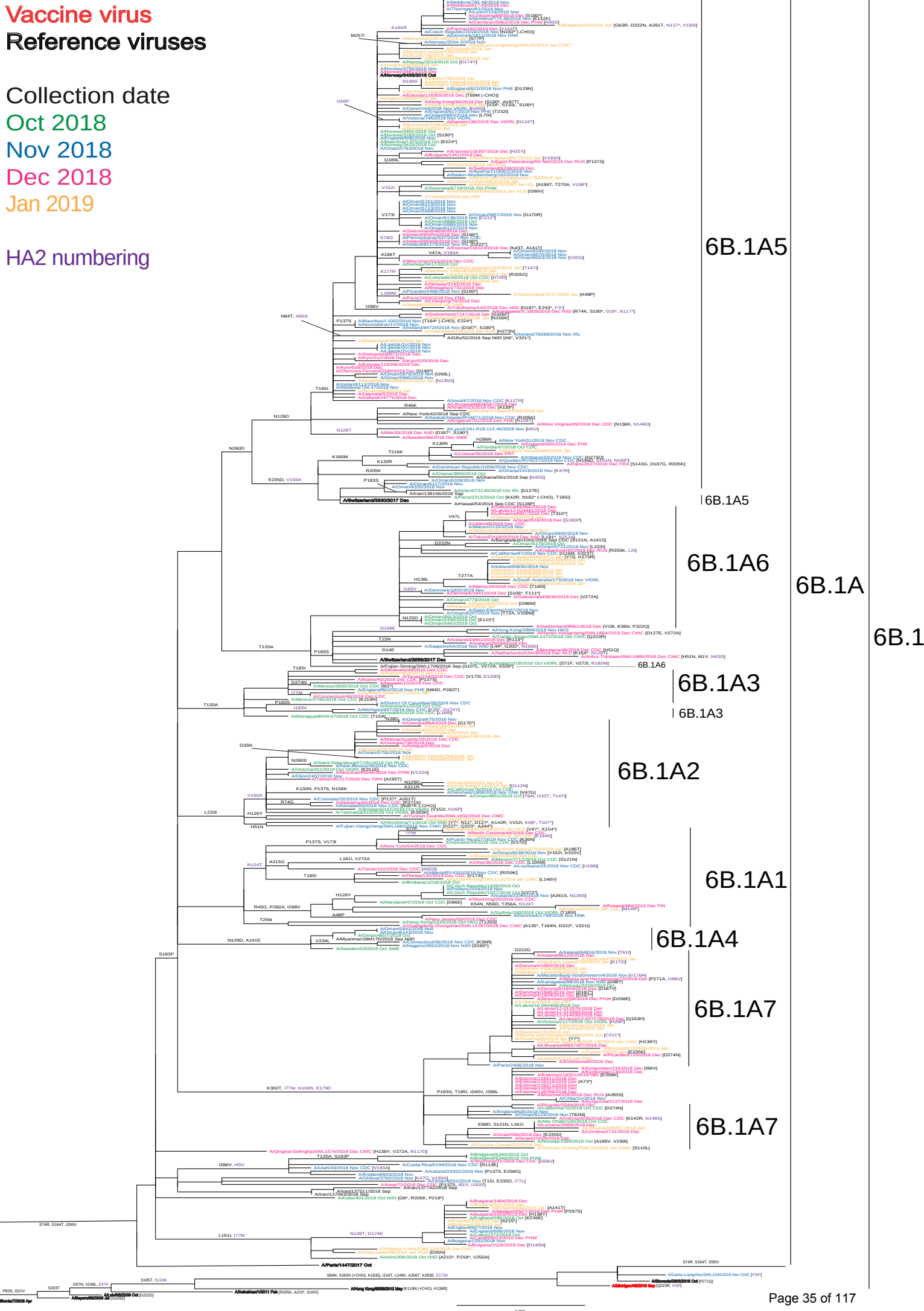
Oct 2018

Nov 2018

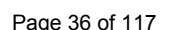
Dec 2018

Jan 2019

HA2 numbering



Vaccine virus: **Reference virus:** **Genetic group:** **HA1/HA2 boundary:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400
A/California/7/2009	GKCPKYVKSTKLR	LATGLRNIPSIQSR	GLFGAIA	GFIEGGWTGMVD	GWYGYHHQNEQ	QSGYAADLKSTQ	NAIDEITNKVNS	VIEKMNTQFTAV	GKEFNHL	
A/Bayern/69/2009	V									
A/Lviv/N6/2009	V									
A/Astrakhan/1/2011	V						K			5
A/Hong Kong/5659/2012	V						K			6A
A/Michigan/45/2015	V		X				K			6B.1
A/Slovenia/2903/2015	V						K			6B.1
A/Paris/1447/2017	V						K			6B.1A
A/Bulgaria/1533/2018	V						K			6B.1A
A/Iceland/24352/2018	V						V.K			6B.1A
A/Lithuania/MB357/2019	T						K			6B.1A7
A/Estonia/118371/2018	T						K			6B.1A
A/Denmark/1849/2018	T						K			6B.1A7
A/Oman/5941/2018	V						K			6B.1A4
A/Poitiers/2104/2018	V						K			6B.1A1
A/Northern Ireland/763/2019	V						K			6B.1A1
A/Hong Kong/95/2019	V						K			6B.1A2
A/Georgia/197/2019	V						K			6B.1A2
A/Northern Ireland/771/2019	V						K			6B.1A3
A/England/602/2018	V						K			6B.1A3
A/Switzerland/2656/2017	V						K			6B.1A
A/Iceland/25090/2018	V						K			6B.1A
A/Poland/50/2019	V						K			6B.1A6
A/Latvia/12-024461/2018	V						K			6B.1A6
A/Switzerland/3330/2017	V						K			6B.1A5
A/England/690/2018	V						K			6B.1A5
A/Israel/553/2018	V						K			6B.1A5
A/Slovenia/69/2019	V						K			6B.1A5
A/Oman/6242/2018	V						K			6B.1A5
A/Norway/3433/2018	V						K			6B.1A5
A/Parma/182/2018	V						K			6B.1A5
A/Bulgaria/043/2019 Jan	V						K			6B.1A5
	410	420	430	440	450	460	470	480	490	500
A/California/7/2009	EKRIENLNKKVDD	GLDIWTYNAELL	VLLLENERTLDY	HDSNVKNLYEK	VRSQLKNNAKEI	GNCGCFEFYHK	DNTCMESVKN	NGTYDPKYSEE	AKLNREEI	
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Astrakhan/1/2011										
A/Hong Kong/5659/2012				N						
A/Michigan/45/2015				N					K	
A/Slovenia/2903/2015				N					K	
A/Paris/1447/2017				N					K	
A/Bulgaria/1533/2018	M			N	T				K	
A/Iceland/24352/2018				N					K	
A/Lithuania/MB357/2019	M			N					S.K	
A/Estonia/118371/2018	M			N					S.K	
A/Denmark/1849/2018	M			N					S.K	
A/Oman/5941/2018				N					K	
A/Poitiers/2104/2018				N					K	
A/Northern Ireland/763/2019				T					K	
A/Hong Kong/95/2019				N					K	
A/Georgia/197/2019				N					K	
A/Northern Ireland/771/2019	M			N					K	
A/England/602/2018				N					K	
A/Switzerland/2656/2017				N					K	
A/Iceland/25090/2018				N					K	
A/Poland/50/2019				N					K	
A/Latvia/12-024461/2018				N					K	
A/Switzerland/3330/2017				N					K	
A/England/690/2018				N					K	
A/Israel/553/2018				N					K	
A/Slovenia/69/2019				N					K	
A/Oman/6242/2018				N					K	
A/Norway/3433/2018				N					K	
A/Parma/182/2018				N		F			R	K
A/Bulgaria/043/2019 Jan				X	N				R	K
	510	520	530	540						
A/California/7/2009	DGVKLESTRIYQ	ILAIYSTVASSL	VLVSLGAI	SFWMCSNGSLQ	CRICI					
A/Bayern/69/2009										
A/Lviv/N6/2009										
A/Astrakhan/1/2011										
A/Hong Kong/5659/2012										
A/Michigan/45/2015										
A/Slovenia/2903/2015										
A/Paris/1447/2017										
A/Bulgaria/1533/2018	E									
A/Iceland/24352/2018										
A/Lithuania/MB357/2019	D									
A/Estonia/118371/2018	D									
A/Denmark/1849/2018	D									
A/Oman/5941/2018										
A/Poitiers/2104/2018										
A/Northern Ireland/763/2019										
A/Hong Kong/95/2019										
A/Georgia/197/2019				A						
A/Northern Ireland/771/2019										
A/England/602/2018										
A/Switzerland/2656/2017										
A/Iceland/25090/2018										
A/Poland/50/2019										
A/Latvia/12-024461/2018										
A/Switzerland/3330/2017				A						
A/England/690/2018				A						
A/Israel/553/2018										
A/Slovenia/69/2019										
A/Oman/6242/2018				A						
A/Norway/3433/2018										
A/Parma/182/2018										
A/Bulgaria/043/2019 Jan				I						

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

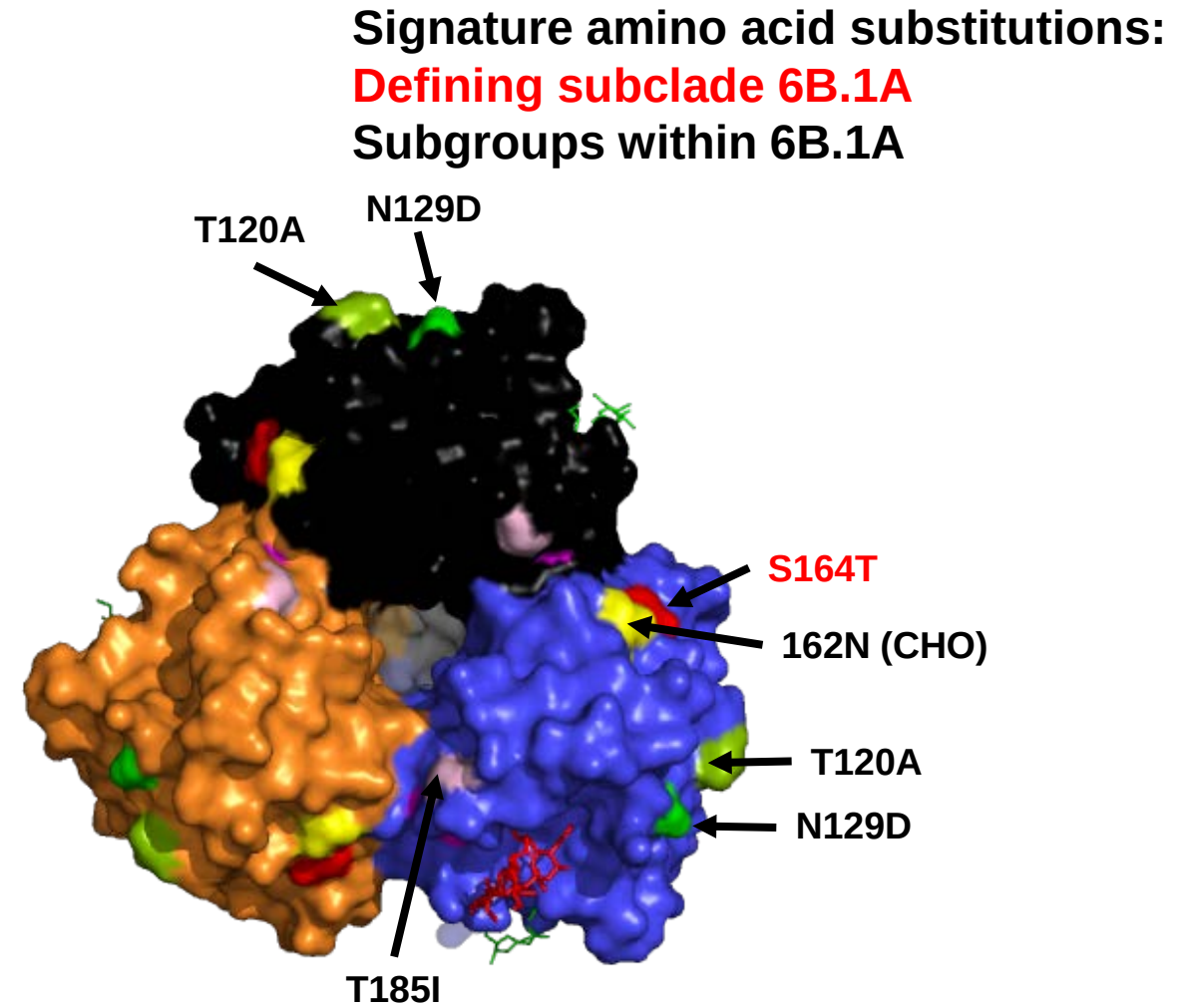
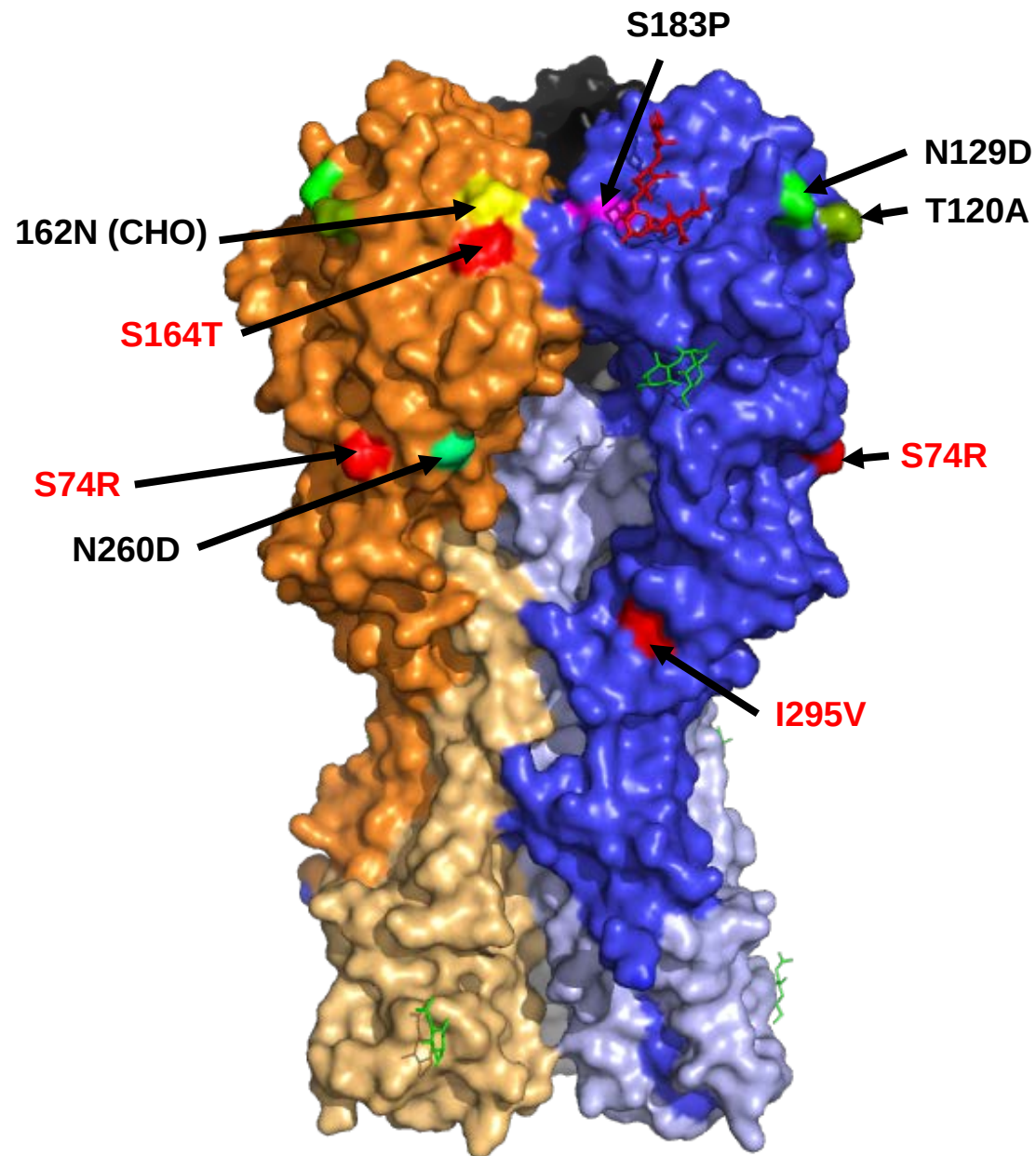
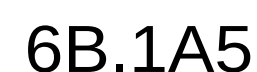


Figure 3. Amino acid substitutions in the HA of recent H1N1pdm09 viruses

Vaccine virus
Reference viruses

	A/Ireland/84720/2018 Nov	A/Switzerland/7347/2018 Dec A/Kyiv/509/2018 Dec	6B.1A5
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6B.1A5

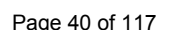
6B.1A

6B.1A5

6B.1A5

A1

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2018-08-31 have been received from NICs in 49 countries or administrative regions in four WHO Regions: African (4: Cameroon, Cote d'Ivoire, Ghana and Mauritius) Eastern Mediterranean (4: Egypt, Iran, Jordan and Oman), Western Pacific (1: Hong Kong SAR) and European (40 countries) (**Table 1**).

Influenza A(H3N2) viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of these RBCs. Of the successfully propagated viruses with collection dates since 2018-08-31, 24% had sufficient HA titre to be analysed by HI assays using guinea pig RBCs in the presence of 20nM oseltamivir, added to circumvent NA-mediated binding to the RBCs (**Tables 1, 2, 6 & 7**), this being a small decrease over the proportion (33%) seen in February 2018. The remainder were unable to agglutinate guinea pig RBCs at all (**Tables 2 & 6**), or were unable to agglutinate RBCs in the presence of 20nM oseltamivir. Neutralisation of selected subsets of viruses, a total of 173 test viruses with collection dates after 2018-08-31 and including many for which HI assay could not be performed, were analysed by plaque reduction neutralisation assay (PRNA: **Tables 9-1 to 9-10**).

HI analyses.

A total of 54 viruses with collection dates after 2018-08-31 could be analysed by HI assay and summaries of the HI results are shown in **Tables 2** and **7**. Individual HI results are shown in **Tables 8-1 to 8-11**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2018-08-31 (**Tables 8-6 to 8-11**). The HA genetic group is indicated for those viruses that have been sequenced. All of the viruses sequenced with collection dates after 2018-08-31 fell into clade 3C.3a or subclades 3C.2a1b, 3C.2a2 and 3C.2a3 of clade 3C.2a.

A summary of the HI results for viruses collected since 2018-08-31 with antisera raised against the egg-propagated cultivars of A/Singapore/INFIMH-16-0019/2016 and A/Switzerland/8060/2017 (the most recently recommended vaccine virus) and cell culture-propagated A/Hong Kong/5738/2014 (a 3C.2a clade virus that is A/Hong Kong/4801/2014-like) and cell culture-propagated A/Stockholm/6/2014 (a clade 3C.3a virus) is shown in **Table 7**. A more detailed summary of the whole panel of antisera is shown in **Table 8-12**.

The antiserum raised against cell culture-propagated A/Hong Kong/5738/2014 recognised 70% of the test viruses collected after 2018-08-31 at titres within 4-fold of the titre of the antiserum with the homologous virus, 43% within 2-fold of the homologous titre.

Antiserum raised against cell culture-propagated A/Stockholm/6/2014, a clade 3C.3a virus, recognised 61% of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, but only 28% at titres within 2-fold of the homologous titre.

Antiserum raised against the egg-propagated cultivar of A/Singapore/INFIMH-16-0019/2016 (propagated with a predominance of glycine at HA1 position 225 [225G]), recommended for use in the Northern Hemisphere 2018/2019, recognised 50% of the test viruses at titres within 4-fold of the homologous titre of the antiserum, but only 13% at a titre within 2-fold of the homologous titre. An antiserum raised against the egg-propagated cultivar of A/Switzerland/8060/2017, the vaccine virus recommended for use in the Southern Hemisphere 2019, recognised the test viruses poorly with only a single virus being recognised at a titre within 4-fold of the titre of the antiserum with the homologous virus.

Antisera raised against four subclade 3C.2a1 and 3C.2a1b viruses were also used in the tests although no homologous titres were available for the antisera since the viruses were unable to agglutinate RBCs. Three of these antisera, raised against A/Norway/4436/2016 and A/Greece/4/2017 (both 3C.2a1 subclade) and A/La Rioja/2202/2017 and A/Norway/3275/2018 (both 3C.2a1b subclade) recognised 47%, 41%, 57% and 35%, respectively, of the test viruses collected after 2018-08-31 at titres ≥ 160 , titres similar to the titres of these antisera with most of the reference viruses.

Antisera were raised against two cell culture-propagated viruses in clade 3C.2a2, A/Bretagne/1413/2017 and A/Hong Kong/656/2018. These antisera recognised, respectively, only 7% and 6% of test viruses well with the others being recognised at titres ≥ 8 -fold lower than the homologous titres of the antisera.

An antiserum raised against the 3C.3a reference virus A/England/538/2018 recognised 46% of test viruses at titres within 4-fold of the homologous titre of the antiserum but gene sequence analysis of most of these viruses has not been completed – although two 3C.3a viruses were recognised at titres 4-fold lower than the titre of the antiserum with the homologous virus.

An antiserum raised against egg-propagated A/Netherlands/10260/2018, a subclade 3C.2a1b virus, recognised none of the test viruses well.

Virus Neutralisation results

The results of Plaque Reduction Neutralisation Assay (PRNA) tests are shown in **Tables 9-1 to 9-10**. Summaries of the results are shown in **Tables 9-S1, 9-S2 and 9-S3**.

To summarise, antisera raised against

- cell culture-propagated A/Singapore/INFIMH-16-0019/2016 (subclade 3C.2a1) recognised 55% of the test viruses (55% of the 3C.2a1b viruses, 79%

- of the 3C.2a2 viruses, and four of six 3C.2a3 viruses) at titres within 4-fold of the homologous titre,
- cell culture-propagated A/La Rioja/2202/2017 (3C.2a1b) recognised 64% of the test viruses (68% of the 3C.2a1b viruses, 79% of the 3C.2a2 viruses and one of five 3C.2a3 viruses) at titres within 4-fold of the homologous titre,
 - cell culture-propagated A/Norway/3275/2018 (3C.2a1b) recognised 87% of the test viruses (86% of the 3C.2a1b viruses) at titres within 4-fold of the homologous titre,
 - cell culture-propagated A/Hong Kong/656/2017 (3C.2a2) recognised 12% of the test viruses (all but one of which were subclade 3C.2a2 viruses) at titres within 4-fold of the homologous titre,
-
- egg-propagated A/Singapore/INFIMH-16-0019/2016, were from two ferrets-antiserum F46/17 recognised none of the 76 test viruses at a titre within 4-fold of the homologous titre, and the antiserum F15/18, made against an earlier passage of egg-propagated A/Singapore/INFIMH-16-0019/2016, recognised 21% of 58 test viruses at titres within 4-fold of the homologous titre – viruses in genetic group 3C.2a2 being recognised best by this antiserum,
 - egg-propagated A/Netherlands/10260/2018 (3C.2a1b) recognised none of the 14 test viruses analysed at titres within 4-fold of the homologous titre
 - egg-propagated A/Switzerland/8060/2017 (3C.2a2) recognised 9% of the test viruses analysed at titres within 4-fold of the homologous titre – all were subclade 3C.2a2 viruses.

Two antisera raised against clade 3C.3a viruses, cell culture-propagated A/England/538/2018 and the previously used vaccine virus, egg-propagated A/Switzerland/9715293/2013, in combination with antisera raised against A/Singapore/INFIMH-16-0019/2016 were used in **Table 9-10**. The ten clade 3C.3a test viruses were poorly recognised by the antiserum raised against cell culture-propagated A/Singapore/INFIMH-16-0019/2016 and by the antiserum F46/17 (raised against egg-propagated A/Singapore/INFIMH-16-0019/2016), but antiserum from ferret F15/18 also raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (see above) recognised three of the five 3C.3a viruses tested with this antiserum at titres within 4-fold of the homologous titre. The antiserum raised against the former vaccine virus egg-propagated A/Switzerland/9715293/2013 recognised only one of the ten 3C.3a viruses at titres within 4-fold of the homologous titre of the antiserum but the antiserum raised against cell culture-propagated A/England/538/2018 recognised all ten tested 3C.3a viruses at titres within 4-fold of the homologous titre.

The HA genes of A/La Rioja/2202/2018 and A/Norway/3275/2018, 3C.2a1b viruses, differed from each other with A/La Rioja/2202/2018 having the substitution **T128A**, resulting in the loss of a potential glycosylation site in **HA1**, and that of A/Norway/3275/2018 having the substitutions **T131K** and **K135T** (resulting in the restoration of the glycosylation motif **NGT** at residues **133** to **135** in HA1). A summary of the reactivity of these two antisera is shown in **Table 9-S3**. A

comparison of the cross-neutralisation of the two reference viruses indicated that they were antigenically different with little reciprocal neutralisation. However, the antiserum raised against A/Norway/3275/2018 recognised all but one of the test viruses with **T135K** in **HA1** and all of the test viruses with **T131K** and **135T**. In contrast the antiserum raised against A/La Rioja/2202/2018 recognised only just over a third of test viruses (36%) encoding **T131K** and **K135T**. These results suggest some limited antigenic diversity within the 3C.2a1b genetic group.

The VN results also indicate that currently circulating viruses in clade 3C.3a are antigenically distinct from the previously used vaccine virus, egg-propagated A/Switzerland/9715293/2013.

Genetic analyses.

Phylogenetic analysis of the HA genes of recently circulating H3N2 viruses analysed at the Francis Crick Institute and other representative viruses is shown in **Figure 6**. The HA genes fell within clade 3C.2a, primarily subclade 3C.2a1b, and an increased proportion in clade 3C.3a, but with small numbers falling in subclades 3C.2a2 and 3C.2a3 and 3C.2a4.

- Clade **3C.2a** viruses are defined by **L3I**, **N128T** (resulting in the gain of a potential glycosylation site), **N144S** (resulting in the loss of a potential glycosylation site), **N145S**, **F159Y**, **K160T** (resulting in the gain of a potential glycosylation site at residue 158), **P198S**, **F219S**, **N225D** and **Q311H** in **HA1** and **D160N** in **HA2**;
- **N121K**, **N171K** in **HA1** and **I77V**, **G155E** and **G160E** in **HA2**, define **subclade 3C.2a1** and many carry the substitution **T135K** in **HA1** which results in the loss of a glycosylation motif at residues **133 to 135** of **HA1**. Within **subclade 3C.2a1** additional groups can be defined with HA substitutions:
 - **K92R** and **H311Q** in **HA1** define the large **subclade 3C.2a1b**, but there is considerable diversity in this subclade with most viruses analysed carrying the substitutions **E62G** and **R142G**, but further subgroups of viruses carry HA genes that encode **T128A** (resulting in the loss of a glycosylation motif) in **HA1** (e.g. A/La Rioja/2202/2018), or **T131K**, **K135T** (restoring a glycosylation motif) in **HA1** and **V200I** in **HA2**;
- **T131K**, **R142K** and **R261Q** in **HA1**, define **subclade 3C.2a2** (e.g. A/Switzerland/8060/2017);
- **N121K**, **T128A**, **T135K** (resulting in the loss of a glycosylation motif) and **S144K** in **HA1**, define **subclade 3C.2a3** with further substitutions **R142G** and **R261Q** seen in many viruses;
- **N31S**, **D53N**, **R142G**, **S144R**, **N171K**, **I192T** and **Q197H** in **HA1**, define **subclade 3C.2a4**;

- **Clade 3C.3a** viruses are defined by the substitutions **T128A**, **A138S** and **R142G** in **HA1**. The HA genes of recent clade 3C.3a viruses encode the substitutions **L3I**, **S91N**, **N144K** (resulting in the loss of a potential glycosylation site), **F193S** and **K326R** in **HA1**, and **D160N** in **HA2** (e.g. A/England/538/2018).

Figure 7 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny. The vaccine viruses A/Singapore/INFIMH-16-0019/2016 and A/Switzerland/8060/2017 are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade to which each of the HA sequences belongs is also shown.

Figure 8a shows the location on a 2005 H3 HA structure of amino acid substitutions that define recent subclades of viruses compared with the vaccine virus A/Singapore/INFIMH-16-0019/2016. **Figure 8b** shows the location of HA amino acid substitutions found in recently circulating clade 3C.3a viruses compared with A/Switzerland/9715293/2013.

Figure 9 shows a NA phylogenetic tree for representative viruses with their HA genetic groups indicated. **Figure 10** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the clade/subclade to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of recent viruses with HA genes in genetic clusters within clade 3C.2a and its subclades, and in subclade 3C.2a1 and its genetic groups, show some generally similar subclade clustering. There are several subgroups defined by amino acid substitutions:

- Recently collected 3C.2a1b viruses share the amino acid substitutions **K220N**, **V303I** and **N329S** (losing a potential glycosylation motif) in the NA and many also have substitutions **S315R**, **E344K** and **G346D**.
- Most recent clade **3C.3a** viruses have the substitutions **K75R**, **Y155H**, **N329T** (resulting in the loss of a potential glycosylation site) and **E344K** and many share the additional substitutions **N161S** and **P386S**, but the NA gene of some recent viruses is distinct with substitutions **P46A**, **E344K**, **G414S** and **P468L**.

Antiviral resistance

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 186 viruses with collection dates after 2018-08-31. All showed normal inhibition by oseltamivir and zanamivir (**Table 14**).

Conclusion.

Viruses in subclade 3C.2a1b were predominant. Of the recovered viruses 24% were able to agglutinate RBCs in the presence of oseltamivir such that they could be characterised by HI assay.

In HI assays the antiserum raised against a cell culture-propagated clade 3C.2a virus, A/Hong Kong/5738/2014 (genetically similar to the vaccine virus A/Hong Kong/4801/2014), recognised the majority of recently collected test viruses well. An antiserum raised against the previously recommended vaccine virus, egg-propagated A/Singapore/INFIMH-16-0019/2016, recognised 50% of test viruses at titres ≤ 4 -fold reduced compared to the homologous titre but an antiserum raised against the recently recommended vaccine virus A/Switzerland/8060/2017 recognised the test viruses poorly.

In PRNA, the antiserum raised against cell-propagated A/Singapore/INFIMH-16-0019/2016 recognised 55% of test viruses at titres within 4-fold of the homologous titre.

In PRNA, two antisera raised against the egg-propagated vaccine virus A/Singapore/INFIMH-16-0019/2016 (3C.2a1) recognised none or only 21% of the test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus. However, an antiserum raised against egg-propagated A/Netherland/10260/2018 (a subclade 3C.2a1b virus) recognised the test viruses poorly.

Viruses in clade 3C.3a have continued to evolve and show evidence of antigenic drift from viruses similar to the former vaccine virus A/Switzerland/9715293/2013.

Table 6. A(H3N2) viruses recovered from specimens collected after 2018-01-31, as assessed by NA activity (MUNANA-based assay)

Situation at the time of the February 2019 VCM (viruses with collection dates Sep 2018-Jan 2019)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3C.2a1	Number in clade 3C.2a1b	Number in clade 3C.2a2	Number in clade 3C.2a3	Number in clade 3C.2a4
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir										
+	+	+	Yes	35	29 (83%)	9	0	0	19	1	0	0
-	+	+	Yes	17	3 (18%)	0	0	0	1	1	1	0
-	-	+	Yes	1	1 (100%)	0	0	0	1	0	0	0
+	-	+	Yes	1	1 (100%)	0	0	0	1	0	0	0
			Totals	54	34 (63%)	9	0	0	22	2	1	0
+	+	-	No	8	8 (100%)	0	0	0	6	1	1	0
+	-	-	No	0	0	0	0	0	0	0	0	0
-	+	-	No	24	7 (29%)	0	0	0	6	0	1	0
-	-	-	No	143	85 (59%)	0	0	0	73	8	3	1
			Totals	175	100 (57%)	0	0	0	85	9	5	1
			Totals	229	134	9	0	0	107	11	6	1

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2019-02-12: a number still in process

Situation at the time of the September 2018 VCM (viruses with collection dates Feb-Aug 2018)

RBC binding phenotype ¹			HI assay possible	Number of isolates	Number sequenced ²	Number in clade 3C.3a	Number in clade 3C.2a	Number in clade 3C.2a1	Number in clade 3C.2a1b	Number in clade 3C.2a2	Number in clade 3C.2a3
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir									
+	+	+	Yes	23	20 (87%)	5	0	0	7	8	0
-	+	+	Yes	43	40 (93%)	0	0	0	24	16	0
-	-	+	Yes	1	1 (100%)	0	0	0	1	0	0
+	-	+	Yes	0	0	0	0	0	0	0	0
			Totals	67	61 (91%)	5	0	0	32	24	0
+	+	-	No	14	14 (100%)	0	0	0	4	10	0
+	-	-	No	0	0	0	0	0	0	0	0
-	+	-	No	15	15 (100%)	0	0	0	8	7	0
-	-	-	No	106	101 (95%)	0	0	0	38	59	4
			Totals	135	130 (96%)	0	0	0	50	76	4
			Totals	202	191	5	0	0	82	100	4

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4

2. As of 2017-09-13: a number still in process

Table 7. Summary of influenza A(H3N2) viruses analysed by HI, collected after 2018-08-31

MONTH	H3N2													
	Country	Number received ¹	Number propagated ²	3C.3a: Cell-propagated A/Stockholm/6/2014		3C.2a: Cell-propagated Kong/5738/2014		A/Hong	3C.2a1: Egg-propagated A/Singapore/INFIMH-16-0019/2016			3C.2a2:Egg-propagated A/Switzerland/8060/2017		
				≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶
2018														
SEPTEMBER														
France	6	3		3		1	2		1	2				3
OCTOBER														
France	7	4			4	1	3			2	2			4
Mauritius	3	3	1		2	1	1	1		2	1			3
NOVEMBER														
Austria	3	1	1			1				1				1
France	7	3	2	1		3				3				3
Iceland	11	7			7	1	6		3	3	1			7
Latvia	2	1	1			1				1			1	
Norway	12	1		1		1			1					1
Russia	2	1			1			1			1			1
Switzerland	2	1	1			1				1				1
DECEMBER														
France	14	4		4		2	2		1	2	1			4
Hungary	2	1	1					1			1			1
Israel	7	3	1	2				3		1	2			3
Latvia	1	1	1			1				1				1
Norway	7	1	1			1			1					1
Sweden	4	3			3		1	2			3			3
Ukraine	21	2			2	2					2			2
2019														
JANUARY														
Albania	3	1		1				1			1			1
Armenia	3	1	1					1		1				1
France	15	2		2		2					2			2
Hungary	2	2	1	1		1		1			2			2
Netherlands	4	2		1	1			2			2			2
Portugal	4	3	2	1		2		1		1	2			3
Spain	43	2	1		1			2			2			2
Ukraine	2	1		1		1					1			1
	187	54	15	18	21	23	15	16	7	21	26	0	1	53
17 Countries			27.8%	33.3%	38.9%	42.6%	27.8%	29.6%	13.0%	38.9%	48.1%	0.0%	1.9%	98.1%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

3. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Stockholm/6/2014 (3C.3a)

4. Compared to homologous titres for ferret antisera raised against tissue culture-propagated A/Hong Kong/5738/2014 (3C.2a)

5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Singapore/INFIMH-16-0019/2016 (3C.2a1)

6. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/8060/2017 (3C.2a2)

Table 8-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-09-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				A/Stock	A/HK	A/Bretagne	A/Nor	A/Greece	A/Singapore	A/HK	A/La Rioja	A/Switz	A/Nor	A/Nor	A/Eng	A/HK
				6/14	5738/14	1413/17	4436/16	4/17	0019/16	656/18	2202/18	8060/17	2516/18	2620/18	538/18	675/18
	Passage history			SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg	SIAT	SIAT	SIAT	SIAT
	Ferret number			F14/14 ⁻¹	F30/14 ⁻¹	F01/18 ⁻¹	F03/17 ⁻¹	F27/17 ⁻¹	F41/17 ⁻¹	F25/18 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F29/18 ⁻¹	F30/18 ⁻¹	F31/18 ⁻¹	F32/18 ⁻¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2	3C.2a2	3C.2a1b	3C.3a	3C.2a2
REFERENCE VIRUSES																
A/Stockholm/6/2014	3C.3a	2017-11-20	SIAT1/SIAT3	320	160	80	320	160	160	160	160	320	160	160	160	160
A/Hong Kong/5738/2014	3C.2a	2017-11-16	MDCK1/MDCK2/SIAT2	160	160	160	320	320	320	320	160	320	160	80	80	160
A/Bretagne/1413/2017	3C.2a2	2018-06-28	MDCK1/SIAT4	160	80	1280	320	160	160	1280	160	1280	160	80	80	160
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2017-10-05	E5/E1	40	80	80	160	160	640	80	320	160	80	160	160	160
A/Hong Kong/656/2018	3C.2a2	2018-07-30	MDCK1/SIAT3	320	640	1280	640	320	320	2560	640	2560	320	160	320	640
A/Switzerland/8060/2017	3C.2a2	2018-08-13	E7 (AM2AL5)	40	80	1280	160	320	640	1280	160	1280	80	160	160	160
A/Norway/2620/2018	3C.2a1b	2018-04-16	SIAT3	ND	ND	ND	ND	ND	320	ND	ND	ND	ND	320	ND	ND
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT2	40	40	80	160	80	40	80	<	40	160	80	320	80
TEST VIRUSES																
A/Beijing-Huairou/1557/2018	3C.2a1b		C2/SIAT1/SIAT1	80	40	40	80	160	80	40	160	80	<	80	40	40
A/Guizhou-Nanming/1958/2018	3C.2a2		C2/SIAT1/SIAT1	80	40	160	160	80	80	160	80	160	40	40	40	40

Table 8-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-09-18

Viruses	Other information	Passage history	Collection date	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock	A/HK	A/Bretagne	A/Nor	A/Greece	A/Singapore	A/HK	A/La Rioja	A/Switzer	
				6/14	5738/14	1413/17	4436/16	4/17	0019/16	656/18	2202/18	8060/17	
	Passage history			SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg	
	Ferret number			F14/14 ⁻¹	F30/14 ⁻¹	F01/18 ⁻¹	F03/17 ⁻¹	F27/17 ⁻¹	F41/17 ⁻¹	F25/18 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2	
REFERENCE VIRUSES													
A/Stockholm/6/2014			2017-11-20	SIAT1/SIAT3	320	80	160	320	320	160	160	320	
A/Hong Kong/5738/2014			2017-11-20	MDCK1/MDCK2/SIAT2	40	80	160	160	160	160	80	160	
A/Bretagne/1413/2017			2018-06-28	MDCK1/SIAT4	160	160	640	320	320	160	1280	160	
A/Singapore/INFIMH-16-0019/2016			2017-10-05	E5/E2	40	80	80	160	320	1280	160	320	
A/Hong Kong/656/2018			2018-07-30	MDCK1/SIAT3	320	320	640	320	320	320	1280	160	
A/Switzerland/8060/2017	clone 57		2018-08-13	E7 (AM2AL5)	40	160	1280	160	320	640	2560	320	
TEST VIRUSES													
A/Kanagawa/IC1618/2017				NIID-MDCK3/E4/E1	<	80	1280	80	160	320	1280	80	
A/Kanagawa/AC1624/2017				NIID-MDCK3/E7/E1	160	320	640	640	320	320	640	320	
A/Brisbane/190/2017			2017-10-18	E4/E1	160	320	>5120	320	640	1280	>5120	320	
A/Brisbane/01/2018			2018-01-02	E4/E1	40	160	1280	160	320	640	1280	160	
A/Singapore/GP0454/2018			2018-02-19	E5/E1	40	<	<	<	40	40	40	80	
A/Sydney/22/2018			2018-03-03	E5/E1	80	80	80	160	640	1280	160	320	
A/Singapore/KK0259/2018			2018-03-09	E5/E1	<	80	1280	80	160	320	2560	160	
A/Brisbane/34/2018			2018-03-17	E2/E1	40	<	<	40	<	<	40	<	
A/Brisbane/43/2018			2018-04-07	E5/E1	40	160	1280	160	320	640	1280	160	
A/Zaragoza/490/2018			2018-05-17	SIAT1/SIAT1	160	160	320	320	320	320	320	160	
A/Zaragoza/491/2018			2018-05-28	SIAT1/SIAT1	80	80	160	160	80	160	160	80	
A/Tyumen/RII-01/2018			2018-06-11	MDCK1/SIAT1	320	320	320	320	320	160	320	160	
A/Zaragoza/489/2018			2018-07-16	SIAT1/SIAT1	160	80	160	160	160	160	160	80	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-09-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock	A/HK	A/Bretagne	A/Nor	A/Greece	A/Singapore	A/HK	A/La Rioja	A/Switz
				6/14	5738/14	1413/17	4436/16	4/17	0019/16	656/18	2202/18	8060/17
	Passage history			SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Ferret number			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F41/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
REFERENCE VIRUSES												
A/Stockholm/6/2014		2017-11-20	SIAT1/SIAT3	320	160	160	160	160	160	160	160	320
A/Hong Kong/5738/2014		2017-11-16	MDCK1/MDCK2/SIAT2	160	160	160	160	160	160	160	160	320
A/Bretagne/1413/2017		2018-06-28	MDCK1/SIAT4	160	160	1280	320	320	160	1280	160	1280
A/Singapore/INFIMH-16-0019/2016		2017-10-05	E5/E2	40	80	80	160	160	640	80	320	160
A/Hong Kong/656/2018		2018-07-30	MDCK1/SIAT3	160	320	640	320	320	160	1280	160	640
A/Switzerland/8060/2017	clone 57	2018-08-13	E7 (AM2AL5)	40	160	1280	160	320	640	2560	320	2560
TEST VIRUSES												
A/Zaragoza/1476697/2018		2018-04-09	SIAT1	160	160	160	320	160	160	80	80	160
A/Zaragoza/1485189/2018		2018-04-24	SIAT1	80	160	160	320	160	160	160	80	160
A/Zaragoza/1496248/2018		2018-05-14	SIAT1	320	320	320	640	320	320	320	160	640
A/Zaragoza/490/2018		2018-05-17	SIAT1	160	160	320	320	160	160	160	80	320
A/Zaragoza/491/2018		2018-05-28	SIAT1	80	80	160	320	160	160	80	80	320
A/Zaragoza/1512154/2018		2018-06-09	SIAT1	80	160	160	320	160	160	160	80	320
A/Zaragoza/492/2018		2018-07-12	SIAT1	160	320	320	640	320	320	320	160	640
A/Zaragoza/489/2018		2018-07-16	SIAT2	160	160	320	320	160	160	160	80	320

Table 8-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-09-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock	A/HK	A/Bretagne	A/Nor	A/Greece	A/Singapore	A/HK	A/La Rioja	A/Switz
				6/14	5738/14	1413/17	4436/16	4/17	0019/16	656/18	2202/18	8060/17
	Passage history			SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Ferret number			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F41/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
REFERENCE VIRUSES												
A/Stockholm/6/2014		2017-11-20	SIAT1/SIAT2	320	80	80	160	160	160	80	80	160
A/Hong Kong/5738/2014		2017-11-16	MDCK1/MDCK2/SIAT2	160	160	160	320	160	160	160	160	160
A/Bretagne/1413/2017		2018-06-28	MDCK1/SIAT4	160	160	2560	160	160	160	1280	160	1280
A/Singapore/INFIMH-16-0019/2016		2017-10-05	E5/E2	40	40	80	80	160	640	80	320	160
A/Hong Kong/656/2018		2018-07-30	MDCK1/SIAT3	160	320	2560	320	320	160	1280	160	1280
A/Switzerland/8060/2017	clone 57	2018-08-13	E7 (AM2AL5)	40	160	2560	80	320	640	1280	160	1280
TEST VIRUSES												
A/Zaragoza/1528894/2018		2018-07-09	SIAT3	80	80	160	160	80	160	160	80	160
A/Zaragoza/493/2018		2018-08-23	SIAT3	40	80	160	160	80	160	160	40	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-10-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Nor 4436/16	A/Greece 4/17	A/Singapore 0019/16	A/HK 656/18	A/La Rioja 2202/18	A/Switz 8060/17
				SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Passage history			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F41/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
	Genetic group											
REFERENCE VIRUSES												
A/Stockholm/6/2014	3C.3a	2017-11-20	SIAT1/SIAT2	320	160	160	320	320	160	160	160	320
A/Hong Kong/5738/2014	3C.2a	2017-11-16	MDCK1/MDCK2/SIAT2	160	160	320	320	320	320	320	160	320
A/Bretagne/1413/2017	3C.2a2	2018-06-28	MDCK1/SIAT4	160	160	1280	320	320	160	640	160	1280
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2017-10-05	E5/E2	40	40	40	160	160	640	80	160	160
A/Hong Kong/656/2018	3C.2a2	2018-07-30	MDCK1/SIAT3	320	320	640	320	320	160	1280	160	640
A/Switzerland/8060/2017	clone 57 3C.2a2	2018-08-13	E7 (AM2AL5)	40	160	1280	160	320	320	1280	160	2560
TEST VIRUSES												
A/Oman/3542/2018	3C.2a1b	2018-06-01	SIAT1/SIAT1	160	80	80	160	320	80	80	320	80
A/Oman/3329/2018	3C.2a1b	2018-06-04	SIAT1/SIAT1	80	80	<	80	160	40	<	160	40
A/Oman/3356/2018	3C.2a1b	2018-06-07	SIAT1/SIAT1	160	80	40	160	160	80	40	160	80
A/Oman/3340/2018	3C.2a1b	2018-06-08	SIAT1/SIAT1	160	80	40	160	320	80	40	320	80
A/Oman/3372/2018	3C.2a1b	2018-06-13	SIAT1/SIAT1	80	80	<	80	160	40	<	160	80
A/Oman/3407/2018	3C.2a1b	2018-06-16	SIAT1/SIAT1	40	80	<	80	320	40	<	160	40
A/Oman/3577/2018	3C.2a1b	2018-07-04	SIAT1/SIAT1	160	80	40	160	320	40	80	160	80
A/Oman/3610/2018	3C.2a1b	2018-07-06	SIAT0	80	80	40	160	160	40	40	160	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2018-11-28

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Nor 4436/16	A/Greece 4/17	A/Singapore 0019/16	A/HK 656/18	A/La Rioja 2202/18	A/Switz 8060/17
				SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Passage history			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F41/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
	Genetic group											
REFERENCE VIRUSES												
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT2	320	80	80	160	320	160	80	80	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT2	80	160	160	160	160	160	160	80	160
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	80	80	640	160	160	160	640	80	640
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E2	<	40	40	80	160	640	80	160	80
A/Hong Kong/656/2018		2018-04-07	MDCK1/SIAT3	160	160	1280	320	320	160	1280	80	640
A/Switzerland/8060/2017	clone 57	2017-12-12	E7 (AM2AL5)	<	160	2560	160	320	640	2560	80	1280
TEST VIRUSES												
A/Mauritius/I-906/2018		2018-10-02	MDCK1/SIAT1	160	160	40	160	320	160	40	320	80
A/Mauritius/I-925/2018		2018-10-08	MDCK1/SIAT1	40	<	<	40	80	80	<	80	<
A/Mauritius/I-967/2018		2018-10-22	MDCK2/SIAT1	40	40	40	80	160	160	<	160	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-01-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Nor 4436/16	A/Greece 4/17	A/Singapore 0019/16	A/HK 656/18	A/La Rioja 2202/18	A/Switz 8060/17
				SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Passage history			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F41/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
	Genetic group											
REFERENCE VIRUSES												
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	320	160	160	320	320	320	160	160	320
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT2	160	160	160	320	320	320	320	160	320
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	160	160	1280	320	320	320	1280	160	1280
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E3	<	40	40	160	160	640	40	160	80
A/Hong Kong/656/2018		2018-04-07	MDCK1/SIAT3	160	160	640	320	320	320	1280	80	640
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	40	160	2560	320	160	640	2560	160	2560
TEST VIRUSES												
A/Latvia/11-019324/2018		2018-11-07	MDCK2/SIAT1	160	160	640	320	160	160	640	80	640
A/Clermont-Ferrand/2062/2018		2018-11-07	MDCK3/SIAT1	160	160	80	320	320	160	80	160	160
A/Iceland/108/2018		2018-11-13	MDCK1/SIAT1	40	40	<	80	160	80	<	160	40
A/Austria/1102969/2018		2018-11-16	SIAT1/SIAT1	160	80	40	320	320	160	40	320	80
A/Lyon/2106/2018		2018-11-20	MDCK3/SIAT1	160	80	80	160	160	160	40	320	80
A/Paris/2511/2018		2018-11-29	MDCK1/SIAT1	80	80	80	320	160	160	80	40	160
A/Latvia/12-005233/2018		2018-12-03	MDCK1/SIAT1	160	160	640	320	320	160	640	160	320
A/Paris/2538/2018		2018-12-04	MDCK1/SIAT1	80	80	80	320	160	160	160	40	80
A/Paris/2544/2018		2018-12-04	MDCK1/SIAT1	80	80	80	320	160	160	80	80	160

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-01-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Nor 4436/16	A/Greece 4/17	A/Singapore 0019/16	A/HK 656/18	A/La Rioja 2202/18	A/Switz 8060/17
				SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg
	Passage history			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F15/18 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2
	Genetic group											
REFERENCE VIRUSES												
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	320	160	160	320	160	160	160	160
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	160	160	160	320	160	160	160	160
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	160	640	320	160	160	640	640
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E3	<	40	40	80	80	160	40	80
A/Hong Kong/656/2018		3C.2a2	2018-04-07	MDCK1/SIAT3	320	320	640	320	320	1280	160	640
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	40	160	1280	160	160	1280	160	1280
TEST VIRUSES												
A/Saint-Etienne/1912/2018		3C.2a1b	2018-09-07	MDCK2/SIAT1	80	40	40	160	80	40	<	320
A/La Reunion/1970/2018		3C.2a1b	2018-09-11	MDCK2/SIAT1	80	80	40	80	80	80	80	320
A/La Reunion/1967/2018		3C.2a1b	2018-09-15	MDCK2/SIAT1	80	40	40	80	80	40	40	320
A/Poitiers/1978/2018		3C.2a1b	2018-10-08	MDCK2 /SIAT1	40	40	<	80	80	40	<	160
A/Poitiers/1976/2018		3C.2a1b	2018-10-09	MDCK3/SIAT1	40	40	<	80	80	<	<	160
A/Poitiers/2003/2018		3C.2a1b	2018-10-20	MDCK2/SIAT1	40	80	40	80	80	40	160	320
A/Lorraine/2365/2018		3C.2a1b	2018-10-22	MDCK2/SIAT2	<	40	<	40	80	<	<	80
A/Iceland/101/2018		3C.2a1b	2018-11-01	MDCK1/SIAT1	<	40	<	40	40	40	<	80
A/Iceland/100/2018		3C.2a1b	2018-11-01	MDCK1/SIAT1	40	80	<	40	80	40	160	160
A/Iceland/102/2018		3C.2a1b	2018-11-02	MDCK1/SIAT1	40	40	<	40	40	80	<	160
A/Iceland/103/2018		3C.2a1b	2018-11-03	MDCK1/SIAT1	40	40	<	80	80	80	<	160
A/Iceland/104/2018			2018-11-05	MDCK1/SIAT1	<	40	<	40	80	80	<	160
A/Iceland/107/2018		3C.2a1b	2018-11-09	MDCK1/SIAT2	40	40	<	80	80	40	<	160
A/Norway/3668/2018		3C.2a1b	2018-11-29	SIAT1	80	80	80	80	160	160	40	320
A/Norway/3735/2018		3C.2a1b	2018-12-03	SIAT1	160	80	320	320	160	160	80	320
A/Paris/2572/2018		3C.3a	2018-12-06	SIAT1	80	40	<	160	40	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40Vaccine
SH 2018
NH 2018-19Vaccine
SH 2019

Table 8-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-01-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Stock 6/14	A/HK 5738/14	A/Bretagne 1413/17	A/Nor 4436/16	A/Greece 4/17	A/Singapore 0019/16	A/HK 656/18	A/La Rioja 2202/18	A/Switz 8060/17	
				SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg	
	Passage history			F14/14 ¹	F30/14 ¹	F01/18 ¹	F03/17 ¹	F27/17 ¹	F46/17 ¹	F25/18 ¹	F26/18 ¹	F27/18 ¹	
	Ferret number			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2	
	Genetic group												
REFERENCE VIRUSES													
A/Stockholm/6/2014		3C.3a	2014-02-06	SIAT1/SIAT3	320	80	80	320	160	320	80	80	160
A/Hong Kong/5738/2014		3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	320	160	160	320	160	640	160	160	320
A/Bretagne/1413/2017		3C.2a2	2017-10-09	MDCK1/SIAT4	160	160	1280	320	160	160	640	80	640
A/Singapore/INFIMH-16-0019/2016		3C.2a1	2016-04-14	E5/E3	<	40	40	80	40	640	40	160	80
A/Hong Kong/656/2018		3C.2a2	2018-04-07	MDCK1/SIAT3	160	320	640	320	160	320	1280	160	640
A/Switzerland/8060/2017	clone 57	3C.2a2	2017-12-12	E7/E1	<	160	1280	160	80	640	1280	80	1280
TEST VIRUSES													
A/Trollhattan/3/2018		3C.2a1b	2018-12-01	MDCK0/SIAT1	40	40	40	80	80	80	40	160	40
A/Paris/2671/2018		3C.3a	2018-12-17	MDCK1/SIAT1	80	40	40	160	40	80	40	<	40
A/Bretagne/026/2019		3C.3a	2019-01-02	MDCK1/SIAT1	80	80	40	160	80	80	40	40	80
A/Paris/029/2019		3C.3a	2019-01-03	MDCK1/SIAT1	80	80	40	160	40	80	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-02-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock	A/HK	A/Singapore	A/La Rioja	A/Switz	A/Eng	A/Neth	A/Nor	
				6/14	5738/14	0019/16	2202/18	8060/17	538/18	10260/18	3275/18	
	Passage history			SIAT	MDCK	Egg 10-4	SIAT	Egg	SIAT	Egg	SIAT	
	Ferret number			F14/14 ¹	F30/14 ¹	F46/17*1	F26/18 ¹	F27/18 ¹	F31/18 ¹	F02/19 ¹	F03/19 ¹	
	Genetic group			3C.3a	3C.2a	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.2a1b	
REFERENCE VIRUSES												
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	320	40	160	80	160	320	<	80	
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT2	160	80	320	160	320	320	40	320	
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E3	40	40	640	160	80	80	40	40	
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	40	80	640	160	1280	160	40	80	
A/England/538/2018		2018-02-26	MDCK1/SIAT1	80	ND	ND	ND	40	1280	ND	ND	
A/Netherlands/10260/2018		2018-02-15	E5	40	40	160	320	80	80	2560	160	
TEST VIRUSES												
A/Sundsvall/2/2018		2018-12-01	MDCK0/SIAT1	40	<	<	<	<	320	ND	ND	
A/Sundsvall/1/2018		2018-12-03	MDCK0/SIAT1	40	<	<	<	<	320	ND	ND	
A/Israel/10464/2018		2018-12-21	C1/SIAT1	80	<	80	40	40	640	<	40	
A/Israel/603/2018		2018-12-23	C1/SIAT1	160	<	160	160	160	640	<	40	
A/Kyiv/532/2018		2018-12-24	SIAT1/SIAT1	<	40	40	160	<	40	<	160	
A/Israel/604/2018		2018-12-25	C1/SIAT1	80	<	80	40	80	320	<	40	
A/Ukraine/8106/2018		2018-12-27	MDCK2/SIAT1	<	40	40	160	<	40	<	160	
A/Ukraine/8103/2019		2019-01-03	MDCK2/SIAT1	80	40	80	320	40	40	<	160	
A/Portugal/139/2019		2019-01-04	MDCK1/SIAT1	160	<	80	80	80	160	<	80	
A/Portugal/151/2019		2019-01-06	SIAT1/SIAT1	160	80	160	320	80	160	<	320	
A/Portugal/146/2019		2019-01-07	MDCK1/SIAT1	80	40	40	160	40	40	<	160	
A/Avila/32/2019		2019-01-15	SIAT1/SIAT1	<	<	<	<	<	40	<	<	
A/Burgos/31/2019		2019-01-15	SIAT1/SIAT1	160	<	80	40	80	640	<	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40; ND = Not Done

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-02-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Stock	A/HK	A/Bretagne	A/Singapore	A/La Rioja	A/Switz	A/Eng	A/Neth	A/Nor
				6/14	5738/14	1413/17	0019/16	2202/18	8060/17	538/18	10260/18	3275/18
	Passage history			SIAT	MDCK	SIAT	Egg 10 ⁻⁴	SIAT	Egg	SIAT	Egg	SIAT
	Ferret number			F14/14 ¹	F30/14 ¹	F01/18 ¹	F46/17 ¹	F26/18 ¹	F27/18 ¹	F31/18 ¹	F02/19 ¹	F03/19 ¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.2a1b
REFERENCE VIRUSES												
A/Stockholm/6/2014		2014-02-06	SIAT1/SIAT3	320	40	80	320	80	160	320	<	80
A/Hong Kong/5738/2014		2014-04-30	MDCK1/MDCK2/SIAT1	160	80	160	320	160	160	160	40	160
A/Bretagne/1413/2017		2017-10-09	MDCK1/SIAT4	160	40	640	320	80	640	160	40	160
A/Singapore/INFIMH-16-0019/2016		2016-04-14	E5/E3	40	<	40	640	160	80	80	<	40
A/Switzerland/8060/2017	clone 57	2017-12-12	E7/E1	80	80	2560	1280	160	1280	160	40	160
A/England/538/2018		2018-02-26	MDCK1/SIAT3	160	640	80	320	40	80	1280	<	40
A/Netherlands/10260/2018		2018-02-15	E5	40	<	80	160	320	160	80	1280	160
TEST VIRUSES												
A/Switzerland/293/2018		2018-11-01	MDCK1/SIAT3	160	40	80	160	160	160	160	<	320
A/Nizhny Novgorod/14645V/2018		2018-11-20	MDCK1/SIAT2	<	<	<	40	80	<	40	<	40
A/Hungary/5/2019		2018-12-19	Cx/SIAT1	160	<	<	80	160	<	40	<	160
A/Hungary/28/2019		2019-01-03	Cx/SIAT1	80	<	<	80	80	<	40	<	80
A/Hungary/26/2019		2019-01-07	Cx/SIAT1	160	40	40	80	320	80	80	40	80
A/Netherlands/10002/2019		2019-01-07	MDCK-MIX2/SIAT1	80	<	40	40	<	<	640	<	<
A/Armenia/91/2019		2019-01-14	SIAT1	160	<	40	160	160	40	80	<	80
A/Neth/10006/2019		2019-01-14	MDCK-MIX2/SIAT1	40	<	40	40	<	<	640	<	<
A/Albania/9142/2019		2019-01-19	SIAT1	80	<	<	80	80	40	40	<	40

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used) ¹ < = <40

Vaccine
SH 2018
NH 2018-19

Vaccine
SH 2019

Table 8-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2019-01-25

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				A/Stock	A/HK	A/Bretagne	A/Nor	A/Greece	A/Singapore	A/HK	A/La Rioja	A/Switz	A/Eng	NEW	NEW
				6/14	5738/14	1413/17	4436/16	4/17	0019/16	656/18	2202/18	8060/17	538/18	A/Neth	A/Nor
	Passage history			SIAT	MDCK	SIAT	SIAT	SIAT	Egg 10 ⁻⁴	SIAT	SIAT	Egg	SIAT	Egg	SIAT
	Ferret number			F14/14 ⁻¹	F30/14 ⁻¹	F01/18 ⁻¹	F03/17 ⁻¹	F27/17 ⁻¹	F46/17 ⁻¹	F25/18 ⁻¹	F26/18 ⁻¹	F27/18 ⁻¹	F31/18 ⁻¹	F02/19 ⁻¹	F03/19 ⁻¹
	Genetic group			3C.3a	3C.2a	3C.2a2	3C.2a1	3C.2a1a	3C.2a1	3C.2a2	3C.2a1b	3C.2a2	3C.3a	3C.2a1b	3C.2a1b
REFERENCE VIRUSES															
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	320	80	80	320	160	320	80	80	160	320	<	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT2	320	160	160	320	160	640	160	160	320	320	40	320
A/Bretagne/1413/2017	3C.2a2	2017-10-09	MDCK1/SIAT4	160	160	1280	320	160	160	640	80	640	ND	ND	ND
A/Singapore/INFIMH-16-0019/2016	3C.2a1	2016-04-14	E5/E3	<	40	40	80	40	640	40	160	80	80	40	40
A/Hong Kong/656/2018	3C.2a2	2018-04-07	MDCK1/SIAT3	160	320	640	320	160	320	1280	160	640	ND	ND	ND
A/Switzerland/8060/2017	3C.2a2	2017-12-12	E7/E1	<	160	1280	160	80	640	1280	80	1280	160	40	80
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT1	80	ND	ND	ND	ND	ND	ND	ND	40	1280	ND	ND
A/Netherlands/10260/2018	3C.2a1b	2018-02-15	E5	40	40	ND	ND	ND	160	ND	320	80	80	2560	160
TEST VIRUSES															
Number of viruses tested				54	54	41	32*	32*	54	32	54*	54	22	20	20*
No with titre reduction ≤2-fold				15	23	3	15	13	7	2	31		5		7
%				27.8	42.6	7.3	46.9	40.6	13.0	6.2	57.4		22.7		35.0
No with titre reduction =4-fold				18	30				21			1	3		
%				33.3	55.6				38.9			1.8	13.7		
No with titre reduction ≥8-fold				21	1	38			26	30		53	14	20	
%				38.9	1.8	92.7			48.1	93.8		98.2	63.6	100	

* Homologous HI titres not available - only results for viruses yielding HI titres of ≥160 with the respective antisera are shown

Vaccine
NH 2018-19

Vaccine
SH 2019

Reference virus results are taken from individual tables as examples. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Table 9-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-07

Viruses	Passage history	Collection Date	Neutralisation Titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
			Post-infection ferret antisera											
			A/Singapore INFIMH-16-0019/16 Egg 10 ⁻⁴ F15/18 3C.2a1		A/Singapore INFIMH-16-0019/16 SIAT F45/17 3C.2a1		A/Switzerland 8060/17 Egg F27/18 3C.2a2		A/HK 656/18 SIAT F25/18 3C.2a2		A/La Rioja 2202/18 SIAT F26/18 3C.2a1b			
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES														
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	320	442	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO) , N171K, L194P, D225G
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	119	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K
A/Switzerland/8060/2017	3C.2a2		E6(Am2AI4)c57 10 ⁻⁵	320	315	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO) , L194P
A/Hong Kong/656/2018	3C.2a2		MDCK1/SIAT1 10 ⁻¹	160	134	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	70	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q
TEST VIRUSES														
A/Egypt/5816/2018	3C.2a1b	2018-09-28	SIAT1	80	86	320	395	40	46	80	91	320	312	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q
A/Mauritius/I-906/2018	3C.2a1b	2018-10-02	MDCK2/SIAT1	80	60	320	267	40	20	40	43	160	177	E62G, Q75H, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q
A/Norway/3275/2018	3C.2a1b	2018-10-04	SIAT1	80	82	320	290	160	121	160	132	80	65	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3288/2018	3C.2a1b	2018-10-05	SIAT1/SIAT1	80	80	320	248	80	76	80	100	80	62	E62G, K92R, A106V, N121K, T131K, T135X (-CHO), R142G, N171K, H311Q
A/Mauritius/I-925/2018	3C.2a1b	2018-10-08	MDCK1/SIAT1	40	20	80	80	<	<	<	<	80	66	E62G, Q75H, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160X (-CHO), N171K, H311Q
A/Norway/3284/2018	3C.2a1b	2018-10-09	SIAT1	80	110	320	272	160	127	160	122	80	66	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q
A/Norway/3313/2018	3C.2a1b	2018-10-18	SIAT1/SIAT1	160	122	320	341	80	117	160	129	80	66	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q
A/Norway/3393/2018	3C.2a1b	2018-10-20	SIAT1	160	126	320	420	160	170	160	146	80	99	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q
A/Egypt/6213/2018	3C.2a1b	2018-10-22	SIAT1	40	40	160	201	40	20	40	20	80	70	E62G, K83E, K92R, N121K, T131K, T135A (-CHO), R142G, N171K, H311Q
A/Mauritius/I-967/2018	3C.2a1b	2018-10-22	MDCK2/SIAT1	40	20	80	63	40	20	<	10	40	49	E62G, Q75H, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160R (-CHO), N171K, H311Q
A/Norway/3444/2018	3C.2a1b	2018-11-06	SIAT1	160	186	320	426	40	48	80	80	320	253	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), N171K, S198P, H311Q
A/Mauritius/I-888/2018	3C.2a2	2018-09-28	SIAT1	160	191	640	788	5120	3870	5120	3996	160	138	T131K, R142K, A212T, R261Q
A/Norway/3294/2018	3C.2a2	2018-10-10	SIAT1/SIAT1	160	126	640	560	5120	5120	5120	7111	80	74	E62K, T131K, R142K, , A212T, R261Q
A/Norway/3384/2018	3C.2a2	2018-10-24	SIAT2	320	242	640	887	5120	3921	2560	3603	80	108	R33Q, T131K, R142K, A212T, R261Q, D291E
A/Ghana/3294/2018	3C.2a3	2018-10-28	SIAT1	80	90	320	411	80	68	160	126	160	224	N121K, T128A (-CHO), T135K (-CHO), S144K, I192V
			Vaccine SH 2018 NH 2018-19				Vaccine SH 2019							

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-10

Viruses	Collection Date	Passage History	Neutralisation Titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate		
			Post-infection ferret antisera												
			A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18				
			Egg 10 ⁻⁴		SIAT		Egg		SIAT		SIAT				
Passage history	Ferret number	Genetic group	3C.2a1		3C.2a1		3C.2a2		3C.2a2		3C.2a2		3C.2a1b		
			2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES															
A/Singapore/INFIMH-16-0019/2016	3C.2a1	E5/E2 10 ⁻⁴	320	442	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO), N171K, L194P, D225G		
A/Singapore/INFIMH-16-0019/2016	3C.2a1	MDCK1/SIAT3/SIAT3	80	119	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K		
A/Switzerland/8060/2017	3C.2a2	E6(Am2A14)c57 10 ⁻⁵	320	315	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO), L194P		
A/Hong Kong/656/2018	3C.2a2	MDCK1/SIAT1 10 ⁻¹	160	134	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V		
A/La Rioja/2202/2018	3C.2a1b	SIAT1/SIAT1 10 ⁻¹	80	70	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q		
TEST VIRUSES															
A/England/e2Rpa2huY2hobQ/2018	3C.2a1b	SIAT1/SIAT1	40	55	320	301	40	40	80	61	320	263	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, R261Q, H311Q		
A/Ireland/71305/2018	3C.2a1b	SIAT1/SIAT1	40	40	320	273	40	20	40	55	160	200	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, I214T, H311Q		
A/Lorraine/2365/2018	3C.2a1b	MDCK2/SIAT1	80	68	160	195	40	40	40	44	160	147	D53N, E62G, K82T, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160X (-CHO), N171K, R208K, H311Q		
A/Bayern/49/2018	3C.2a1b	C1/SIAT1	80	107	320	371	80	98	160	130	80	71	E62G, K92R, N121K, S124N (-CHO), T131K, R142G, N171K, I214T, H311Q		
A/England/lemhqaG1sYmxkZg/2018	3C.2a1b	SIAT1/SIAT1	80	67	640	905	40	20	80	77	640	498	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, S198P, H311Q		
A/Ireland/77960/2018	3C.2a1b	SIAT1/SIAT1	80	86	320	293	80	106	160	139	80	86	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q		
A/Ireland/78114/2018	3C.2a1b	SIAT1/SIAT1	80	107	640	529	40	40	80	87	320	297	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, R261Q, H311Q		
A/Bremen/22/2018	3C.2a1b	C2/SIAT1	160	125	640	659	40	40	80	71	320	367	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q		
A/Iceland/106/2018	3C.2a1b	MDCK1/SIAT1	40	40	160	160	40	40	40	44	80	112	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, P221X, H311Q		
A/Iceland/107/2018	3C.2a1b	MDCK1/SIAT1	80	68	160	220	40	40	80	69	160	211	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, F193X, H311Q		
A/Iceland/108/2018	3C.2a1b	MDCK1/SIAT1	40	55	80	74	40	20	40	20	80	73	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, H311Q		
A/Ireland/81183/2018	3C.2a1b	SIAT3/SIAT1	40	53	320	418	40	40	80	59	160	215	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, R261Q, H311Q		
A/Austria/1102969/2018	3C.2a1b	SIAT1/SIAT1	40	43	320	320	<	5	40	20	80	73	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), S137F, A138S, R142G, N171K, F193S, I214T, H311Q		
A/Iceland/109/2018	3C.2a1b	MDCK1/SIAT1	80	114	320	320	80	78	160	165	80	79	E62G, K92R, N121K, T131K, R142G, N171K, H311Q		
A/Iceland/110/2018	3C.2a1b	MDCK1/SIAT1	640	530	1280	1211	640	528	640	784	640	495	E62G, K92R, N121K, T131K, R142G, N171K, H311Q		
A/Berlin/57/2018	3C.2a1b	C2/SIAT1	80	111	640	672	40	40	80	69	320	347	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, S198P, H311Q		
A/Hessen/44/2018	3C.2a2	C1/SIAT1	320	335	1280	1394	5120	6919	5120	7111	160	137	C52X, P103X, T131K, R142K, V204I, A212T, R261Q		
A/Hamburg/4/2018	3C.2a3	C1/SIAT1	<	5	320	291	40	20	80	61	160	181	D53N, N121K, T128A (-CHO), T135K (-CHO), S144K, I192V, S262N		
A/Paris/2513/2018	3C.3a	MDCK1/SIAT1	80	63	40	46	40	40	40	40	40	20	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R		
A/Paris/2511/2018	3C.3a	MDCK1/SIAT1	80	102	40	58	40	51	40	44	40	42	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R		
A/Paris/2538/2018	3C.3a	MDCK1/SIAT1	80	77	40	54	40	46	40	59	40	50	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R		
A/Paris/2544/2018	3C.3a	MDCK1/SIAT1	80	73	80	83	80	63	80	66	80	61	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, I267V, H311Q, K326R		
			Vaccine SH 2018 NH 2018-19				Vaccine SH 2019								

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-14

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18					
				Egg 10 ⁻⁴ F15/18 3C.2a1		SIAT F45/17 3C.2a1		Egg F27/18 3C.2a2		SIAT F25/18 3C.2a2		SIAT F26/18 3C.2a1b					
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read				
REFERENCE VIRUSES																	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	320	442	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO), N171K, L194P, D225G	NA Polymorphism		
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	119	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K			
A/Switzerland/8060/2017	3C.2a2		E6(Am2A4)c57 10 ⁻⁵	320	315	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO), L194P			
A/Hong Kong/656/2018	3C.2a2		MDCK1/SIAT1 10 ⁻¹	160	134	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V			
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	70	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q			
TEST VIRUSES																	
A/Iceland/98/2018	3C.2a1b	2018-10-11	MDCK1/SIAT1	<	10	40	40	<	1	<	3	40	40	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160I (-CHO), N171K, P221L, H311Q	D151X		
A/Iceland/101/2018	3C.2a1b	2018-11-01	MDCK1/SIAT1	40	20	40	40	<	<	<	<	40	40	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, H311Q			
A/Iceland/100/2018	3C.2a1b	2018-11-01	MDCK1/SIAT1	<	10	80	107	<	<	40	20	80	80	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N158K (-CHO), N171K, H311Q			
A/Iceland/102/2018	3C.2a1b	2018-11-02	MDCK1/SIAT1	40	40	40	40	<	<	<	<	40	45	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, H311Q			
A/Iceland/103/2018	3C.2a1b	2018-11-03	MDCK1/SIAT1	<	10	80	60	<	<	<	<	80	67	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160R (-CHO), N171K, H311Q			
A/Iran/143227/2018	3C.2a1b	2018-11-05	SIAT2/SIAT1	40	49	320	338	40	20	40	55	320	250	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q			
A/Poitiers/2105/2018	3C.2a1b	2018-11-17	MDCK3/SIAT1	40	52	320	347	<	0	40	42	160	199	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, K276R, I300V, H311Q			
A/Austria/1105514/2018	3C.2a1b	2018-11-18	SIAT1/SIAT1	80	61	320	294	80	64	80	67	40	57	E62G, K92R, N121K, T131K, R142G, N171K, I214T, H311Q			
A/Lyon/2106/2018	3C.2a1b	2018-11-20	MDCK3/SIAT1	80	64	160	160	<	<	<	<	160	130	E62G, D85X, K92R, N121K, T135K (-CHO), I140M, R142G, T160X (-CHO), N171K, K310T, H311Q			
A/Iran/145771/2018	3C.2a1b	2018-11-24	SIAT2/SIAT1	40	40	320	292	<	<	40	40	160	156	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q			
A/Iran/145723/2018	3C.2a1b	2018-11-24	SIAT2/SIAT1	40	50	320	274	<	5	40	40	160	128	E62G, K92R, N121K, T128A (-CHO)T135K (-CHO), R142G, N171K, I214T, H311Q			
A/Austria/1108601/2018	3C.2a1b	2018-11-29	SIAT1/SIAT1	80	64	320	427	80	68	80	83	40	50	E62G, K92R, N121K, T131K, R142G, N171K, H311Q			
A/Austria/1108689/2018	3C.2a1b	2018-12-01	SIAT1/SIAT1	40	48	640	551	<	<	80	68	320	422	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, S198P, H311Q			
A/Norway/3735/2018	3C.2a1b	2018-12-03	SIAT1	40	40	80	82	40	20	<	10	<	1	E62G, K92R, A106V, N121K, T131K, R142G, T160X (-CHO), N171K, H311Q			
A/Latvia/11-019324/2018	3C.2a2	2018-11-07	MDCK2/SIAT1	80	70	80	119	320	364	1280	978	40	40	T131K, R142K, N158K (-CHO), A212T, R261Q	D151X		
A/Latvia/11-058643/2018	3C.2a2	2018-11-22	MDCK2/SIAT1	80	79	320	260	320	467	640	564	40	20	T131K, R142K, T160X (-CHO), A212T, R261Q			
A/Iran/145740/2018	3C.2a2	2018-11-24	SIAT2/SIAT1	320	353	640	615	2560	2472	5120	4096	80	113	T131K, R142G, A212T, R261Q			
A/Latvia/12-005233/2018	3C.2a2	2018-12-03	MDCK1/SIAT1	160	107	320	288	1280	1209	1280	1164	40	52	T131K, R142K, T160A (-CHO), A212T, R261Q			
A/Clermont-Ferrand/2062/2018	3C.2a3	2018-11-07	MDCK3/SIAT1	40	40	320	341	<	10	40	43	80	112	N121K, N126K (-CHO), T135K (-CHO), R142G, S144K, R150K, R261Q			
A/Pays de Loire/2512/2018	3C.2a3	2018-11-28	SIAT1	40	40	640	480	<	10	40	58	160	178	N121K, N126K (-CHO), T135K (-CHO), I140T, R142G, S144K, R150K, R261Q			
A/Paris/2572/2018	3C.3a	2018-12-06	SIAT1	<	10	<	0	<	<	<	1	<	<	S91N, T128A (-CHO)A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R			
A/Iceland/104/2018	Pending	2018-11-05	MDCK1/SIAT1	40	20	40	20	<	<	<	<	40	20	Sequence incomplete			
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019									

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-17

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18			
				Egg 10 ⁻⁴ F46/17 3C.2a1		SIAT F45/17 3C.2a1		Egg F27/18 3C.2a2		SIAT F25/18 3C.2a2		SIAT F26/18 3C.2a1b			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES															
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO) , N171K, L194P, D225G	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K	
A/Switzerland/8060/2017	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO) , L194P	
A/Hong Kong/656/2018	3C.2a2		MDCK1/SIAT1 10 ⁻¹	80	70	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V	
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q	
TEST VIRUSES															
A/Denmark/1787/2018	3C.2a1b	2018-10-02	SIAT3/SIAT1	<	0	320	309	<	<	40	20	160	149	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, I214T, H331Q	
A/Poitiers/1976/2018	3C.2a1b	2018-10-09	MDCK3/SIAT1	40	50	160	193	<	<	<	1	160	234	E62G, K92R, N121K, T135K (-CHO), R142G, T160X (-CHO), N171K, K276R, H311Q	
A/Norway/3419/2018	3C.2a1b	2018-10-29	SIAT1/SIAT1	40	20	320	400	80	91	160	137	80	68	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Denmark/1793/2018	3C.2a1b	2018-10-31	SIAT3/SIAT1	40	40	640	480	160	129	160	160	80	71	E62G, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3411/2018	3C.2a1b	2018-11-01	SIAT1/SIAT1	160	155	640	640	40	51	160	130	320	250	K92R, N121K, T135N (-/+CHO), N171K, D188N, I214T, H311Q	
A/Norway/3398/2018	3C.2a1b	2018-11-01	SIAT1/SIAT1	40	40	320	265	80	71	80	83	40	58	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3569/2018	3C.2a1b	2018-11-08	SIAT1/SIAT1	40	50	320	357	80	79	80	116	80	60	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3570/2018	3C.2a1b	2018-11-13	SIAT1/SIAT1	40	20	320	387	80	88	80	107	80	61	E62G, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3541/2018	3C.2a1b	2018-11-13	SIAT1/SIAT1	40	47	320	302	80	86	80	116	40	51	E62G, K83E, K92R, N121K, T131K, R142G, T160X (-CHO), N171K, H311Q	
A/Norway/3545/2018	3C.2a1b	2018-11-15	SIAT1/SIAT1	<	5	320	251	80	64	80	77	40	40	E62G, K83E, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Norway/3589/2018	3C.2a1b	2018-11-18	SIAT1/SIAT1	40	20	320	301	<	<	<	10	160	144	I48R, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, V309I, H311Q	
A/Denmark/1822/2018	3C.2a1b	2018-11-25	SIAT3/SIAT1	40	42	640	590	<	<	40	49	160	232	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, S198P, H311Q	
A/Denmark/1829/2018	3C.2a1b	2018-11-26	SIAT3/SIAT1	40	44	640	543	<	<	40	58	320	257	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, S198P, H311Q	
A/Norway/3670/2018	3C.2a1b	2018-11-28	SIAT1	40	40	640	499	<	0	40	49	160	222	I48R, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, V309I, H311Q	
A/Norway/3668/2018	3C.2a1b	2018-11-29	SIAT1	40	20	80	72	<	<	<	<	80	65	I48R, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160X (-CHO), P162X, N171K, V309I, H311Q	
A/Norway/3677/2018	3C.2a1b	2018-12-01	SIAT1	80	65	640	610	40	20	40	55	320	274	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q	
A/Norway/3739/2018	3C.2a1b	2018-12-01	SIAT1	40	44	320	333	40	52	80	77	80	70	E62G, K92R, N121K, T131K, R142G, N171K, S219F, H311Q	
A/Denmark/1840/2018	3C.2a1b	2018-12-02	SIAT3/SIAT1	40	47	320	464	<	10	80	77	320	259	E62G, K92R, N121K, T135K (-CHO), R142G, N171K, N216X, H311Q	
A/Denmark/1843/2018	3C.2a1b	2018-12-05	SIAT3/SIAT1	<	0	320	398	80	80	80	89	40	42	D53N, E62G, K83E, K92R, N121K, T131K, R142G, N171K, H311Q	
A/Lyon/1986/2018	3C.2a3	2018-10-15	MDCK2/SIAT1	40	20	320	276	<	<	40	58	160	127	N121K, N126K (-CHO), T135K (-CHO), R142G, S144K, R150K, R261Q	
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019							

Vaccine
SH 2018
NH 2018-19Vaccine
SH 2019¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-21

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18			
				Egg 10 ⁻⁴ F46/17		SIAT F45/17		Egg F27/18		SIAT F25/18		SIAT F26/18			
				3C.2a1	Read	3C.2a1	Read	3C.2a2	Read	3C.2a2	Read	3C.2a1b	Read		
REFERENCE VIRUSES															
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO) , N171K, L194P , D225G	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K	
A/Switzerland/8060/2017	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO) , L194P	
A/Hong Kong/656/2018	3C.2a2		MDCK1/SIAT1 10 ⁻¹	80	70	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V	
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q	
TEST VIRUSES															
A/Myanmar/18M139/2018	3C.2a1b	2018-08-20	SIAT2/SIAT1	80	103	640	499	80	60	80	80	320	335	I48R, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, V309I, H311Q	
A/La Reunion/1970/2018	3C.2a1b	2018-09-11	MDCK2/SIAT1	40	59	160	140	40	20	40	40	160	154	D151X	
A/La Reunion/1967/2018	3C.2a1b	2018-09-15	MDCK2/SIAT1	40	54	160	195	40	20	40	40	160	141	D151X	
A/La Reunion/1965/2018	3C.2a1b	2018-09-17	MDCK2/SIAT1	80	72	320	265	40	40	40	58	160	158	D151X	
A/La Reunion/1972/2018	3C.2a1b	2018-09-19	MDCK2/SIAT1	80	73	320	386	40	40	80	60	160	192	D151X	
A/Poitiers/1978/2018	3C.2a1b	2018-10-08	MDCK2/SIAT1	40	40	80	64	<	<	<	3	80	70	D151X	
A/Poitiers/1975/2018	3C.2a1b	2018-10-09	MDCK2/SIAT1	80	68	160	220	40	40	40	53	160	133	D151X	
A/Oman/4923/2018	3C.2a1b	2018-10-11	CS/SIAT1	40	47	320	298	40	20	40	57	160	220	D151X	
A/Oman/5086/2018	3C.2a1b	2018-10-17	CS/SIAT1	40	47	320	268	40	20	80	66	160	211	D151X	
A/Oman/5972/2018	3C.2a1b	2018-11-12	CS/SIAT1	80	70	320	269	<	10	40	59	160	156	D151X	
A/Oman/6007/2018	3C.2a1b	2018-11-14	CS/SIAT1	80	62	640	500	40	40	80	60	160	186	D151X	
A/Oman/6131/2018	3C.2a1b	2018-11-18	CS/SIAT1	40	40	320	303	<	10	40	42	160	210	D151X	
A/Denmark/1828/2018	3C.2a1b	2018-11-27	SIAT3/SIAT1	80	60	320	455	40	40	80	66	320	254	D151X	
A/Norway/3736/2018	3C.2a1b	2018-11-30	CS/SIAT3	40	47	320	389	40	40	80	80	320	240	D151X	
A/Yokohama/133/2018	3C.2a1b	2018-12-06	MDCK1/MDCK1/SIAT1	80	114	640	610	160	156	160	217	160	155	D151X	
A/Yokohama/136/2018	3C.2a1b	2018-12-07	MDCK1/MDCK1/SIAT1	80	86	640	702	80	117	160	148	160	165	D151X	
A/Louisiana/49/2017	3C.2a2	2017-08-21	SIAT2/SIAT1	80	65	640	788	2560	2211	2560	2786	80	107	D151X	
A/Louisiana/50/2017	3C.2a2	2017-08-21	SIAT2/SIAT1	80	93	640	480	2560	2560	2560	3142	80	110	D151X	
A/Nagano/2531/2018	3C.2a2	2018-09-13	AX-4 1/SIAT2/SIAT1	80	66	320	369	2560	2088	2560	2314	80	100	D151X	
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019							

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-24

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate		
				Post-infection ferret antisera												
				A/Singapore INFIMH-16-0019/16 EGG 10 ⁻⁴ F46/17 3C.2a1	A/Singapore INFIMH-16-0019/16 SIAT F45/17 3C.2a1	A/Switzerland 8060/17 Egg F27/18 3C.2a2	A/HK 656/18 SIAT F25/18 3C.2a2	A/La Rioja 2202/18 SIAT F26/18 3C.2a1b								
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			2-fold
REFERENCE VIRUSES																
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	160	182	640	590	N121K, R142G, T160K (-CHO), N171K, L194P, D225G		
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	156	160	120	N121K, R142G, N171K		
A/Switzerland/8060/2017 DEF1	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	5120	3907	160	124	N96S, T131K, R142K, R261Q, T160K (-CHO), L194P		
A/Hong Kong/656/2018	3C.2a2		MDCK1/SIAT1 10 ⁻¹	80	70	640	576	1280	1807	2560	2255	40	46	T131K, R142K, N158X (-CHO), A212T, R261Q, I300V		
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	80	63	320	291	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q	NA	
TEST VIRUSES																
A/Switzerland/293/2018	3C.2a1b	2018-11-01	MDCK1/SIAT1	<	10	160	206	40	49	80	67	40	46	E62G, K92R, N121K, T131K, R142G, N171K, H311Q	D151X	
A/Switzerland/4912/2018	3C.2a1b	2018-11-28	MDCK1/SIAT1	40	20	640	495	80	99	160	135	80	68	E62G, K92R, N121K, T131K, R142G, N171K, H311Q		
A/Hong Kong/3165/2018	3C.2a1b	2018-12-06	MDCK1/SIAT1	40	40	320	313	<	0	<	3	40	20	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), S137F, A138S, R142G, N171K, F193S, H311Q		
A/Switzerland/4791/2018	3C.2a1b	2018-12-10	MDCK1/SIAT1	40	47	640	498	<	3	40	46	320	254	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q		
A/Bulgaria/1493/2018	3C.2a1b	2018-12-14	SIAT1	40	43	320	471	160	125	160	137	80	103	I34T, E62G, K82E, K92R, N121K, T131K, R142G, N171K, H311Q		
A/Switzerland/90/2018	3C.2a1b	2018-12-20	MDCK1/SIAT1	40	57	640	744	160	131	80	63	640	501	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q		
A/Hong Kong/46/2019	3C.2a1b	2018-12-24	MDCK1/SIAT1	<	10	160	122	640	712	<	10	80	103	D53N, E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), I140L, R142G, N171K, P221L, H311Q		
A/Hong Kong/126/2019	3C.2a1b	2018-12-28	MDCK1/SIAT1	40	48	320	280	80	77	80	99	160	113	E62G, N81D, K92R, N121K, T131K, R142G, N171K, S219F, H311Q		
A/Hong Kong/118/2019	3C.2a1b	2018-12-28	MDCK1/SIAT1	80	91	320	458	80	107	160	140	160	147	E62G, K92R, N121K, T131K, R142G, N171K, Q197R, S219F, H311Q		
A/Hong Kong/123/2019	3C.2a1b	2019-01-02	MDCK1/SIAT1	40	56	320	348	<	1	<	<	160	237	D53N, E62G, K92R, N121K, T135K (-CHO), R142G, N171K, S209N, H311Q		
A/Hong Kong/122/2019	3C.2a1b	2019-01-02	MDCK1/SIAT1	40	44	320	340	<	0	40	40	320	244	D53N, E62G, K92R, N121K, T135K (-CHO), R142G, N171K, S209N, H311Q		
A/Bulgaria/018/2019	3C.2a1b	2019-01-04	SIAT1	40	40	320	403	160	117	160	125	80	87	E62G, K83E, K92R, N121K, T131K, R142G, N171K, H311Q		
A/Bulgaria/1344/2018	3C.2a2	2018-12-01	SIAT1	80	94	1280	960	5120	6400	5120	5954	80	76	R33Q, T131K, R142K, A212T, R261Q, D291E		
A/Bulgaria/1409/2018	3C.2a2	2018-12-04	SIAT1	160	123	1280	1167	5120	5565	5120	6244	160	150	R33Q, N126D (-CHO), T131K, R142K, A212T, R261Q, D291E		
A/Lithuania/MB35528/2018	3C.2a4	2018-12-07	SIAT2/SIAT1	80	61	640	498	80	62	80	71	320	437	N31S, I34V, D53N, R142G, S144R, Q173H, I192T, Q197H, I242L		
A/Ghana/792/2018	Pending	2018-12-04	SIAT1	80	70	320	351	160	196	40	43	320	262	HA incomplete		
A/Ghana/4101/2018	Pending	2018-12-08	SIAT1	80	79	320	356	40	40	40	58	320	251	HA incomplete		
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019								

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-7 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-28

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate		
				Post-infection ferret antisera												
				A/Singapore INFIMH-16-0019/16 EGG 10 ⁻⁴ F46/17 3C.2a1	A/Singapore INFIMH-16-0019/16 SIAT F45/17 3C.2a1	A/Switzerland 8060/17 Egg F27/18 3C.2a2	A/La Rioja 2202/18 SIAT F26/18 3C.2a1b	A/Norway 3275/18 SIAT F03/19 3C.2a1b	A/Netherlands 10260/18 Egg F02/19 3C.2a1b							
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		2-fold	Read
REFERENCE VIRUSES																
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	640	590	40	40	<	10	N121K.R142G.T160K (-CHO), N171K.L194P.D225G N121K.R142G.N171K N96S, T131K, R142K, T160K (-CHO), L194P, R261Q E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, G186V, R261L, S219F, D225G, H311Q
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	120	1280	1222	40	20	
A/Switzerland/8060/2017 DEF1	3C.2a2		E6(Am2A4)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	160	124	160	151	40	40	
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	320	291	80	72	40	40	
A/Norway/3275/2018	3C.2a1b		SIAT1	<	10	320	271	80	90	40	54	640	524	<	6	
A/Netherlands/10260/2018	3C.2a1b		E5 10 ⁻⁵	80	78	320	313	80	82	320	359	160	160	5120	4998	
TEST VIRUSES																
A/Brasov/239416/2018	3C.2a1b	2018-12-18	SIAT1/SIAT1	80	112	1280	997	80	95	320	303	1280	1564	80	116	K2E, E62G, K92R, N121K, T131K, R142G, N171K, P227S, H311Q
A/Brasov/239414/2018	3C.2a1b	2018-12-18	SIAT1/SIAT1	40	59	640	498	160	172	320	281	320	254	<	5	Q57R, E62G, K92R, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Bucuresti/239376/2018	3C.2a1b	2018-12-18	SIAT1/SIAT1	40	56	320	433	<	0	320	395	320	351	<	10	E62G, K92R, A106V, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Linkoping/72/2018	3C.2a1b	2018-12-21	MDCK0/SIAT1	40	20	320	309	80	65	40	54	640	556	<	0	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q
A/Bucuresti/239696/2018	3C.2a1b	2018-12-23	SIAT1/SIAT1	80	61	320	451	<	10	160	169	160	217	<	0	E62G, K92R, A106V, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Switzerland/5733/2018	3C.2a1b	2018-12-27	SIAT1	80	70	1280	1157	320	411	320	267	2560	3151	80	70	K2E, E62G, K92R, N121K, T131K, R142G, N171K, D188N, H311Q
A/Switzerland/5699/2018	3C.2a1b	2018-12-27	SIAT1	40	54	640	558	<	10	320	384	160	204	40	20	E62G, K92R, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Bucuresti/3586-4138/2018	3C.2a1b	2018-12-27	SIAT1/SIAT1	<	10	320	253	40	58	40	50	640	667	<	10	K2E, E62G, K92R, N121K, T131K, R142G, N171K, H311Q
A/Switzerland/7145/2019	3C.2a1b	2019-01-03	SIAT1	40	40	320	384	<	5	320	292	160	158	<	10	E62G, K92R, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Slovenia/11/2019	3C.2a1b	2019-01-03	SIATx/SIAT1	40	56	640	486	<	<	320	256	160	227	<	10	E62G, K92R, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, H311Q
A/Iasi/239544/2019	3C.2a1b	2019-01-04	SIAT1/SIAT1	80	63	640	512	<	5	320	473	160	237	40	20	E62G, K92R, A106V, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, S198P, H311Q
A/Slovenia/71/2019	3C.2a1b	2019-01-07	SIATx/SIAT1	80	77	640	563	<	1	640	524	160	219	40	20	E62G, K92R, N121K, T128A (-CHO), T135K(-CHO), R142G, N171K, I192V, A304T, H311Q
A/Slovenia/99/2019	3C.2a1b	2019-01-08	SIATx/SIAT1	40	20	320	290	40	45	40	40	640	786	40	20	D53N, E62G, K83E, K92R, N121K, T131K, R142G, N171K, K276N, H311Q
A/Bosnia and Herzegovina/69/2018	3C.2a2	2018-12-27	SIAT1	40	51	640	587	1280	1690	80	80	1280	1171	40	40	N122T, T131K, R142K, R261Q
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019								

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-01-31

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/La Rioja 2202/18		A/Norway 3275/18		A/Netherlands 10260/18			
				EGG 10 ⁻⁴ F46/17 3C.2a1		SIAT F45/17 3C.2a1		Egg F27/18 3C.2a2		SIAT F26/18 3C.2a1b		SIAT F03/19 3C.2a1b		Egg F02/19 3C.2a1b			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	640	590	40	40	<	10	N121K, R142G, T160K (-CHO), N171K, L194P, D225G	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	120	1280	1222	40	20	N121K, R142G, N171K	
A/Switzerland/8060/2017 DEF1	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	160	124	160	151	40	40	N96S, T131K, R142K, R261Q, T160K (-CHO), L194P	
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	320	291	80	72	40	40	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q	
A/Norway/3275/2018	3C.2a1b		SIAT1	<	10	320	271	80	90	40	54	640	524	<	6	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Netherlands/10260/2018	3C.2a1b		E5 10 ⁻⁵	80	78	320	313	80	82	320	359	160	160	5120	4998	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, G186V, R261L, S219F, D225G, H311Q	
TEST VIRUSES																	
A/Palermo/331/2018	Pending	2018-11-29	SIAT3/SIAT1	160	133	1280	1021	5120	4876	160	205	1280	1755	80	74	PENDING	
A/Palermo/334/2018	Pending	2018-11-30	SIAT3/SIAT1	80	105	1280	1024	5120	4267	80	114	1280	1707	40	20	PENDING	
A/Palermo/333/2018	Pending	2018-11-30	SIAT3/SIAT1	160	127	640	747	2560	3559	160	139	1280	1112	<	10	PENDING	
A/Tomsk/19592V/2018	Pending	2018-12-04	MDCK1/SIAT1	40	40	320	274	80	84	40	59	640	683	40	40	PENDING	
A/Irkutsk/9356V/2018	Pending	2018-12-05	MDCK1/SIAT1	40	40	320	305	80	75	80	68	640	613	40	40	PENDING	
A/Georgia/784/2018	Pending	2018-12-12	SIAT1	80	69	640	557	<	0	320	357	320	245	40	40	PENDING	
A/Israel/10464/2018	Pending	2018-12-21	C1/SIAT1	<	1	<	1	<	0	<	5	<	0	<	0	PENDING	
A/Israel/603/2018	Pending	2018-12-23	C1/SIAT1	<	0	<	1	<	0	<	5	<	0	<	0	PENDING	
A/Israel/589/2018	Pending	2018-12-23	C1/SIAT1	40	50	320	423	<	10	320	246	160	187	40	20	PENDING	
A/Israel/604/2018	Pending	2018-12-25	C1/SIAT1	<	1	<	0	<	0	<	0	<	0	<	0	PENDING	
A/Georgia/917/2018	Pending	2018-12-26	SIAT1	80	65	640	640	160	127	80	109	1280	1097	40	47	PENDING	
A/Bolzano/1/2019	Pending	2019-01-03	SIAT1/SIAT1	80	99	640	768	5120	3840	80	105	1280	1638	40	44	PENDING	
A/Bosnia and Herzegovina/84/2018	Pending	2019-01-07	SIAT2	40	54	320	436	1280	1336	80	66	1280	1224	40	20	PENDING	
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019									

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-02-04

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate	
				A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/La Rioja 2202/18		A/Norway 3275/18			
				EGG 10 ⁻⁴ F46/17		SIAT F45/17		Egg F27/18		SIAT F26/18		SIAT F03/19			
				3C.2a1		3C.2a1		3C.2a2		3C.2a1b		3C.2a1b			
2-fold		Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read					
REFERENCE VIRUSES															
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	640	590	40	40	N121K, R142G, T160K (-CHO), N171K, L194P, D225G	
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	160	120	1280	1222	N121K, R142G, N171K	
A/Switzerland/8060/2017 DEF1	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	160	124	160	151	N96S, T131K, R142K, R261Q, T160K (-CHO), L194P	
A/La Rioja/2202/2018	3C.2a1b		SIAT1/SIAT1 10 ⁻¹	80	74	640	567	<	3	320	291	80	72	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, N171K, H311Q	
A/Norway/3275/2018	3C.2a1b		SIAT1	<	10	320	271	80	90	40	54	640	524	E62G, K92R, A106V, N121K, T131K, R142G, N171K, H311Q	
A/Netherlands/10260/2018	3C.2a1b		E5 10 ⁻⁵	80	78	320	313	80	82	320	359	160	160	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, G186V, R261L, S219F, D225G, H311Q	NA
TEST VIRUSES														Polymorphism	
A/Trollhattan/3/2018	3C.2a1b	2018-12-01	MDCK0/SIAT1	40	40	80	74	<	0	80	60	40	20	E62G, K92R, N121K, T128A (-CHO), T135K (-CHO), R142G, T160K (-CHO), N171K, H311Q	
A/Brest/2748/2018	3C.2a1b	2018-12-08	MDCK1/SIAT1	40	55	160	160	<	0	160	130	80	103	E62G, G78S, S91N, K92R, N121K, T135K (-CHO), R142G, N171K, H311Q	D151X
A/Netherlands/10601/2018	Pending	2018-10-16	MMDCCK2/SIAT1	80	66	320	427	40	20	160	229	160	182	PENDING	
A/Finland/972/2018	Pending	2018-10-30	SIAT1/SIAT1	80	60	320	280	40	58	160	142	320	333	PENDING	
A/Finland/973/2018	Pending	2018-11-01	SIAT1/SIAT1	<	0	160	147	40	20	80	90	160	226	PENDING	
A/Nizhny Novgorod/14645V/2018	Pending	2018-11-20	MDCK1/SIAT1	40	20	320	292	<	0	160	174	160	144	PENDING	
A/Nefteyuganski4V/2018	Pending	2018-11-23	MDCK1/SIAT1	40	43	320	429	<	0	320	327	320	243	PENDING	
A/Finland/979/2018	Pending	2018-11-29	SIAT1/SIAT1	40	20	640	489	<	10	320	320	320	276	PENDING	
A/Netherlands/10606/2018	Pending	2018-11-30	MMDCCK2/SIAT1	40	44	320	438	<	0	320	254	160	123	PENDING	
A/Khabarovsk/2204V/2018	Pending	2018-12-04	MDCK1/SIAT1	40	40	320	441	<	0	320	277	160	208	PENDING	
A/Kyiv/515/2018	Pending	2018-12-18	SIAT2/SIAT1	40	42	640	712	<	0	320	406	320	290	PENDING	
A/Kyiv/538/2018	Pending	2018-12-23	SIAT1/SIAT1	40	20	320	301	<	10	320	261	80	117	PENDING	
A/Kyiv/532/2018	Pending	2018-12-24	SIAT1/SIAT1	40	43	320	307	<	0	160	232	160	158	PENDING	
A/Ukraine/8107/2018	Pending	2018-12-26	MDCK2/SIAT1	40	40	320	379	<	0	160	228	160	151	PENDING	
A/Ukraine/8106/2018	Pending	2018-12-27	MDCK2/SIAT1	80	66	320	444	<	0	320	246	160	148	PENDING	
A/Ukraine/8105/2018	Pending	2018-12-27	MDCK2/SIAT1	40	20	320	289	<	0	160	213	80	103	PENDING	
A/Ukraine/8103/2019	Pending	2019-01-03	MDCK2/SIAT1	80	68	320	432	40	20	320	357	160	180	PENDING	
A/Nordrhein-Westfalen/3/2019	Pending	2019-01-07	P1/SIAT1	40	50	320	446	<	0	320	244	160	189	PENDING	
A/Hessen/1/2019	Pending	2019-01-07	P1/SIAT1	80	66	320	409	<	0	320	466	320	265	PENDING	
A/Nordrhein-Westfalen/2/2019	Pending	2019-01-07	P1/SIAT1	40	40	320	373	<	0	160	196	160	148	PENDING	
A/Baden-Wuerttemberg/2/2019	Pending	2019-01-07	P1/SIAT1	40	20	160	227	40	57	40	51	640	780	PENDING	
A/Luxembourg/2574/2019	Pending	2019-01-14	SIAT1	40	20	320	255	80	62	40	52	640	678	PENDING	
A/Luxembourg/2571/2019	Pending	2019-01-14	SIAT1	40	55	640	674	<	0	320	308	320	242	PENDING	
A/Luxembourg/2566/2019	Pending	2019-01-14	SIAT1	40	50	640	489	40	45	40	50	1280	1009	PENDING	
A/Luxembourg/3108/2019	Pending	2019-01-17	SIAT1	40	20	320	320	<	0	160	185	80	104	PENDING	
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019							

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2019-02-07

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera										HA1 substitutions for 3C.2a(1) viruses compared to A/Hong Kong/4801/2014: Egg adaptation HA substitutions compared to the corresponding cell isolate
				A/Singapore INFIMH-16-0019/16 EGG 10 ⁻⁴ F46/17 3C.2a1		A/Singapore INFIMH-16-0019/16 SIAT F45/17 3C.2a1		A/Switzerland 8060/17 Egg F27/18 3C.2a2		A/England 538/18 SIAT F31/18 3C.3a		A/Switzerland 9715293/13 CI123 EGG F29/15 3C.3a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES														
A/Singapore/INFIMH-16-0019/2016	3C.2a1		E5/E2 10 ⁻⁴	1280	1186	320	262	160	222	<	0	<	0	N121K, R142G, T160K (-CHO) , N171K, L194P , D225G
A/Singapore/INFIMH-16-0019/2016	3C.2a1		MDCK1/SIAT3/SIAT3	80	66	1280	1236	80	92	<	0	<	10	N121K, R142G, N171K
A/Switzerland/8060/2017 DEF1	3C.2a2		E6(Am2A14)c57 10 ⁻⁵	1280	1180	320	244	2560	3812	<	0	<	0	N96S, T131K, R142K, R261Q, T160K (-CHO) , L194P
A/England/538/2018	3C.3a	2018-02-26	MDCK1/SIAT1	<	5	<	3	<	0	160	145	<	0	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 CI123 10 ⁻⁴	0	1	<	18	40	20	<	5	160	222	I3L, T128A (-CHO), A138S, I140R , R142G, S144N, Y159S, T160K, G186V , H311Q, K326R
TEST VIRUSES														
A/Paris/2742/2018	3C.2a3	2018-12-27	MDCK1/SIAT1	80	108	320	369	40	20	80	62	40	44	T30A, N121K, T128A (-CHO), T135K (-CHO), R142G, S144K, R261Q, R299K
A/Paris/2513/2018	3C.3a	2018-11-26	MDCK1/SIAT1	80	63	40	46	40	40	640	588	<	0	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R
A/Paris/2511/2018	3C.3a	2018-11-29	MDCK1/SIAT1	80	102	40	58	40	51	640	491	<	0	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R
A/Sundsvall/2/2018	3C.3a	2018-12-01	MDCK0/SIAT1	<	0	<	0	<	0	320	334	<	0	G5R, N45D,D53N, E62G, S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, I214S, S219Y, H311Q, K326R
A/Sundsvall/1/2018	3C.3a	2018-12-03	MDCK0/SIAT1	<	0	<	1	<	0	320	459	<	0	G5R, N45D, D53N, E62G, S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, I214S, S219Y, H311Q, K326R
A/Paris/2538/2018	3C.3a	2018-12-04	MDCK1/SIAT1	80	77	40	54	40	46	320	464	<	2.5	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R
A/Paris/2544/2018	3C.3a	2018-12-04	MDCK1/SIAT1	80	73	80	83	80	63	640	525	<	0	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, I267V, H311Q, K326R
A/Paris/2572/2018	3C.3a	2018-12-06	SIAT1	<	10	<	0	<	<	320	431	<	0	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K (-CHO), F193S, H311Q, K326R
A/Paris/2671/2018	3C.3a	2018-12-17	MDCK1/SIAT1	40	52	40	20	40	40	640	788	<	5	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
A/Bretagne/026/2019	3C.3a	2019-01-02	MDCK1/SIAT1	40	40	40	40	<	0	640	580	<	10	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
A/Paris/029/2019	3C.3a	2019-01-03	MDCK1/SIAT1	40	40	40	40	40	40	640	553	40	40	S91N, T128A (-CHO), A138S, R142G, S144K, Y159S, T160K, F193S, H311Q, K326R
A/Israel/10464/2018	Pending	2018-12-21	C1/SIAT1	<	1	<	1	<	0	320	442	<	0	PENDING
A/Israel/603/2018	Pending	2018-12-23	C1/SIAT1	<	0	<	1	<	0	160	216	<	0	PENDING
A/Israel/604/2018	Pending	2018-12-25	C1/SIAT1	<	1	<	0	<	0	160	140	<	0	PENDING
				Vaccine SH 2018 NH 2018-19				Vaccine SH 2019						

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Repeated data

Table 9-S1 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary (2-fold)

Neutralisation Titre ¹																							
Post-infection ferret antisera																							
Viruses	A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18		A/Norway 3275/18		A/Netherlands 10260/18		A/England 538/18		A/Switzerland 9715293/13				
	Passage history		Egg 10 ⁻⁴		SIAT		Egg		SIAT		SIAT		SIAT		Egg		SIAT		CI123 Egg				
	Ferret number		F15/18		F46/17		F45/17		F27/18		F25/18		F26/18		F03/19		F02/19		F31/18				
	Genetic group		3C.2a1		3C.2a1		3C.2a1		3C.2a2		3C.2a2		3C.2a1b		3C.2a1b		3C.2a1b		3C.3a				
		2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																							
A/Singapore/INFIMH-16-0019/2016-E	3C.2a1	320	442	1280	1186	320	262	160	222	160	182	640	590	40	40	<	10	<	0	<	0		
A/Singapore/INFIMH-16-0019/2016-C	3C.2a1	80	119	80	66	1280	1236	80	92	160	156	160	120	1280	1222	40	20	<	0	<	10		
A/Switzerland/8060/2017	3C.2a2	320	315	1280	1180	320	244	2560	3812	5120	3907	160	124	160	151	40	40	<	0	<	0		
A/Hong Kong/656/2018	3C.2a2	160	134	80	70	640	576	1280	1807	2560	2255	40	46	NA	NA	NA	NA	<	5	40	20		
A/La Rioja/2202/2018	3C.2a1b	80	70	80	74	640	567	<	3	80	63	320	291	80	72	40	40	<	0	<	10		
A/Norway/3275/2018	3C.2a1b	NA	NA	<	10	320	271	80	90	NA	NA	40	54	640	524	<	6	NA	NA	NA	NA		
A/Netherlands/10260/2018	3C.2a1b	NA	NA	80	78	320	313	80	82	NA	NA	320	359	160	160	5120	4998	NA	NA	NA	NA		
A/England/538/2018	3C.3a	NA	NA	<	5	<	3	<	0	<	0	<	2	NA	NA	NA	NA	160	145	<	0		
A/Switzerland/9715293/2013	3C.3a	NA	NA	0	1	<	18	40	20	<	10	<	14	NA	NA	NA	NA	<	5	160	222		
Genetic subgroup		No tested				Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre																	
		No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%		
3C.2a1b	41	21	51.2																				
3C.2a2	8	8	100.0																				
3C.2a3	4	1	25.0																				
3C.3a	5	3	60.0																				
Total	58	33	56.9																				
3C.2a1b	62			0	0.0																		
3C.2a2	6			0	0.0																		
3C.2a3	2			0	0.0																		
3C.2a4	1			0	0.0																		
3C.3a	5			0	0.0																		
Total	76			0	0.0																		
3C.2a1b	103					79	76.7	2	1.9														
3C.2a2	14					13	92.9	12	85.7														
3C.2a3	6					6	100.0	0	0.0														
3C.2a4	1					1	100.0	0	0.0														
3C.3a	10					0	0.0	0	0.0														
Total	134					99	73.9	14	10.4														
3C.2a1b	88									1	1.1												
3C.2a2	13									13	100.0												
3C.2a3	5									0	0.0												
3C.2a4	1									0	0.0												
3C.3a	5									0	0.0												
Total	112									14	12.5												
3C.2a1b	103											87	84.5										
3C.2a2	14											11	78.6										
3C.2a3	5											2	40.0										
3C.2a4	1											1	100.0										
3C.3a	5											1	20.0										
Total	128											102	79.7										
3C.2a1b	15													13	86.7								
3C.2a2	1													1	100.0								
Total	16													14	87.5								
3C.2a1b	13															0	0.0						
3C.2a2	1															0	0.0						
Total	14															0	0.0						
3C.2a3	1																	1	100	1			
3C.3a	10																	10	100	10.0			
Total	11																	11	100.0	2			
		Vaccine NH 2018-19					Vaccine SH 2019																

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-S2 . Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary (Reading)

Neutralisation Titre ¹																					
Post-infection ferret antisera																					
Viruses	A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Singapore INFIMH-16-0019/16		A/Switzerland 8060/17		A/HK 656/18		A/La Rioja 2202/18		A/Norway 3275/18		A/Netherlands 10260/18		A/England 538/18		A/Switzerland 9715293/13		
Passage history	Egg 10 ⁻⁴		Egg 10 ⁻⁴		SIAT		Egg		SIAT		SIAT		SIAT		Egg		SIAT		CI123 Egg		
Ferret number	F15/18		F46/17		F45/17		F27/18		F25/18		F26/18		F03/19		F02/19		F31/18		F29/15		
Genetic group	3C.2a1		3C.2a1		3C.2a1		3C.2a2		3C.2a2		3C.2a1b		3C.2a1b		3C.2a1b		3C.3a		3C.3a		
	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																					
A/Singapore/INFIMH-16-0019/2016-E	3C.2a1	320	442	1280	1186	320	262	160	222	160	182	640	590	40	40	<	10	<	0	<	0
A/Singapore/INFIMH-16-0019/2016-C	3C.2a1	80	119	80	66	1280	1236	80	92	160	156	160	120	1280	1222	40	20	<	0	<	10
A/Switzerland/8060/2017	3C.2a2	320	315	1280	1180	320	244	2560	3812	5120	3907	160	124	160	151	40	40	<	0	<	0
A/Hong Kong/656/2018	3C.2a2	160	134	80	70	640	576	1280	1807	2560	2255	40	46	NA	NA	NA	NA	<	5	40	20
A/La Rioja/2202/2018	3C.2a1b	80	70	80	74	640	567	<	3	80	63	320	291	80	72	40	40	<	0	<	10
A/Norway/3275/2018	3C.2a1b	NA	NA	<	10	320	271	80	90	NA	NA	40	54	640	524	<	6	NA	NA	NA	NA
A/Netherlands/10260/2018	3C.2a1b	NA	NA	80	78	320	313	80	82	NA	NA	320	359	160	160	5120	4998	NA	NA	NA	NA
A/England/538/2018	3C.3a	NA	NA	<	5	<	3	<	0	<	0	<	2	NA	NA	NA	NA	160	145	<	0
A/Switzerland/9715293/2013	3C.3a	NA	NA	0	1	<	18	40	20	<	10	<	14	NA	NA	NA	NA	<	5	160	222
Genetic subgroup	No tested	Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre																			
		No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%	No	%
3C.2a1b	41	7	17.1																		
3C.2a2	8	5	62.5																		
3C.2a3	4	0	0.0																		
3C.3a	5	0	0.0																		
Total	58	12	20.7																		
3C.2a1b	62			0	0.0																
3C.2a2	6			0	0.0																
3C.2a3	2			0	0.0																
3C.2a4	1			0	0.0																
3C.3a	5			0	0.0																
Total	76			0	0.0																
3C.2a1b	103					57	55.3	0	0.0												
3C.2a2	14					11	78.6	12	85.7												
3C.2a3	6					4	66.7	0	0.0												
3C.2a4	1					1	100.0	0	0.0												
3C.3a	10					0	0.0	0	0.0												
Total	134					73	54.5	12	9.0												
3C.2a1b	88									1	1.1										
3C.2a2	13									13	100.0										
3C.2a3	5									0	0.0										
3C.2a4	1									0	0.0										
3C.3a	5									0	0.0										
Total	112									14	12.5										
3C.2a1b	103											70	68.0								
3C.2a2	14											11	78.6								
3C.2a3	5											1	20.0								
3C.2a4	1											0	0.0								
3C.3a	5											0	0.0								
Total	128											82	64.1								
3C.2a1b	15													13	86.7						
3C.2a2	1													1	100.0						
Total	16													14	87.5						
3C.2a1b	13															0	0.0				
3C.2a2	1															0	0.0				
Total	14															0	0.0				
3C.2a3	1																	1	100	0	0
3C.3a	10																	10	100	0	0
Total	11																	11	100.0	0	0.0
		Vaccine NH 2018-19				Vaccine SH 2019															

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 9-S3. Antigenic analysis of influenza A(H3N2) 3C.2a1b viruses - Plaque Reduction Neutralisation (MDCK-SIAT) - Summary (Reading: antisera raised against genetic subgroup 3C.2a1b viruses)

Viruses			Neutralisation Titre ¹			
			Post-infection ferret antisera			
			A/La Rioja 2202/18 SIAT F26/18 3C.2a1b		A/Norway 3275/18 SIAT F03/19 3C.2a1b	
			2-fold	Read	2-fold	Read
REFERENCE VIRUSES						
A/La Rioja/2202/2018		3C.2a1b	320	291	80	72
A/Norway/3275/2018		3C.2a1b	40	54	640	524
Genetic subgroup	HA amino acid substitutions	No tested	Number and % of viruses reacting ≤4-fold reduced compared to the homologous titre			
			No	%	No	%
3C.2a1b	T135K	52	51	98.1		
	T131K, R142G	33	12	36.4		
	T135K, T160K/X	15	6	40.0		
	T131K, R142G, T160K/X	2	0	0.0		
	T135N (-/+CHO)	1	1	100.0		
3C.2a2		14	11	78.6		
3C.2a3		5	1	20.0		
3C.2a4		1	0	0.0		
3C.3a		5	0	0.0		
Total		128	82	64.1		
3C.2a1b	T135K	9			8	88.9
	T131K, R142G	5			5	100.0
	T135K, T160K/X	1			0	0.0
3C.2a2		1			1	100.0
Total		16			14	87.5

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Vaccine viruses

Reference viruses

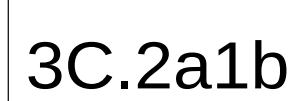
Oct 2018

Nov 2018

Dec 2018

Jan 2019

HA2 numbering



3C.2a1

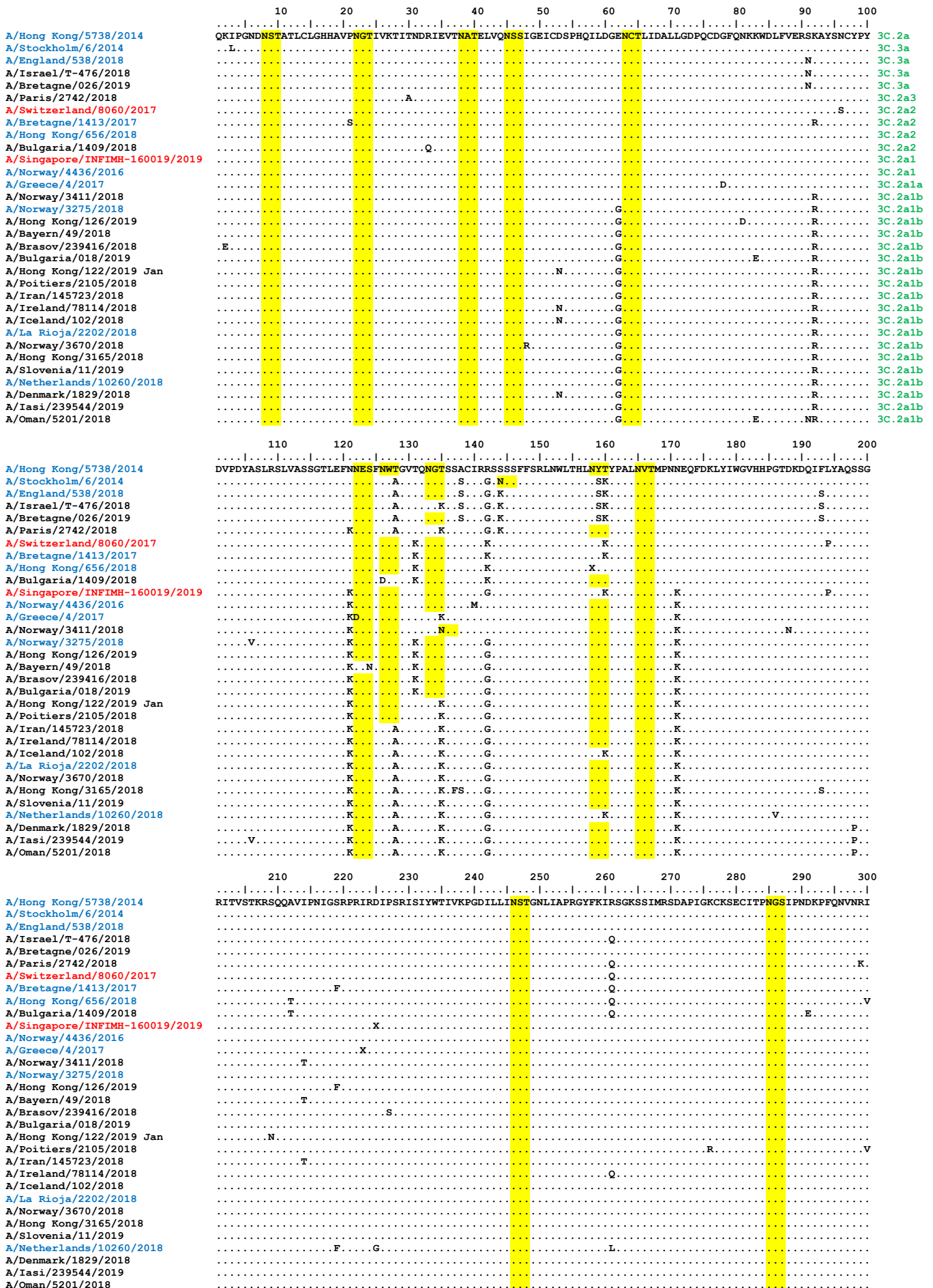
3C.2a3

3C.2a2

3C.3a

Figure 7. HA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



	310	320	330	340	350	360	370	380	390	400	
A/Hong Kong/5738/2014	TYGACPRYVKHXTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRRHONSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS										3C.2a
A/Stockholm/6/2014	QS.....										3C.3a
A/England/538/2018	QS.....R.....										3C.3a
A/Israel/T-476/2018	QS.....R.....										3C.3a
A/Bretagne/026/2019	QS.....R.....										3C.3a
A/Paris/2742/2018	S.....				L.....						3C.2a3
A/Switzerland/8060/2017	S.....										3C.2a2
A/Bretagne/1413/2017	S.....										3C.2a2
A/Hong Kong/656/2018	S.....										3C.2a2
A/Bulgaria/1409/2018	S.....										3C.2a2
A/Singapore/INFIMH-160019/2019	S.....										3C.2a1
A/Norway/4436/2016	S.....										3C.2a1
A/Greece/4/2017	S.....										3C.2a1a
A/Norway/3411/2018	QS.....										3C.2a1b
A/Norway/3275/2018	QS.....										3C.2a1b
A/Hong Kong/126/2019	QS.....M.....										3C.2a1b
A/Bayern/49/2018	QS.....M.....										3C.2a1b
A/Brasov/239416/2018	QS.....										3C.2a1b
A/Bulgaria/018/2019	QS.....										3C.2a1b
A/Hong Kong/122/2019 Jan	QS.....										3C.2a1b
A/Poitiers/2105/2018	QS.....										3C.2a1b
A/Iran/145723/2018	QS.....										3C.2a1b
A/Ireland/78114/2018	QS.....										3C.2a1b
A/Iceland/102/2018	QS.....										3C.2a1b
A/La Rioja/2202/2018	QS.....										3C.2a1b
A/Norway/3670/2018	I.QS.....										3C.2a1b
A/Hong Kong/3165/2018	QS.....										3C.2a1b
A/Slovenia/11/2019	QS.....										3C.2a1b
A/Netherlands/10260/2018	QS.....										3C.2a1b
A/Denmark/1829/2018	QS.....										3C.2a1b
A/Iasi/239544/2019	QS.....										3C.2a1b
A/Oman/5201/2018	QSN.....										3C.2a1b

	410	420	430	440	450	460	470	480	490	500	
A/Hong Kong/5738/2014	EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENOHTIDLTDSSEMNKLFETKTKQLRENAEDMGNGCCFKIYHKCDNACIGSIRNGT										
A/Stockholm/6/2014	D.....										
A/England/538/2018	M.....T.....										
A/Israel/T-476/2018	M.....										
A/Bretagne/026/2019	M.....										
A/Paris/2742/2018	I.....										
A/Switzerland/8060/2017	I.....										
A/Bretagne/1413/2017	I.....										
A/Hong Kong/656/2018	T.....										
A/Bulgaria/1409/2018	V.....E.....E.....										
A/Singapore/INFIMH-160019/2019	V.....E.....E.....										
A/Norway/4436/2016	V.....E.....E.....										
A/Greece/4/2017	V.....E.....E.....										
A/Norway/3411/2018	V.....E.....E.....										
A/Norway/3275/2018	V.....E.....E.....										
A/Hong Kong/126/2019	V.....E.....E.....										
A/Bayern/49/2018	V.....E.....E.....										
A/Brasov/239416/2018	V.....E.....E.....										
A/Bulgaria/018/2019	V.....E.....E.....										
A/Hong Kong/122/2019 Jan	V.....E.....E.....										
A/Poitiers/2105/2018	V.....E.....E.....										
A/Iran/145723/2018	V.....E.....E.....										
A/Ireland/78114/2018	V.....E.....E.....										
A/Iceland/102/2018	V.....E.....E.....										
A/La Rioja/2202/2018	V.....E.....E.....										
A/Norway/3670/2018	V.....A.....Q.....										
A/Hong Kong/3165/2018	V.....A.....Q.....										
A/Slovenia/11/2019	V.....E.....E.....										
A/Netherlands/10260/2018	V.....E.....E.....										
A/Denmark/1829/2018	V.....E.....E.....										
A/Iasi/239544/2019	V.....E.....E.....										
A/Oman/5201/2018	V.....E.....E.....										

	510	520	530	540	550	
A/Hong Kong/5738/2014	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI					
A/Stockholm/6/2014	V.....					
A/England/538/2018	V.....					
A/Israel/T-476/2018	V.....					
A/Bretagne/026/2019	V.....					
A/Paris/2742/2018	V.....					
A/Switzerland/8060/2017	V.....					
A/Bretagne/1413/2017	V.....					
A/Hong Kong/656/2018	V.....					
A/Bulgaria/1409/2018	V.....					
A/Singapore/INFIMH-160019/2019	V.....					
A/Norway/4436/2016	V.....					
A/Greece/4/2017	V.....					
A/Norway/3411/2018	I.....					
A/Norway/3275/2018	I.....					
A/Hong Kong/126/2019	IV.....					
A/Bayern/49/2018	I.....					
A/Brasov/239416/2018	I.....					
A/Bulgaria/018/2019	I.....					
A/Hong Kong/122/2019 Jan	L.....					
A/Poitiers/2105/2018	L.....					
A/Iran/145723/2018	L.....					
A/Ireland/78114/2018	L.....					
A/Iceland/102/2018	L.....					
A/La Rioja/2202/2018	L.....					
A/Norway/3670/2018	I.....					
A/Hong Kong/3165/2018	I.....					
A/Slovenia/11/2019	I.....					
A/Netherlands/10260/2018	I.....					
A/Denmark/1829/2018	I.....					
A/Iasi/239544/2019	I.....					
A/Oman/5201/2018	I.....					

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

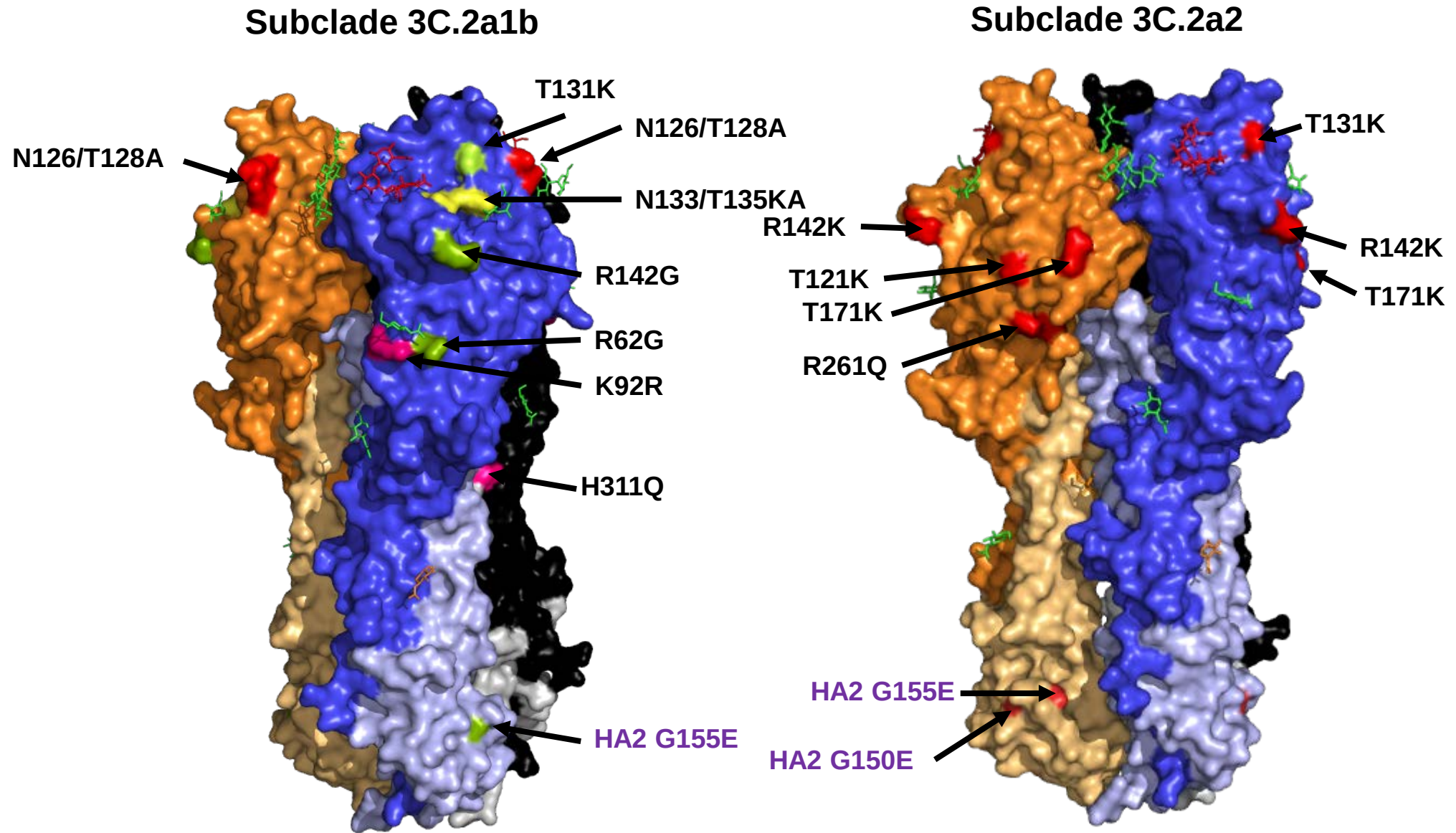


Figure 8a. Amino acid substitutions in the HA of recent clade 3C.2 H3N2 viruses compared with A/Singapore/INFIMH-16-0019/2016

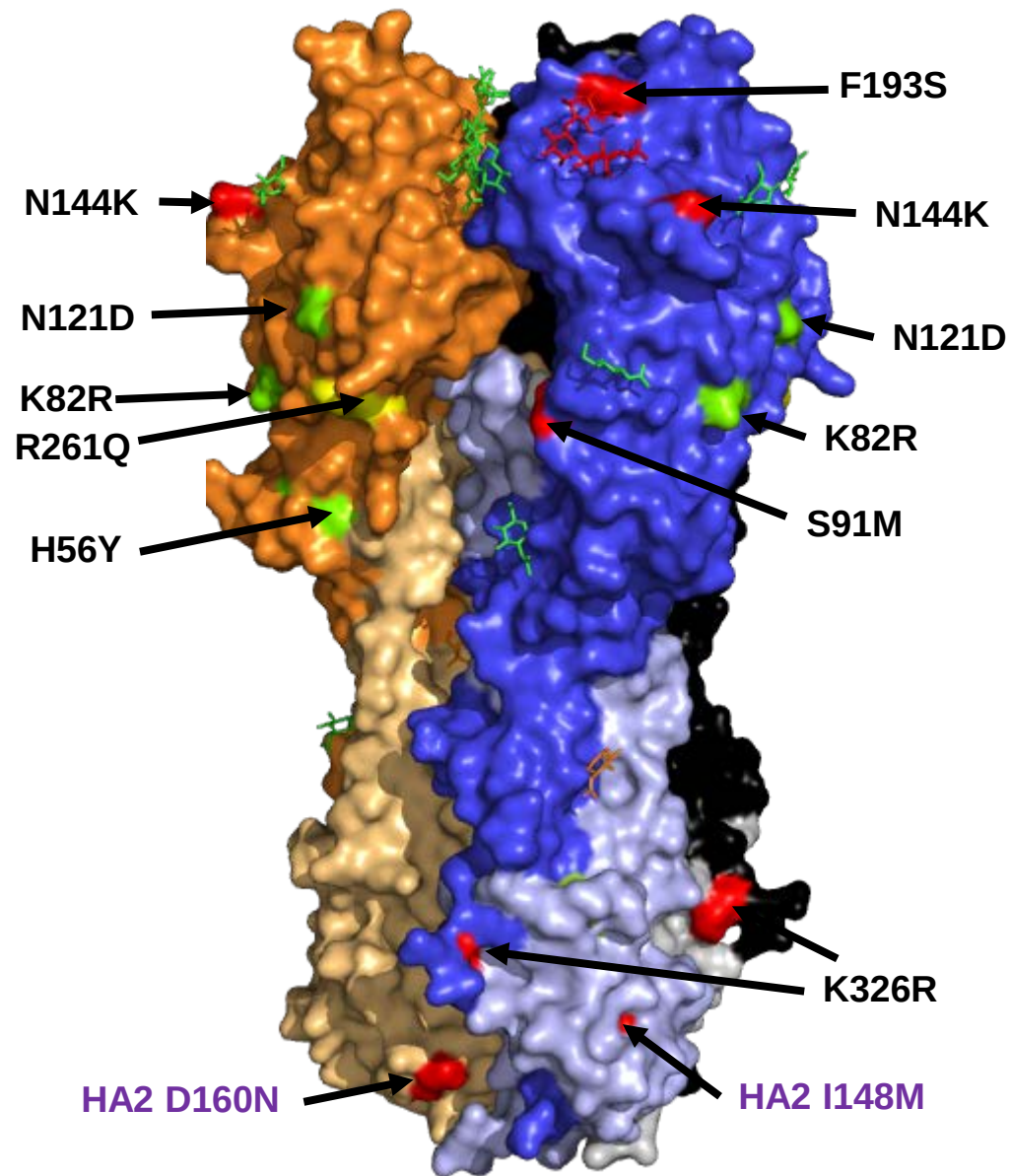


Figure 8b. Amino acid substitutions in the HA of recent clade 3C.3a H3N2 viruses compared with A/Switzerland/9715293/2013

Reference viruses

Collection date

Oct 2018

Nov 2018

Dec 2018

Jan 2019

D151* if N (+CHO)

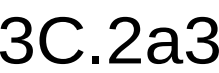


Figure 10. NA protein alignment for influenza A(H3N2) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100					
A/Hong Kong/5738/2014	MNP	NQKIITIGSV	SLTISTICFF	MQIAILLIT	TVTLHFKQ	YEFNSPP	NNQVM	LCEPTI	IERNITE	IVLYLTNT	TTIEKEIC	PKPAEYRN	WSKPCQ	GGIKGFAPF	3C.2a
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.3a
A/England/538/2018	.	.	.	.	.	.	.	.	.	R.	.	.	.	T.	3C.3a
A/Montana/01/2019	.	.	.	.	.	.	.	.	.	R.	.	.	.	T.	3C.3a
A/Bretagne/026/2019	.	.	.	.	.	M.	.	.	.	R.	.	.	.	T.	3C.3a
A/Paris/2742/2018	.	.	.	.	.	.	.	.	.	.	.	.	D.T.	.	3C.2a3
A/Switzerland/8060/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a2
A/Bretagne/1413/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a2
A/Hong Kong/656/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a2
A/Bulgaria/1409/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a2
A/Singapore/INFIMH-160019/2016	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1
A/Norway/4436/2016	.	.	.	.	P.	.	.	.	.	M.	.	.	.	T.	3C.2a1
A/Greece/4/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1a
A/Norway/3411/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Norway/3275/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Hong Kong/126/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Bayern/49/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Brasov/239416/2018	.	.	.	.	.	.	.	.	.	.	.	V.	.	T.	3C.2a1b
A/Bulgaria/018/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Hong Kong/122/2019	.	.	.	.	.	.	.	.	.	V.	.	.	.	T.	3C.2a1b
A/Poitiers/2105/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Iran/145723/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Ireland/78114/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Iceland/102/2018	.	.	T.	.	.	.	.	.	.	.	.	.	.	S.T.	3C.2a1b
A/La Rioja/2202/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Norway/3670/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Hong Kong/3165/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Slovenia/11/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Netherlands/10260/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Denmark/1829/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Iasi/239544/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b
A/Oman/5201/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	T.	3C.2a1b

	110	120	130	140	150	160	170	180	190	200									
A/Hong Kong/5738/2014	SKD	NSIRLS	AGGDI	WVTRE	FYVSCD	PDKCYQ	FALGQ	GTTLNN	VHSNNT	IVRXRT	PYRTL	LMNEL	GVFFHL	GTRQV	CIAWS	SSSCHD	GKAWL	HVCIT	GDDKN
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	D	.	.	.	.	.	.	.	.	.
A/England/538/2018	.	.	.	.	.	.	.	.	.	A.D.	H.	.	.	.	.	.	.	.	.
A/Montana/01/2019	.	.	.	.	.	.	.	.	.	A.D.	H.	.	.	.	.	.	.	.	.
A/Bretagne/026/2019	.	.	.	.	.	.	.	.	.	A.D.	H.	S.	.	.	.	.	.	.	.
A/Paris/2742/2018	.	.	.	.	.	.	.	.	.	X.	D.	H.	S.	.	.	.	.	.	.
A/Switzerland/8060/2017	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	V.	.
A/Bretagne/1413/2017	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	.	.
A/Hong Kong/656/2018	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	.	.
A/Bulgaria/1409/2018	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	.	.
A/Singapore/INFIMH-160019/2016	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Norway/4436/2016	.	.	.	.	.	I.	.	.	.	A.D.	.	.	.	.	.	.	.	.	.
A/Greece/4/2017	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	.	.
A/Norway/3411/2018	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Norway/3275/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Hong Kong/126/2019	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Bayern/49/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Brasov/239416/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Bulgaria/018/2019	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Hong Kong/122/2019	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Poitiers/2105/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Iran/145723/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Ireland/78114/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	S.	.	.	.	.	.
A/Iceland/102/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/La Rioja/2202/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Norway/3670/2018	.	.	.	L.	.	.	.	.	.	D.	.	H.	.	.	.	.	.	.	.
A/Hong Kong/3165/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Slovenia/11/2019	.	.	.	.	.	.	.	.	.	D.	.	.	.	.	M.	.	.	.	X.
A/Netherlands/10260/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Denmark/1829/2018	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Iasi/239544/2019	.	.	.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.
A/Oman/5201/2018	.	.	X.	L.	.	.	.	.	.	D.	.	.	.	.	.	.	.	.	.

	210	220	230	240	250	260	270	280	290	300																	
A/Hong Kong/5738/2014	AT	ASF	IYNG	RLVDS	VVWS	SKDIL	RLTQ	SECE	VCING	CT	VTVM	TDGS	ASGK	ADTK	LILF	IEEG	KIVHT	STLS	GSQA	QHVEEC	SCYP	PRYP	GVRC	VC	CRDN	WKS	NSNR
A/Stockholm/6/2014	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/England/538/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Montana/01/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Bretagne/026/2019	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Paris/2742/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Switzerland/8060/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Bretagne/1413/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Hong Kong/656/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Bulgaria/1409/2018	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Singapore/INFIMH-160019/2016	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Norway/4436/2016	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Greece/4/2017	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Norway/3411/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Norway/3275/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Hong Kong/126/2019	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Bayern/49/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Brasov/239416/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Bulgaria/018/2019	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Hong Kong/122/2019	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Poitiers/2105/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Iran/145723/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Ireland/78114/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Iceland/102/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/La Rioja/2202/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Norway/3670/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Hong Kong/3165/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Slovenia/11/2019	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Netherlands/10260/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Denmark/1829/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Iasi/239544/2019	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.
A/Oman/5201/2018	.	.	.	.	N.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.	.

	310	320	330	340	350	360	370	380	390	400	
A/Hong Kong/5738/2014	PIVDINIKDHSIVSSYVCSGLV	GT	PKRNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVMMGR	TINET	SR	LG	YETFKVIEGWSNPKSLQTN	RQVI	VD	R	3C.2a
A/Stockholm/6/2014						G					3C.3a
A/England/538/2018			T	N	K			V		I	3C.3a
A/Montana/01/2019			T	N	K	V		V		I	3C.3a
A/Bretagne/026/2019			T	N	K			V	S	I	3C.3a
A/Paris/2742/2018	A		T	N		T		V		I	3C.2a3
A/Switzerland/8060/2017			S	N				V	S	I	3C.2a2
A/Bretagne/1413/2017			S	R	N			V	S	I	3C.2a2
A/Hong Kong/656/2018			S	N				V	S	I	3C.2a2
A/Bulgaria/1409/2018			S	N				V	S	I	3C.2a2
A/Singapore/INFIMH-160019/2016				N				X	V	I	3C.2a1
A/Norway/4436/2016				N				V		I	3C.2a1
A/Greece/4/2017				N				V		I	3C.2a1a
A/Norway/3411/2018	I		S	N	K			V		I	3C.2a1b
A/Norway/3275/2018	I	R	S	N	K	D		V		I	3C.2a1b
A/Hong Kong/126/2019	I	R	S	N	K			V		I	3C.2a1b
A/Bayern/49/2018	I	R	S	N	K	D		V		I	3C.2a1b
A/Brasov/239416/2018	I	R	S	N	K	D		V		I	3C.2a1b
A/Bulgaria/018/2019	I	R	S	N	K	D		V		I	3C.2a1b
A/Hong Kong/122/2019	I	M	S	G	N			V		I	3C.2a1b
A/Poitiers/2105/2018	I	M	S	N				V		I	3C.2a1b
A/Iran/145723/2018	I		S	N	K			V		I	3C.2a1b
A/Ireland/78114/2018	I		S	N	FN			V		I	3C.2a1b
A/Iceland/102/2018	I		S	FN				V		I	3C.2a1b
A/La Rioja/2202/2018	I		S	FN				V		I	3C.2a1b
A/Norway/3670/2018	I		S	N				V		I	3C.2a1b
A/Hong Kong/3165/2018	I		S	N				V		I	3C.2a1b
A/Slovenia/11/2019	I		S	N					S	I	3C.2a1b
A/Netherlands/10260/2018	I		S	FN				V		I	3C.2a1b
A/Denmark/1829/2018	I		S	N				V		I	3C.2a1b
A/Iasi/239544/2019	I		S	N				V		I	3C.2a1b
A/Oman/5201/2018	I		S	N				V	F	I	3C.2a1b
	410	420	430	440	450	460					
A/Hong Kong/5738/2014	GDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTS	SGTYGTG	SWP	DGADL	NLM	PI					
A/Stockholm/6/2014											
A/England/538/2018											
A/Montana/01/2019											
A/Bretagne/026/2019											
A/Paris/2742/2018											
A/Switzerland/8060/2017											
A/Bretagne/1413/2017											
A/Hong Kong/656/2018											
A/Bulgaria/1409/2018											
A/Singapore/INFIMH-160019/2016											
A/Norway/4436/2016											
A/Greece/4/2017											
A/Norway/3411/2018											
A/Norway/3275/2018											
A/Hong Kong/126/2019											
A/Bayern/49/2018											
A/Brasov/239416/2018											
A/Bulgaria/018/2019											
A/Hong Kong/122/2019											
A/Poitiers/2105/2018											
A/Iran/145723/2018											
A/Ireland/78114/2018		X									
A/Iceland/102/2018											
A/La Rioja/2202/2018											
A/Norway/3670/2018											
A/Hong Kong/3165/2018											
A/Slovenia/11/2019											
A/Netherlands/10260/2018											
A/Denmark/1829/2018											
A/Iasi/239544/2019											
A/Oman/5201/2018											

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza B virus analyses.

Influenza B viruses represented only 7.6% of the samples received with collection dates after 2018-08-31 and were received from NICs in 23 countries or administrative regions in four WHO Regions: African (5: Cameroon, Cote d'Ivoire, Ghana, Madagascar, and Mauritius), Eastern Mediterranean (5: Egypt, Iran, Oman, Saudi Arabia and Tunisia), European (12) and Western Pacific (1: Hong Kong SAR) (**Table 1**). B/Yamagata lineage viruses and B/Victoria lineage viruses were received in similar numbers.

B/Victoria lineage.

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 10**.

HI results for the propagated viruses are shown in **Tables 11-1 to 11-6**: test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 11-4 to 11-6** are those with collection dates after 2018-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees are highlighted. A summary of the recognition of the test viruses collected after 2018-08-31 is shown in **Table 11-7**. Three distinct patterns of antigenicity were detected.

A total of 27 viruses with collection dates after 2018-08-31 were analysed, of these gene sequencing has been completed for 20. Three viruses, two from Hong Kong and one from Denmark, had HA genes with a deletion of two amino acids in HA1 (Δ 162-163) and all three were recognised by the antiserum raised against cell culture-propagated B/Norway/2409/2017 (Δ 162-163) at titres within 2-fold of the titre of the antiserum with the homologous virus, and similar results were seen for the antiserum raised against the cell culture-propagated cultivar of B/Colorado/06/2017 (Δ 162-163). However, the antiserum raised against the egg-propagated cultivar of B/Colorado/06/2017 recognised only a single virus, B/Denmark/3938/2018, at a titre within 4-fold of the titre of the antiserum for the homologous virus. The Δ 162-163 test viruses were recognised generally poorly by antisera raised against egg-propagated or cell culture-propagated viruses without the 162-163 deletion (cell culture-propagated B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016, and egg-propagated B/Brisbane/60/2008, B/Malta/636714/2011 and B/South Australia/81/2012) and by antisera raised against viruses with a deletion of three amino acids (Δ 162-164) in HA1, egg-propagated B/Hong Kong/269/2017 and cell culture-propagated B/Cote d'Ivoire/1662/2018.

Seven viruses were identified as having HA genes encoding an HA with a three amino acid deletion (Δ 162-164) in HA1. These viruses were recognised by an antiserum raised against B/Cote d'Ivoire/1662/2018 (Δ 162-164) at titres equal to or higher than the low titre of the antiserum for the homologous virus. These viruses were recognised poorly by an antiserum raised against another virus with the three amino acid deletion, egg-propagated B/Hong Kong/269/2017, as they were by the antisera raised against viruses with no deletion in HA1, and by the antisera raised against the cell culture-propagated

cultivar of B/Norway/2409/2017 and the egg-propagated cultivar of B/Colorado/06/2017 (Δ 162-163 viruses). However, the antiserum raised against the cell culture-propagated cultivar of B/Colorado/06/2017 recognised five of these seven test viruses at titres 4-fold lower than the homologous titre of the antiserum.

Seven viruses collected after 2018-08-31 had no deletion in HA1; these viruses, from Ghana and Hong Kong, were recognised poorly by antisera raised against cell culture-propagated B/Norway/2409/2017 and both the cell culture-propagated and the egg-propagated cultivars of B/Colorado/06/2017 (Δ 162-163 viruses) and were also recognised poorly by antisera raised against egg-propagated B/Hong Kong/269/2017 and B/Cote d'Ivoire/1662/2018 (Δ 162-164 viruses) at titres $\geq$ 8-fold lower than the homologous titres of the antisera. However, the antisera raised against cell culture-propagated B/Hong Kong/514/2009, B/Ireland/3154/2016 and B/Nordrhein-Westfalen/1/2016 recognised these seven test viruses at titres within 4-fold of the homologous titres of the antisera and none of seven, all seven and all seven, respectively, at titres within 2-fold of the homologous titres.

Five viruses, from Mauritius and Montenegro, have yet to be genetically characterised. The HI results indicate that the viruses from Madagascar are Δ 162-163 viruses, sharing the same pattern of reactivity as known Δ 162-163 test viruses, and that the virus from Montenegro is a Δ 162-164 virus, being recognised only by the antiserum raised against B/Cote d'Ivoire/1662/2018 (Δ 162-164),

Phylogenetic analysis of the HA genes of representative B/Victoria lineage viruses is shown in **Figure 11** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 12**. All recently circulating viruses analysed carried HA genes that fell into genetic clade 1A, the B/Brisbane/60/2008 genetic clade. All recently circulating viruses fell in a genetic group defined by three **HA1** amino acid substitutions compared with B/Brisbane/60/2008: **I117V**, **N129D** and **V146I**. Three virus clusters have emerged within clade 1A with deletions in HA1. Test viruses analysed that have HA genes encoding the two amino acid deletion in **HA1** Δ 162-163 also encode **D129G** and **I180V** amino acid substitutions in **HA1** and **R151K** in **HA2**. The HA genes of most viruses that have a three amino acid deletion in **HA1** Δ 162-164 fall into two genetic groups: a now minor group also encodes substitutions **I180T** and **K209N** in **HA1**, in addition to **I117V**, **N129D** and **V146I**, and a more recently emerging group which also encodes **K136E** substitution in **HA1** with many in this group also encoding **K52N** and **E198G** in **HA1**. **Figure 13** shows the locations of amino acid substitutions that define the three groups of deletion variants on a B/Brisbane/60/2008 HA structure.

Figure 14 shows a phylogenetic tree for the NA gene and **Figure 15** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins. The NA gene phylogeny is largely congruent with the HA phylogeny and the Δ 162-163 group of viruses are defined by **NA K371Q** amino acid substitution. Viruses in the more recently emerging Δ 162-164 genetic group encode a **A395T** substitution in the NA.

B/Yamagata lineage.

The collection details and summary results for viruses that were recovered and assayed by HI are shown in **Table 12**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 13-1 to 13-10**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 13-7 to 13-9** are viruses with collection dates after 2018-08-31. The clade into which the HA gene falls is shown. A summary of recognition of test viruses by all of the post-infection ferret antisera is shown in **Table 13-10**.

Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013, recommended for use in quadrivalent vaccines for the Northern Hemisphere 2018-2019 season and the Southern Hemisphere 2019 season, recognised all but one of the 22 (95%) test viruses collected after 2018-08-31 at titres within 4-fold of the titre with the homologous virus, 82% within 2-fold. However, a post-infection ferret antiserum raised against a cell culture-propagated cultivar of B/Phuket/3073/2013 recognised only 50% of the recently collected test viruses at titres within 4-fold, and only 9% within 2-fold, of its titre with the homologous virus.

Antisera raised against other viruses from the B/Wisconsin/1/2010-B/Phuket/3073/2013 clade (clade 3), egg-propagated B/Wisconsin/1/2010, B/Stockholm/12/2011 and cell culture-propagated B/Mauritius/1791/2017, recognised well the test viruses collected after 2018-08-31. Respectively, 100%, 86% and 95% were recognised at titres within 4-fold of the titres of the antisera with their homologous viruses and 91%, 45% and 91% were recognised at titres within 2-fold of the homologous titres of the antisera.

An antiserum raised against egg-propagated B/Massachusetts/02/2012 recognised only 18% of recently collected test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus, and only a single virus (5%) at a titre within 2-fold of the homologous titre. An antiserum raised against the cell culture-propagated cultivar of B/Massachusetts/02/2012 recognised 77% of recently collected test viruses at titres within 4-fold, and 59% within 2-fold, of its homologous titre. Higher levels of test virus recognition were seen with an antiserum raised against another cell culture-propagated virus from the B/Massachusetts/02/2012 clade (clade 2), B/Estonia/55669/2011: 86% of viruses were recognised at titres $\leq$ 4-fold lower than, and 73% at titres within 2-fold of, the titre of the antiserum with the homologous virus.

Figure 16 shows a phylogenetic analysis of the HA genes of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 17**. All tested viruses fall in clade 3, the B/Phuket/3073/2013 clade, within, like the vast majority of recently circulating viruses, a genetic group defined by two **HA1** amino acid substitutions compared with B/Phuket/3073/2013: **L172Q** and **M251V**.

Figure 18 shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 19** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. As for the HA gene, the vast majority of NA

genes of recently collected viruses fell into the B/Phuket/3073/2013 clade (clade 3) with the great majority of viruses falling in a group defined by **NA I49M, I171M, D342K** and **K373Q** amino acid substitutions compared to the B/Phuket/3073/2013 sequence. A number of subgroups have emerged within this group, notably those defined by **R65H** and **S402P** amino acid substitutions.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAI) zanamivir and oseltamivir was assessed phenotypically and by full NA gene sequencing, for 27 B/Victoria-lineage and 21 B/Yamagata-lineage viruses with collection dates after 2018-08-31 (**Table 14**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

Conclusion.

Viruses of the B/Yamagata and the B/Victoria lineages were received in equally small numbers. Viruses of the B/Yamagata lineage fell within genetic clade 3, the B/Phuket/3073/2013 clade, and the majority were recognised well by antisera raised against egg-propagated clade 3 vaccine/reference viruses. Viruses of the B/Victoria lineage fell into four genetic groups. One cluster had HA genes encoding a **deletion** of two amino acids in **HA1, Δ162-163** and two genetic groups had HAs with a three amino acid deletion in **HA1, Δ162-164**. Viruses in the fourth group, lacking amino acid deletions, were received in similar numbers. Viruses with either deletion are antigenically distinct from the other viruses in clade 1A, the **Δ162-163** and **Δ162-164** variants are antigenically distinct from each other and the **Δ162-164** viruses fall into two antigenically distinguishable groups.

Table 10. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2018-08-31

MONTH	B Victoria lineage												
	Country	Number received ¹	Number propagated ²	Post infection ferret antisera raised against									
				Clade 1A viruses			Clade 1A(Δ2) virus			Clade 1A(Δ3) virus			
			≤2 fold ³	4 fold ³	≥8 fold ³		≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴		≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵
2018 SEPTEMBER Ghana OCTOBER Ghana Oman NOVEMBER Denmark Ghana Hong Kong Oman DECEMBER France Ghana Hong Kong Madagascar Montenegro Slovenia	1	1			1				1		1		
	5	2			2				2		2		
	3	3	3						3				3
	1	1			1		1						1
	5	3			3			1	2	3			
	5	5	2		3	3			2				5
	2	2	2						2				2
	1	1			1			1			1		
	4	1			1			1		1			
	1	1			1			1					1
	5	4			4	4							4
	1	1			1			1		1			
	2	2			2			2		2			
	8 Countries	36	27	7	0	20	8	7	12	11	0	16	
				25.9%	0.0%	74.1%	29.6%	25.9%	44.5%	40.7%	0.0%	59.3%	

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Compared to homologous titres for ferret antisera raised against at least one of two tissue-culture propagated B/Brisbane/60/2008-like equivalents (B/Ireland/3154/2016 & B/Nordrhein-Westfalen/1/2016) - Clade 1A

4 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Colorado/06/2017 - Clade 1A(Δ2)

5 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Cote D'Ivoire/1662/2018 - Clade 1A(Δ3) 'African'

Table11-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2018-09-12 + 2018-12-19

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												NEW
				Post-infection ferret antisera												
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/HK 269/17 Egg	B/CIV 1662/18 MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	NIB F52/16 ^{*2}	F29/13 ^{*2}	F25/16 ^{*2}	NIB F47/16 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	F40/17 ^{*2}	F09/18 ^{*2}	F10/18 ^{*2}	F50/17 ^{*2}	F37/18 ^{*2}	
	Passage history			1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	1A(Δ3)	
REFERENCE VIRUSES																
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	320	320	320	40	40	<	40	80	10	<	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	320	320	640	320	80	40	<	20	80	10	<	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	320	160	640	160	40	40	<	40	80	<	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	40	80	80	160	80	80	<	20	<	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	20	40	40	80	160	80	<	10	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	10	20	20	40	80	80	<	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK2	80	<	<	10	<	10	10	40	80	40	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	10	<	<	<	40	80	40	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E1	640	80	80	80	20	<	<	40	160	160	10	<	
B/Hong Kong/269/2017	1A(Δ3)	2017-05-24	E5/E2	640	80	80	80	40	<	<	<	20	80	80	20	
B/Cote d'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK1	80	10	10	20	<	<	10	<	40	20	<	40	
TEST VIRUSES																
B/Zhejiang-Haishu/1280/2018	1A		C3/C1/MDCK1	320	<	<	<	<	<	<	<	10	<	ND	ND	
B/Liaoning-Yinzhou/1202/2018	1A		C3/C1/MDCK1	320	<	<	<	<	10	<	<	10	<	ND	ND	
B/Qinghai-Gangcha/1226/2018	1A		C1/C1/MDCK1	320	<	<	<	<	10	<	<	10	<	ND	ND	
B/Qinghai-Gangcha/1209/2018	1A		C1/C1/MDCK1	640	<	<	<	<	10	<	<	10	<	ND	ND	
B/Hunan-Beihu/1222/2018	1A(Δ3)		C2/C1/MDCK1	160	<	<	10	<	10	10	<	40	20	ND	ND	
B/Guangxi-Lingshan/359/2018	1A(Δ3)		C2/C1/MDCK1	160	<	<	20	<	20	10	<	40	20	ND	ND	
B/Hubei-Maojian/1219/2018	1A(Δ3)		C2/C2/MDCK1	80	<	<	10	<	<	<	<	40	10	ND	ND	
B/Singapore/EN0958/2017	1A	2017-12-13	E3/E1	640	80	80	160	80	20	20	<	10	40	ND	ND	
B/Cote d'Ivoire/342/2018	1A(Δ3)	2018-03-12	P0/MDCK1	40	<	10	20	<	10	10	<	40	<	<	40	
B/Cote d'Ivoire/754/2018	1A	2018-04-16	P2/MDCK1	2560	20	40	40	40	80	80	<	<	<	<	<	
B/Cote d'Ivoire/1238/2018	1A(Δ3)	2018-06-07	P0/MDCK1	80	10	20	40	<	<	<	<	40	20	<	20	
B/Cote d'Ivoire/1291/2018	1A(Δ3)	2018-06-18	P1/MDCK1	40	10	10	20	<	10	10	<	40	<	<	20	
B/Cote d'Ivoire/1391/2018	1A(Δ3)	2018-06-27	P0/MDCK1	80	10	10	40	<	20	20	<	40	10	<	40	
B/Cote d'Ivoire/1404/2018	1A	2018-06-29	P0/MDCK1	2560	20	80	40	40	160	80	<	<	<	<	<	
B/Cote d'Ivoire/1423/2018	1A(Δ3)	2018-07-03	P0/MDCK1	40	10	10	20	<	10	10	<	40	10	<	20	
B/Cote d'Ivoire/1456/2018	1A(Δ3)	2018-07-03	P0/MDCK1	80	10	20	20	<	20	10	<	40	10	<	20	
B/Cote d'Ivoire/1478/2018	1A(Δ3)	2018-07-06	P1/MDCK1	40	10	10	20	<	10	10	<	40	10	<	40	
B/Cote d'Ivoire/1629/2018	1A(Δ3)	2018-07-23	P0/MDCK1	80	10	10	20	<	10	10	<	40	20	<	20	
B/Cote d'Ivoire/1691/2018	1A(Δ3)	2018-07-26	P1/MDCK1	80	<	<	20	<	<	10	<	40	10	<	20	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]Vaccine[§]

Table 11-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2018-09-18 + 2018-12-19

Viruses	Other information	Haemagglutination inhibition titre															
		Collection date	Passage history	Post-infection ferret antisera													NEW
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/HK 269/17 Egg	B/CIV 1662/18 MDCK		
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	NIB F52/16 ²	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ²	F10/18 ²	F50/17 ²	F37/18 ²		
				Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	1A(Δ3)	
REFERENCE VIRUSES																	
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	320	320	160	40	40	<	40	80	10	<		
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	320	320	320	160	40	40	<	20	80	10	<		
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	320	160	640	160	40	40	<	40	40	<	<		
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	80	80	80	160	80	80	<	20	<	<	<		
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	40	40	40	80	80	80	<	20	<	<	<		
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	20	20	20	40	80	80	<	10	<	<	<		
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK2	80	<	<	10	<	10	10	40	80	20	<	<		
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	10	<	<	10	20	40	20	<	<		
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E1	640	160	40	40	20	<	<	20	160	160	10	<		
B/Hong Kong/269/2017	1A(Δ3)	2017-05-24	E5/E2	640	80	80	80	40	<	<	<	20	80	80	20		
B/Cote d'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK1	80	10	10	20	<	<	10	<	40	20	<	40		
TEST VIRUSES																	
B/Nagasaki/635/2018	1A		MDCK2/MDCK1/MDCK1	1280	160	40	40	10	<	<	<	10	40	ND	ND		
B/Gangwon/1116/2017	1A		MDCK2/MDCK1/MDCK1	640	10	10	20	<	<	<	<	20	20	ND	ND		
B/Kagawa/18162/2018	1A(Δ2)		MDCK1/MDCK1/MDCK1	1280	20	20	40	10	10	<	20	80	80	ND	ND		
B/Cote d'Ivoire/1334/2018	1A(Δ3)	2018-06-22	P2/MDCK1	40	10	10	20	<	<	<	<	20	10	ND	20		
B/Cote d'Ivoire/1564/2018	1A(Δ3)	2018-07-16	P1/MDCK2	1280	40	40	80	20	<	<	<	40	40	ND	20		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadrivalent vaccines SH 2018[§] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Table 11-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2018-10-04 + 2018-10-09 + 2018-10-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											NEW	
				Post-infection ferret antisera												
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/HK 514/09 MDCK		
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ²	F10/18 ²	F36/18 ²		
				1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)			
REFERENCE VIRUSES																
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	160	320	80	20	20	<	20	40	80		
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	320	320	160	80	20	20	<	10	40	80		
B/South Australia/81/2012	1A	2012-11-28	E4/E2	2560	320	160	640	160	40	40	<	40	40	ND		
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	2560	80	80	80	160	80	80	<	20	<	160		
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	2560	40	40	20	40	80	40	<	<	<	80		
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	20	40	20	40	40	40	<	<	<	40		
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK2	40	<	<	<	<	<	<	40	40	20	<		
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	40	<	<	<	<	<	<	20	80	20	<		
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E1	640	80	40	40	10	<	<	20	80	80	10		
TEST VIRUSES																
B/Cameroon/466/2018	1A	2018-01-04	MDCK1	1280	20	40	20	40	40	40	<	<	<	80		
B/Cameroon/797/2018	1A	2018-01-29	MDCK1	2560	40	40	20	40	80	40	<	<	<	80		
B/Cameroon/2620/2018	1A	2018-02-22	MDCK1	2560	40	40	20	40	80	40	<	<	<	80		
B/Oman/3311/2018	1A	2018-05-22	SIAT1/SIAT1	2560	40	40	40	40	40	40	<	10	<	ND		
B/Qatar/10-VI-18-0043316/2018	1A	2018-08-13	MDCK1	1280	<	40	<	40	80	40	<	<	<	ND		
B/Qatar/10-VI-18-0043325/2018	1A	2018-08-13	MDCK1	1280	<	40	<	40	80	40	<	10	<	ND		
B/Qatar/10-VI-18-0043448/2018	1A	2018-08-14	MDCK1	1280	<	40	<	20	80	40	<	<	<	ND		

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

Vaccine[#]Vaccine[§]¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadrivalent vaccines SH 2018[§] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Table 11-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2018-11-21

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											NEW
				Post-infection ferret antisera											
				B/Bris 60/08	B/Bris 60/08	B/Malta 636714/11	B/Sth Aus 81/12	B/HK 514/09	B/Ireland 3154/16	B/Nord-West 1/16	B/Nor 2409/17	B/Colorado 06/17	B/Colorado 06/17	B/CIV 1662/18	
				Egg	Egg	Egg	Egg	MDCK	MDCK	MDCK	MDCK	MDCK	Egg	MDCK	
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F29/13 ²	F25/16 ²	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ²	F10/18 ⁴	F37/18 ²	
Passage history															
Ferret number															
Genetic group															
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	160	320	80	20	20	<	20	40	<	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	320	320	320	160	20	20	<	<	40	<	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	320	160	320	80	20	10	<	<	40	<	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	80	80	40	80	80	40	<	<	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	20	20	10	40	80	40	<	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK3	640	10	10	10	20	<	80	<	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK2	40	<	<	<	<	<	<	40	40	20	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	40	<	<	<	<	<	<	40	40	20	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	320	80	40	40	10	<	<	20	80	80	<	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK2	<	<	<	10	<	<	<	<	<	<	20	
TEST VIRUSES															
B/Ghana/3074/2018	1A(Δ3)	2018-09-17	C1/MDCK1	<	<	<	<	<	<	<	<	<	<	40	
B/Ghana/3257/2018	1A(Δ3)	2018-10-01	C1/MDCK1	<	<	<	<	<	<	<	<	<	<	20	
B/Ghana/3258/2018	1A(Δ3)	2018-10-01	C1/MDCK1	<	<	<	10	<	<	<	<	<	<	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

§ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Vaccine[§]

Table 11-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-01-30

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre											
					Post-infection ferret antisera											
					B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK	B/HK 269/17 Egg
		Passage history			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F44/17 ⁴	F29/13 ¹	F25/16 ¹⁴	NIB F47/16 ²	F15/16 ²	F16/16 ²	F40/17 ²	F09/18 ^{1,5}	F10/18 ^{4,5}	F37/18 ¹	F50/17 ¹
		Genetic group			1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	1A(Δ3)
REFERENCE VIRUSES																
B/Brisbane/60/2008		1A	2008-08-04	E4/E4	2560	320	320	640	320	40	40	<	20	80	<	10
B/Malta/636714/2011		1A	2011-03-07	E4/E1	2560	320	640	640	320	80	40	<	40	80	<	20
B/South Australia/81/2012		1A	2012-11-28	E4/E2	2560	320	160	640	320	40	40	<	20	40	<	10
B/Hong Kong/514/2009		1B	2009-10-11	MDCK1/MDCK2	2560	40	<	40	320	80	80	<	20	<	<	<
B/Ireland/3154/2016		1A	2016-01-14	MDCK1/MDCK4	2560	10	<	10	80	160	80	<	<	<	<	<
B/Nordrhein-Westfalen/1/2016		1A	2016-01-04	C2/MDCK2	2560	10	<	10	80	80	80	<	<	<	<	<
B/Norway/2409/2017		1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	<	<	<	<	80	40	40	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	MDCK1/MDCK2	160	<	<	<	<	<	<	40	40	10	<	<
B/Colorado/06/2017		1A(Δ2)	2017-02-05	E5/E2	1280	160	40	80	20	<	<	40	80	160	<	10
B/Cote D'Ivoire/1662/2018		1A(Δ3)	2018-07-25	P0/MDCK3	160	<	<	<	<	<	<	<	20	<	40	<
B/Hong Kong/269/2017		1A(Δ3)	2017-05-24	E5	1280	ND	ND	ND	ND	ND	ND	ND	ND	ND	80	160
TEST VIRUSES																
B/Norway/3241/2018		1A(Δ2)	2018-08-21	MDCK1/MDCK1	160	<	<	<	<	<	<	40	40	<	<	<
B/Oman/5030/2018		1A	2018-10-11	MDCK1	2560	20	<	<	80	160	80	<	<	<	<	<
B/Oman/5112/2018		1A	2018-10-17	MDCK1	2560	<	<	<	80	160	80	<	<	<	<	<
B/Oman/5214/2018		1A	2018-10-22	MDCK1	2560	<	<	<	80	80	80	<	<	<	<	<
B/Hong Kong/1393/2018			2018-11-07	MDCK1/MDCK1	80	<	<	<	<	<	<	40	20	<	<	<
B/Hong Kong/1395/2018		1A(Δ2)	2018-11-08	MDCK1/MDCK1	160	<	<	<	<	<	<	40	40	<	<	<
B/Oman/6001/2018		1A	2018-11-11	MDCK1	2560	10	<	<	80	80	80	<	<	<	<	<
B/Hong Kong/1397/2018		1A(Δ2)	2018-11-12	MDCK1/MDCK1	80	<	<	<	<	<	<	40	40	<	<	<
B/Ghana/3847/2018		1A(Δ3)	2018-11-13	C3/MDCK1	640	<	<	<	<	<	<	<	10	<	80	<
B/Oman/6271/2018		1A	2018-11-19	MDCK1	2560	10	<	<	80	80	40	<	<	<	<	<
B/Ghana/3942/2018		1A(Δ3)	2018-11-21	C2/MDCK1	160	<	<	<	<	<	<	<	<	<	40	<
B/Ghana/3994/2018		1A(Δ3)	2018-11-27	C2/MDCK1	40	<	<	<	<	<	<	<	<	<	40	<
B/Hong Kong/1403/2018		1A	2018-11-27	MDCK1/MDCK1	2560	10	<	<	80	80	80	<	<	<	<	<
B/Denmark/3938/2018		1A(Δ2)	2018-11-29	MDCK2/MDCK1	1280	40	<	40	10	<	<	40	80	40	<	<
B/Hong Kong/1404/2018		1A	2018-11-30	MDCK1/MDCK1	2560	10	<	<	80	80	80	<	<	<	<	<
B/Slovenia/3020/2018		1A(Δ3)	2018-12-18	SIATx/MDCK1	<	<	<	<	<	<	<	<	10	<	40	<
B/Paris/2659/2018		1A(Δ3)	2018-12-20	MDCK1/MDCK1	40	<	<	<	<	<	<	<	10	<	40	<
B/Ghana/4207/2018		1A(Δ3)	2018-12-21	MDCK1	640	<	<	<	<	<	<	10	10	<	80	20
B/Hong Kong/1412/2018			2018-12-22	MDCK1/MDCK1	80	<	<	<	<	<	<	20	10	<	<	<
B/Slovenia/3103/2018		1A(Δ3)	2018-12-30	SIATx/MDCK1	40	<	<	<	<	<	<	<	10	<	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ND = Not Done

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

⁵ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]Vaccine⁵

Table 11-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2019-02-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Bris 60/08 Egg	B/Bris 60/08 Egg	B/Malta 636714/11 Egg	B/Sth Aus 81/12 Egg	B/HK 514/09 MDCK	B/Ireland 3154/16 MDCK	B/Nord-West 1/16 MDCK	B/Nor 2409/17 MDCK	B/Colorado 06/17 MDCK	B/Colorado 06/17 Egg	B/CIV 1662/18 MDCK
				Sh 539, 540, 543, 544, 570, 571, 574 ^{*1,3}	F44/17 ^{*4}	F29/13 ^{*2}	F25/16 ^{*4}	NIB F47/16 ^{*2}	F15/16 ^{*2}	F16/16 ^{*2}	F40/17 ^{*2}	F09/18 ^{*1,5}	F10/18 ^{*4,5}	F37/18 ^{*4}
				Genetic group	1A	1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)
REFERENCE VIRUSES														
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	1280	320	160	320	160	40	40	<	20	40	<
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	320	320	320	160	40	40	<	20	40	<
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	320	160	640	160	40	40	<	20	40	<
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	40	40	40	160	80	80	<	10	<	<
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	1280	20	20	20	40	80	80	<	10	<	<
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	1280	20	20	20	80	80	80	<	10	<	<
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	80	<	<	10	<	<	<	40	80	20	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	80	<	<	10	<	<	<	40	40	20	<
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	640	80	40	80	20	<	<	40	80	160	<
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	320	10	<	20	10	<	<	<	20	10	40
TEST VIRUSES														
B/Antsirabe/6397/2018		2018-12-03	MDCK1/MDCK1	160	<	<	10	<	<	<	40	80	40	<
B/Mahajanga/6396/2018		2018-12-05	MDCK1/MDCK1	160	<	<	10	<	<	<	40	40	40	<
B/Antsirabe/6674/2018		2018-12-10	MDCK1/MDCK1	320	<	<	10	<	<	<	40	80	40	<
B/Antsirabe/6672/2018		2018-12-10	MDCK1/MDCK1	160	<	<	10	<	<	<	40	40	40	<
B/Montenegro/53/2018		2018-12-18	MDCKx/MDCK1	160	<	<	<	<	<	<	<	10	<	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ 1:5 serum haemadsorbed with turkey RBCs

B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

\$ B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Table 11-7. Antigenic analyses of influenza B viruses (Victoria lineage) with collection dates after 2018-08-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre											
				Post-infection ferret antisera											
				B/Bris 60/08 Egg F44/17 ^{*4}	B/Malta 636714/11 Egg F29/13 ^{*2}	B/Sth Aus 81/12 Egg F25/16 ^{*4}	B/HK 514/09 MDCK NIB F47/16 ^{*2}	B/Ireland 3154/16 MDCK F15/16 ^{*2}	B/Nord-West 1/16 MDCK F16/16 ^{*2}	B/Nor 2409/17 MDCK F40/17 ^{*2}	B/Colorado 06/17 MDCK F09/18 ^{*1,5}	B/Colorado 06/17 Egg F10/18 ^{*4,5}	B/CIV 1662/18 MDCK F37/18 ^{*4}	B/HK 269/17 Egg F50/17 ^{*1}	
				1A	1A	1A	1B	1A	1A	1A(Δ2)	1A(Δ2)	1A(Δ2)	1A(Δ3)	1A(Δ3)	
REFERENCE VIRUSES															
B/Brisbane/60/2008	1A	2008-08-04	E4/E4	320	160	320	160	40	40	<	20	40	<	10	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	320	320	320	160	40	40	<	20	40	<	20	
B/South Australia/81/2012	1A	2012-11-28	E4/E2	320	160	640	160	40	40	<	20	40	<	10	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	40	40	40	160	80	80	<	10	<	<	<	
B/Ireland/3154/2016	1A	2016-01-14	MDCK1/MDCK4	20	20	20	40	80	80	<	10	<	<	<	
B/Nordrhein-Westfalen/1/2016	1A	2016-01-04	C2/MDCK2	20	20	20	80	80	80	<	10	<	<	<	
B/Norway/2409/2017	1A(Δ2)	2017-04-27	MDCK1/MDCK3	<	<	10	<	<	<	40	80	20	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	MDCK1/MDCK2	<	<	10	<	<	<	40	40	20	<	<	
B/Colorado/06/2017	1A(Δ2)	2017-02-05	E5/E2	80	40	80	20	<	<	40	80	160	<	10	
B/Cote D'Ivoire/1662/2018	1A(Δ3)	2018-07-25	P0/MDCK3	10	<	20	10	<	<	<	20	10	40	<	
B/Hong Kong/269/2017	1A(Δ3)	2017-05-24	E5	ND	ND	ND	ND	ND	ND	ND	ND	ND	80	160	
TEST VIRUSES															
Number of viruses tested*				27	27	27	27	27	27	27	27	27	27	19	
No with titre reduction ≤2-fold								7	7	8	8		11		
%								25.9	25.9	29.6	29.6		40.7		
No with titre reduction =4-fold							7			1	7	5			
%							25.9			3.7	25.9	18.5			
No with titre reduction ≥8-fold				27	27	27	20	20	20	18	12	22	16	19	
%				100	100	100	74.1	74.1	74.1	66.7	44.5	81.5	59.3	100	

* All viruses sequenced carried HA genes that fell in the B/Brisbane/60/2008 clade (clade 1A)

Vaccine[#]

Vaccine^{\$}

[#] B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2017-18 and quadravalent vaccines SH 2018

^{\$} B/Victoria-lineage virus recommended for use in trivalent vaccines NH 2018-19 and SH 2019

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Figure 11. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

Vaccine virus
Reference viruses

Collection date
Oct 2018
Nov 2018
Dec 2018
Jan 2019

HA2 numbering
in HI tables

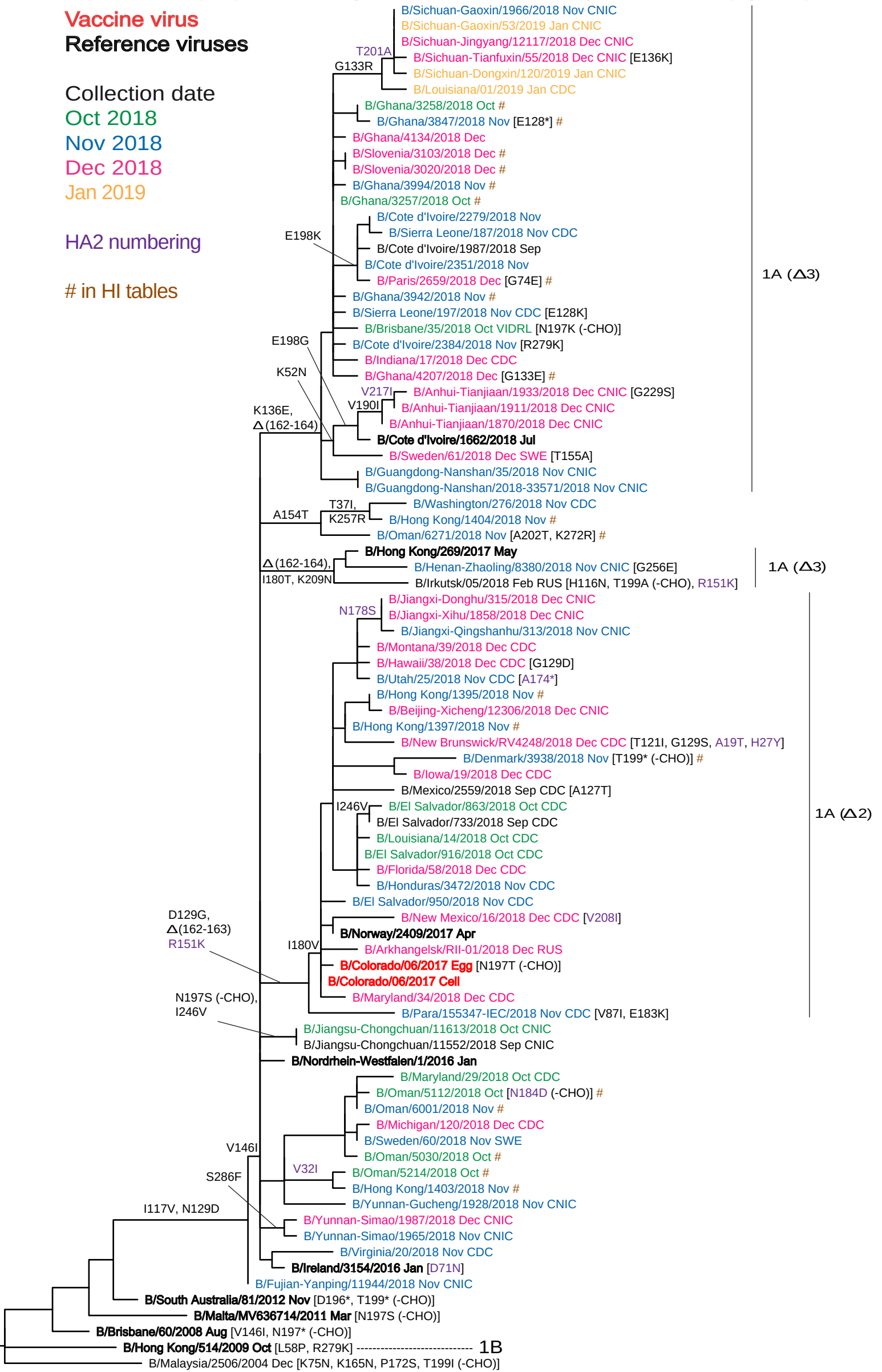
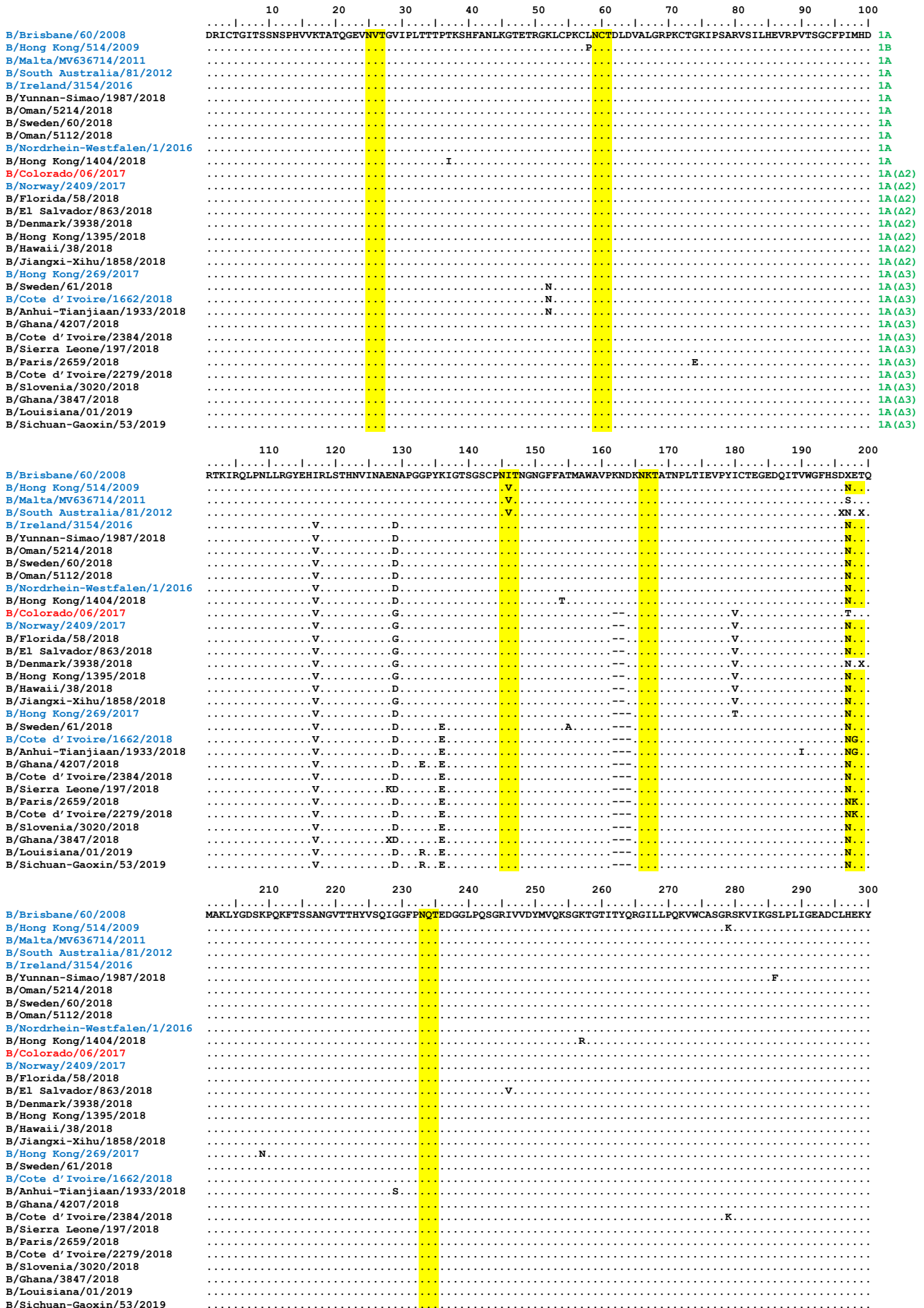


Figure 12. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

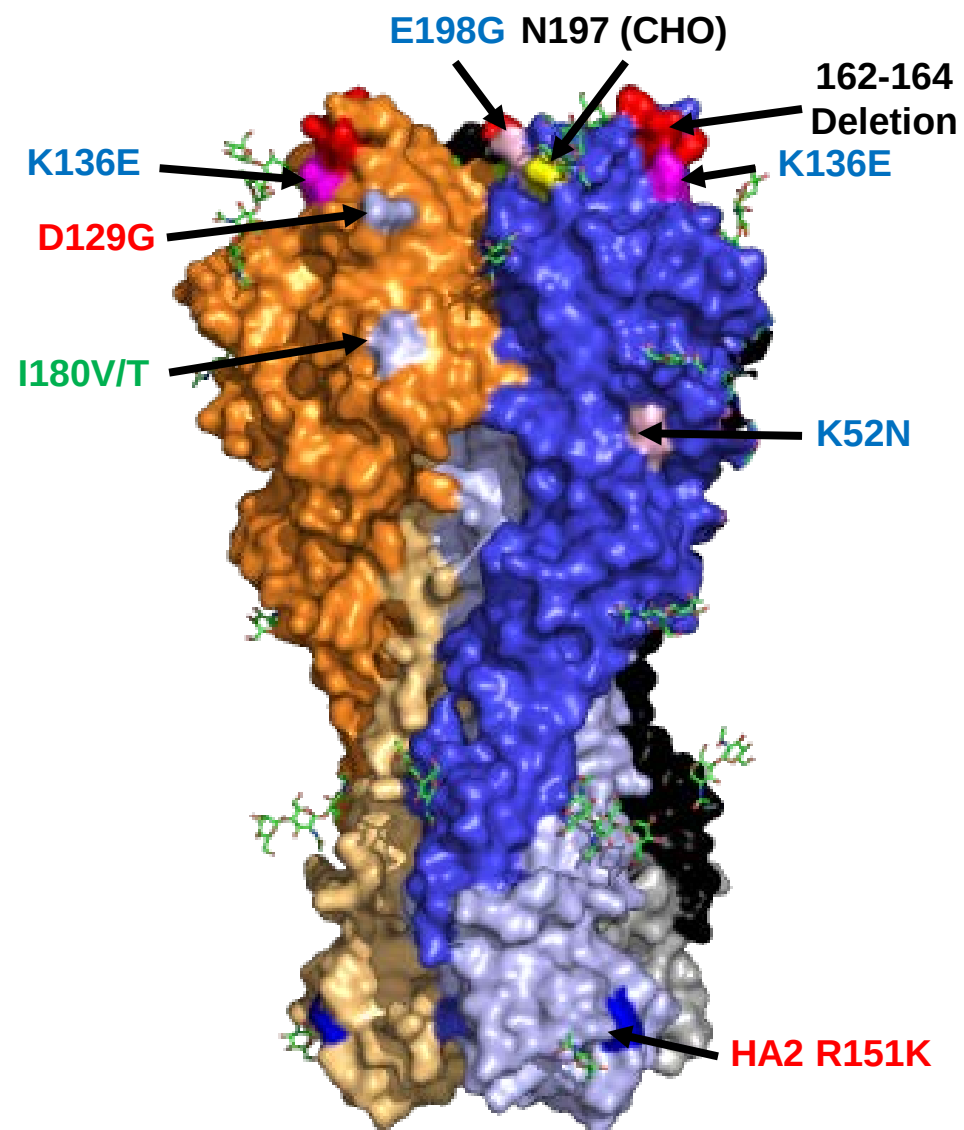


	310	320	330	340	350	360	370	380	390	400													
B/Brisbane/60/2008	GGLN	KSKP	YTTGE	HAKA	IGNC	PIWVK	TPKLAN	TKYR	PPAK	LLKE	RGFF	GAIAG	FLEGG	WEGMI	AGWH	GYTSH	GAGV	AVAAD	LKST	QEA	INKIT	KNLN	1A
B/Hong Kong/514/2009																							1B
B/Malta/MV636714/2011																							1A
B/South Australia/81/2012																							1A
B/Ireland/3154/2016																							1A
B/Yunnan-Simao/1987/2018																							1A
B/Oman/5214/2018																	I						1A
B/Sweden/60/2018																							1A
B/Oman/5112/2018																							1A
B/Nordrhein-Westfalen/1/2016																							1A
B/Hong Kong/1404/2018																							1A
B/Colorado/06/2017																							1A (Δ2)
B/Norway/2409/2017																							1A (Δ2)
B/Florida/58/2018																							1A (Δ2)
B/El Salvador/863/2018																							1A (Δ2)
B/Denmark/3938/2018																							1A (Δ2)
B/Hong Kong/1395/2018																							1A (Δ2)
B/Hawaii/38/2018																							1A (Δ2)
B/Jiangxi-Xihu/1858/2018																							1A (Δ2)
B/Hong Kong/269/2017																							1A (Δ3)
B/Sweden/61/2018																							1A (Δ3)
B/Cote d'Ivoire/1662/2018																							1A (Δ3)
B/Anhui-Tianjiaan/1933/2018																							1A (Δ3)
B/Ghana/4207/2018																							1A (Δ3)
B/Cote d'Ivoire/2384/2018																							1A (Δ3)
B/Sierra Leone/197/2018																							1A (Δ3)
B/Paris/2659/2018																							1A (Δ3)
B/Cote d'Ivoire/2279/2018																							1A (Δ3)
B/Slovenia/3020/2018																							1A (Δ3)
B/Ghana/3847/2018																							1A (Δ3)
B/Louisiana/01/2019																							1A (Δ3)
B/Sichuan-Gaoxin/53/2019																							1A (Δ3)

	410	420	430	440	450	460	470	480	490	500											
B/Brisbane/60/2008	SLSE	LEVKNL	QRLSGAM	DELHNEI	LELDEK	VDDL	LRADTT	ISSQ	IELAVLL	SNEGTI	INSEDE	HLAL	ERKL	KKML	GPSA	VEIGNG	CGFETK	HKHC	NOT	C	LDRIA
B/Hong Kong/514/2009																					
B/Malta/MV636714/2011																					
B/South Australia/81/2012																					
B/Ireland/3154/2016																					
B/Yunnan-Simao/1987/2018																					
B/Oman/5214/2018																					
B/Sweden/60/2018																					
B/Oman/5112/2018																					
B/Nordrhein-Westfalen/1/2016																					
B/Hong Kong/1404/2018																					
B/Colorado/06/2017																					
B/Norway/2409/2017																					
B/Florida/58/2018																					
B/El Salvador/863/2018																					
B/Denmark/3938/2018																					
B/Hong Kong/1395/2018																					
B/Hawaii/38/2018																					
B/Jiangxi-Xihu/1858/2018																					
B/Hong Kong/269/2017																					
B/Sweden/61/2018																					
B/Cote d'Ivoire/1662/2018																					
B/Anhui-Tianjiaan/1933/2018																					
B/Ghana/4207/2018																					
B/Cote d'Ivoire/2384/2018																					
B/Sierra Leone/197/2018																					
B/Paris/2659/2018																					
B/Cote d'Ivoire/2279/2018																					
B/Slovenia/3020/2018																					
B/Ghana/3847/2018																					
B/Louisiana/01/2019																					
B/Sichuan-Gaoxin/53/2019																					

	510	520	530	540	550	560	570
B/Brisbane/60/2008	AGTFDAGEFSLPTFDS	LNITTAASLNDDGLD	NHTILLYYSTAASSLAVT	LMIAIFVVYMVSRDN	NVSCSICL		
B/Hong Kong/514/2009							
B/Malta/MV636714/2011							
B/South Australia/81/2012							
B/Ireland/3154/2016							
B/Yunnan-Simao/1987/2018							
B/Oman/5214/2018							
B/Sweden/60/2018							
B/Oman/5112/2018			D				
B/Nordrhein-Westfalen/1/2016							
B/Hong Kong/1404/2018							
B/Colorado/06/2017							
B/Norway/2409/2017							
B/Florida/58/2018							
B/El Salvador/863/2018							
B/Denmark/3938/2018							
B/Hong Kong/1395/2018							
B/Hawaii/38/2018							
B/Jiangxi-Xihu/1858/2018			S				
B/Hong Kong/269/2017							
B/Sweden/61/2018							
B/Cote d'Ivoire/1662/2018							
B/Anhui-Tianjiaan/1933/2018							I
B/Ghana/4207/2018							
B/Cote d'Ivoire/2384/2018							
B/Sierra Leone/197/2018							
B/Paris/2659/2018							
B/Cote d'Ivoire/2279/2018							
B/Slovenia/3020/2018							
B/Ghana/3847/2018							
B/Louisiana/01/2019							
B/Sichuan-Gaoxin/53/2019					A		

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site



Signature amino acid substitutions:

- 1A(Δ 2)
- 1A(Δ 3) 'Asian'
- 1A(Δ 3) 'African'

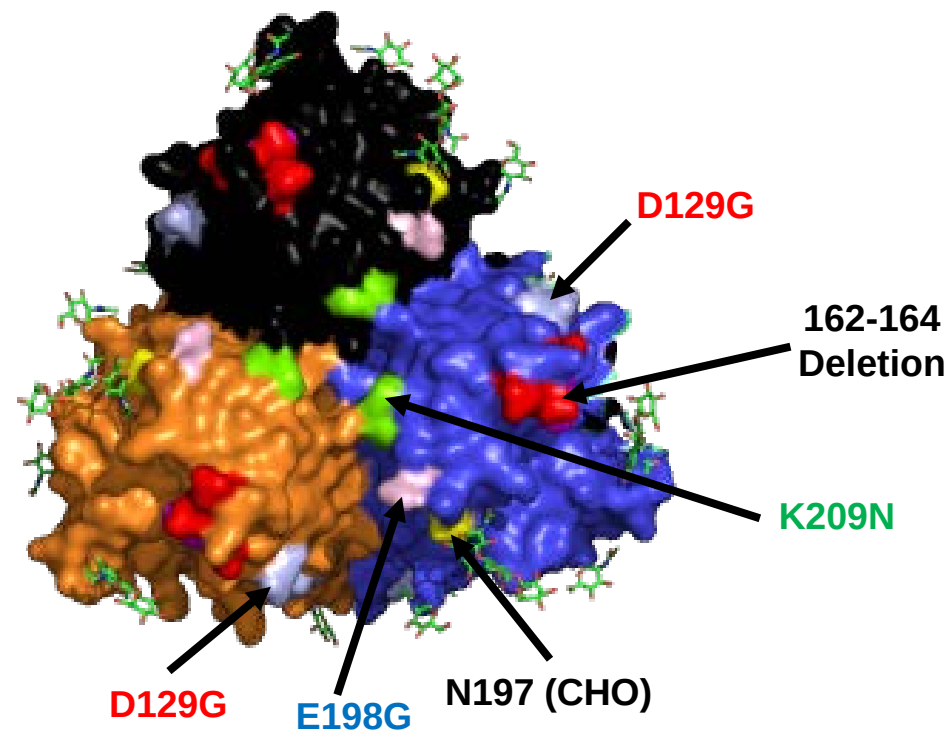


Figure 13. Amino acid substitutions and deletions in the HA of B/Victoria lineage viruses mapped onto B/Brisbane/60/2008

Figure 14. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

Vaccine virus
Reference viruses

Collection date

Oct 2018
Nov 2018
Dec 2018
Jan 2019

in HI tables

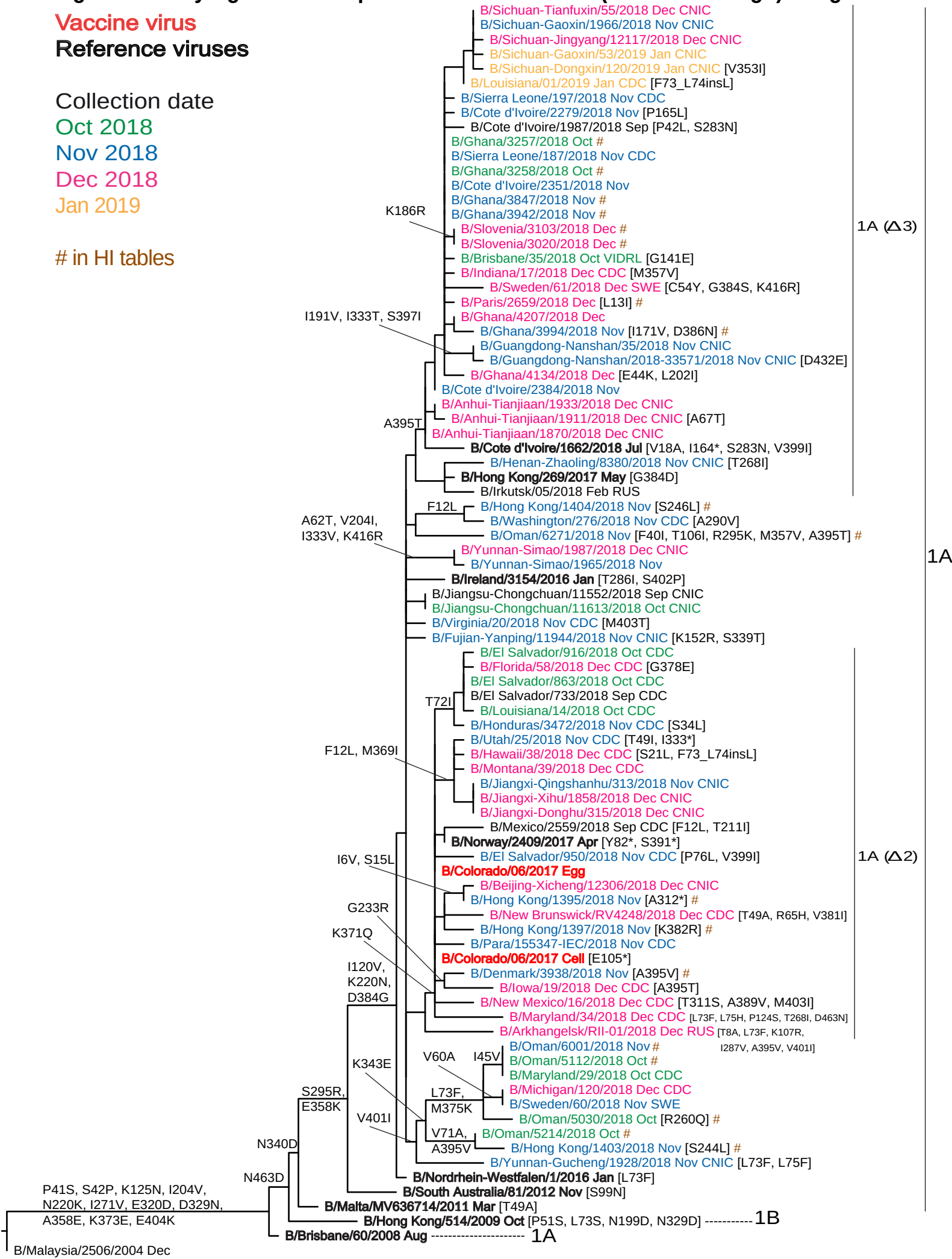
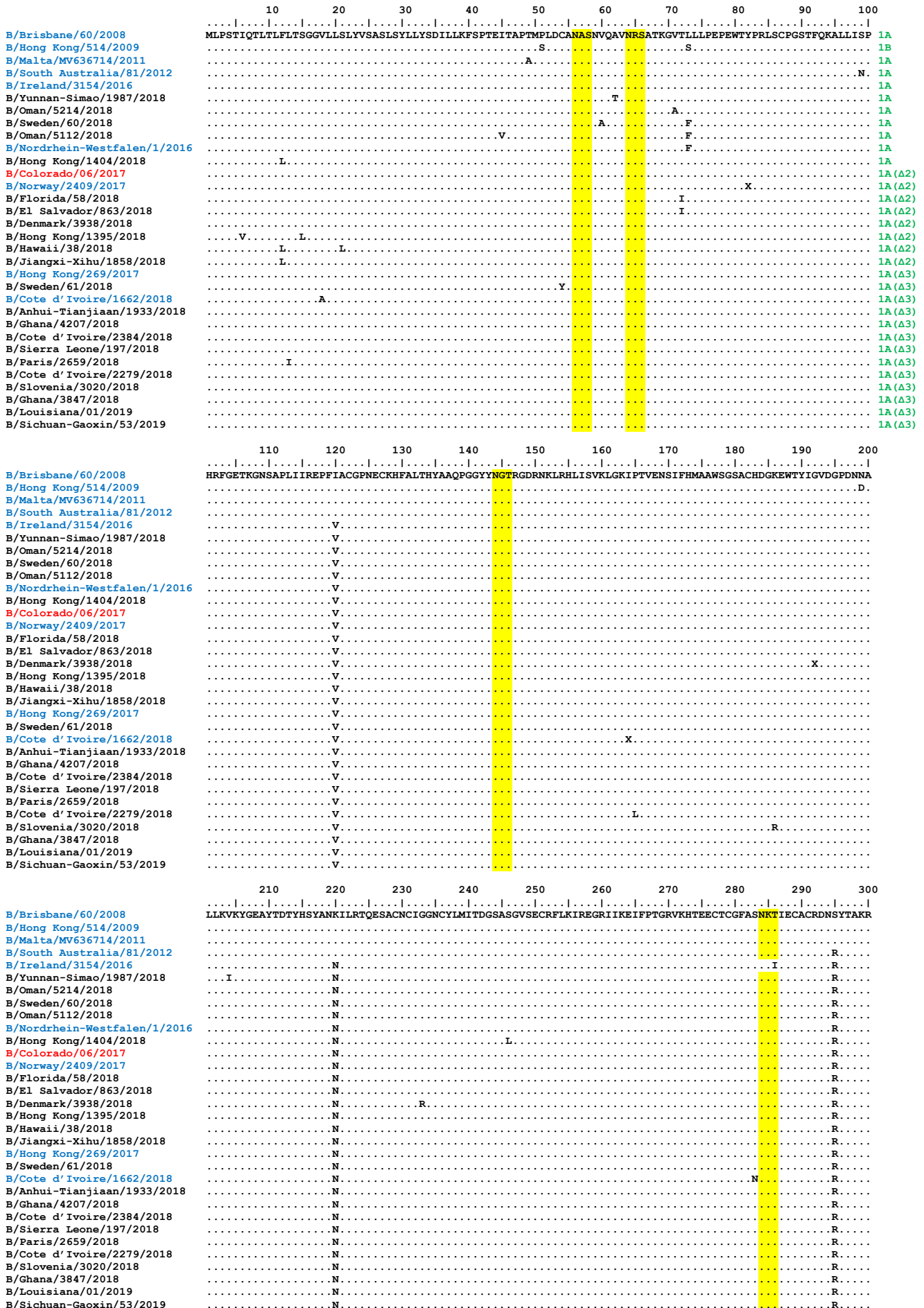


Figure 15. NA protein alignment for B (Victoria-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site



Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**

Table 12. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2018-08-31

MONTH	B Yamagata lineage				
Country	Number received¹	Number propagated²	≤2 fold³	4 fold³	≥8 fold³
2018					
SEPTEMBER					
France	1	1	1		
Madagascar	1	1	1		
Mauritius	2	2	2		
OCTOBER					
Ghana	1	1	1		
Iceland	1	1	1		
Norway	3	1	1		
NOVEMBER					
Ghana	1	1	1		
Hong Kong	1	1	1		
Iran	1	1	1		
Ireland	1	1	1		
Madagascar	2	1	1		
Oman	8	4	3	1	
United Kingdom	1	1	1		
DECEMBER					
Hong Kong	2	2	1	1	
Portugal	1	1			1
2019					
JANUARY					
Croatia	2	1	1		
Luxembourg	1	1		1	
	30	22	18	3	1
14 Countries			81.8%	13.7%	4.5%

¹ Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

² Propagated to sufficient titre to perform HI assay

³ Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 13-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-09-18

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				Ferret number SH614 ^{*1,3}	F27/13 ^{*2}	F10/16 ^{*2}	F16/14 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
				Genetic Group	3	2	2	2	3	3	3	3
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	320	80	80	40	40	40	40	20
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	80	320	80	80	40	160	20
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	80	20	320	80	80	40	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	40	20	320	80	80	40	160	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	40	10	160	80	80	40	80	10
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	160	160	160	160	160	160	160	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	20	10	160	80	80	20	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	40	40	40	40	40	40	80	40
TEST VIRUSES												
B/Jilin-Chaoyang/11723/2018	3		E2/E1/E2	640	20	<	20	20	20	20	20	10
B/Kanagawa/98/2018	3		MDCK2/MDCK1/MDCK1	1280	40	20	40	40	40	40	40	40
B/Saitama/51/2018	3		MDCK1/MDCK1/MDCK1	640	20	20	40	40	40	40	20	20
B/Victoria/540/2017	3	2017-08-25	E2/E2	640	10	<	40	40	20	10	40	10
B/Armenia/4/2018	3	2018-02-16	MDCK1/MDCK1	1280	80	40	80	80	40	80	160	80
B/Armenia/8/2018	3	2018-02-23	MDCK1/MDCK1	1280	40	20	40	40	20	40	40	40
B/Armenia/7/2018	3	2018-02-23	MDCK1/MDCK1	1280	40	20	80	80	40	40	160	40
B/Armenia/11/2018	3	2018-03-06	MDCK1/MDCK1	1280	80	40	40	80	40	80	80	80
B/Armenia/12/2018	3	2018-03-12	MDCK1/MDCK1	1280	40	20	160	80	80	40	160	40
B/Cote d'Ivoire/630/2018	3	2018-04-04	MDCK1	1280	80	40	40	40	40	80	160	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-09-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				SH614 ^{1,3}	F27/13 ^{*2}	F10/16 ^{*2}	F16/14 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
	Passage history			3	2	2	2	3	3	3	3	3
	Ferret number											
	Genetic Group											
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	640	80	80	80	20	40	80	20
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	320	80	320	80	80	40	160	20
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	80	20	320	40	80	20	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	20	10	160	80	80	20	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	20	10	160	80	80	40	80	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	160	160	160	320	160	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	20	10	160	80	80	40	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	80	40	80	80	40	40	80	80
TEST VIRUSES												
B/Armenia/416/2018	3	2018-02-08	MDCK1	2560	40	40	80	80	40	80	80	80
B/Armenia/434/2018	3	2018-02-09	MDCK1	1280	40	20	40	40	40	40	40	40
B/Armenia/578/2018	3	2018-02-23	MDCK1	1280	40	20	40	40	40	40	40	40
B/Armenia/574/2018	3	2018-02-23	MDCK1	1280	40	10	40	40	40	40	40	40
B/Armenia/596/2018	3	2018-03-01	MDCK1	1280	40	20	40	40	40	40	80	80
B/Armenia/618/2018	3	2018-03-02	MDCK1	>5120	160	320	320	320	320	320	320	160
B/Armenia/666/2018	3	2018-03-07	MDCK1	1280	80	40	80	80	40	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-10-05

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK	B/Wis 1/10 Egg	B/Maur 1791/17 MDCK
				Ferret number SH614 ^{1,3}	F27/13 ²	F10/16 ²	F16/14 ²	F36/15 ²	F05/17 ²	F27/15 ²	F25/17 ²	F34/18 ²	F34/15 ²	F35/18 ²
	Passage history			Genetic Group										
REFERENCE VIRUSES														
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	320	40	80	80	40	40	40	10	40	80
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	320	80	320	80	80	40	160	20	160	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	40	20	320	40	40	10	80	<	80	10
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	20	10	160	80	80	20	160	20	160	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	20	10	160	80	160	20	160	20	160	40
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	160	80	160	160	160	160	160	160	160	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	20	10	160	80	80	20	160	20	160	40
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	40	20	40	40	40	40	80	80	40	80
TEST VIRUSES														
B/Cameroon/471/2018	3	2018-01-04	MDCK1	2560	80	80	80	160	160	80	160	160	80	320
B/Cameroon/808/2018	3	2018-01-16	MDCK1	2560	80	80	80	80	80	80	160	160	80	160
B/Cameroon/804/2018	3	2018-01-27	MDCK1	1280	80	80	160	80	80	80	160	80	80	160
B/Cameroon/1452/2018	3	2018-02-01	MDCK1	1280	80	80	160	80	80	80	160	80	80	160
B/Cameroon/2617/2018	3	2018-02-15	MDCK1	2560	80	40	160	160	80	80	160	160	80	160
B/Cameroon/2942/2018	3	2018-03-27	MDCK1	2560	80	80	160	160	160	80	160	160	80	160
B/Cameroon/2944/2018	3	2018-03-28	MDCK1	1280	80	40	160	80	80	80	160	80	80	160
B/Cameroon/4747/2018	3	2018-03-31	MDCK1	1280	80	80	160	80	80	80	160	80	80	160
B/Cameroon/4751/2018	3	2018-04-11	MDCK2	2560	40	40	40	80	80	80	160	80	80	160
B/Cameroon/4900/2018	3	2018-05-08	MDCK1	2560	80	80	80	80	80	80	160	160	160	160
B/Cameroon/4899/2018	3	2018-05-08	MDCK1	2560	80	80	80	160	80	80	160	80	160	160
B/Cameroon/5148/2018	3	2018-05-15	MDCK1	2560	80	80	80	160	80	80	160	160	80	160
B/Cameroon/5147/2018	3	2018-05-15	MDCK2	2560	80	80	160	160	160	160	320	160	160	320
B/Cameroon/5149/2018	3	2018-05-16	MDCK2	2560	80	80	160	160	160	160	320	160	160	160
B/Cameroon/5136/2018	3	2018-05-30	MDCK1	2560	80	80	160	160	80	160	160	160	160	160
B/Cameroon/5535/2018	3	2018-06-09	MDCK1	1280	40	40	40	80	40	80	80	80	80	160
B/Cameroon/5764/2018	3	2018-06-12	MDCK2	1280	40	40	40	80	40	40	80	80	80	80
B/Cameroon/6178/2018	3	2018-07-01	MDCK2	1280	40	20	40	80	80	40	80	80	40	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-10-09

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F16/14 ^{*2}	F36/15 ^{*2}	F05/17 ^{*4}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
	Passage history			3	2	2	2	3	3	3	3	3
	Ferret number											
	Genetic Group											
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	80	80	40	<	20	20	<
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	1280	40	80	320	160	80	40	160	<
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	10	20	320	80	80	20	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	10	160	80	80	40	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	<	20	160	80	160	40	160	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	40	160	160	320	160	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	10	160	80	80	20	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	2560	<	40	80	80	<	40	80	80
TEST VIRUSES												
B/Qatar/10-VI-18-0022755/2018	3	2018-04-07	MDCK1	2560	10	80	80	80	80	80	80	160
B/Qatar/10-VI-18-0029127/2018	3	2018-05-06	MDCK1	1280	<	40	40	40	<	40	80	40
B/Qatar/10-vi-18-0039839/2018	3	2018-07-16	MDCK1	2560	<	80	80	80	80	80	160	80
B/Qatar/10-vi-18-0039836/2018	3	2018-07-16	MDCK1	2560	<	80	80	80	<	80	160	80
B/Qatar/16-vi-18-0040920/2018	3	2018-07-25	MDCK1	1280	<	40	80	80	<	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 < = <20

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-10-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				SH614 ^{*1,3}	F39/17 ^{*2}	F10/16 ^{*2}	F16/14 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
	Passage history			3	2	2	2	3	3	3	3	3
	Ferret number											
	Genetic Group											
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	40	40	40	40	20	20	40	10
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	40	80	320	80	80	40	160	20
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	10	20	320	40	80	10	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	<	10	160	80	80	40	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	<	10	160	80	160	40	160	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK2	2560	20	80	160	160	160	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	<	10	160	80	80	20	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	<	20	40	40	40	40	80	40
TEST VIRUSES												
B/Saudi Arabia/2428983924/2018	3	2018-04-12	MDCK1	1280	<	20	40	40	40	40	40	40
B/Saudi Arabia/1020348783/2018	3	2018-05-30	MDCK1	1280	<	20	40	40	40	40	40	40
B/Oman/3290/2018	3	2018-06-03	SIAT1/SIAT1	640	<	<	<	10	10	<	10	20
B/Cameroon/5777/2018	3	2018-06-07	MDCK3	1280	<	40	40	80	80	40	80	80
B/Saudi Arabia/1142569381/2018	3	2018-06-10	MDCK1	1280	<	20	40	40	40	40	80	40
B/Oman/3437/2018	3	2018-06-13	SIAT1/SIAT1	1280	<	40	40	40	40	40	40	40
B/Oman/3507/2018	3	2018-06-25	SIAT1/SIAT1	1280	<	20	40	40	40	40	40	40
B/Saudi Arabia/1042787406/2018	3	2018-07-08	MDCK1	1280	<	20	40	40	40	40	80	40
B/Saudi Arabia/1026649663/2018	3	2018-07-08	MDCK1	1280	<	40	40	40	40	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-11-21

				Haemagglutination inhibition titre									
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antisera								
					NEW								
					B/Phuket	B/Estonia	B/Mass	B/Mass	B/Wis	B/Stock	B/Phuket	B/Phuket	B/Maur
					3073/13	55669/11	02/12	02/12	1/10	12/11	3073/13	3073/13	1791/17
					Egg	MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK
Ferret number	SH614 ^{1,3}	F40/18 ²	F10/16 ²	F16/14 ²	F36/15 ²	F05/17 ²	F27/15 ²	F25/17 ²	F04/18 ²				
Genetic Group	3	2	2	2	3	3	3	3	3				
REFERENCE VIRUSES													
B/Estonia/55669/2011	2		2011-03-14	MDCK2/MDCK3	640	320	40	80	40	20	40	80	10
B/Massachusetts/02/2012	2		2012-03-13	MDCK1/C2/MDCK3	640	160	40	160	80	20	40	160	10
B/Massachusetts/02/2012	2		2012-03-13	E3/E4	640	40	20	320	40	10	10	80	<
B/Wisconsin/1/2010	3		2010-02-20	E3/E2	2560	80	20	320	160	40	160	320	40
B/Stockholm/12/2011	3		2011-03-28	E4/E1	1280	40	20	160	80	40	80	160	20
B/Phuket/3073/2013	3		2013-11-21	MDCK2/MDCK3	2560	640	80	80	160	160	160	160	160
B/Phuket/3073/2013	3		2013-11-21	E4/E3	1280	40	10	160	80	20	20	160	40
B/Mauritius/1791/2017	3		2017-09-20	MDCK1/MDCK4	1280	80	20	40	40	40	40	80	80
TEST VIRUSES													
B/Saudi Arabia/18/2018	3		2018-05-02	MDCK1	1280	160	80	40	80	80	80	80	80
B/Ghana/2607/2018	3		2018-07-30	C1/MDCK1	2560	320	80	80	160	80	80	160	160
B/Ghana/2857/2018	3		2018-08-22	C2/MDCK1	2560	320	80	80	80	80	80	160	160
B/Ghana/2946/2018	3		2018-08-28	C1/MDCK1	1280	160	80	40	80	80	80	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadrivalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2018-11-27 + 2018-12-12

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				SH614 ^{*1,3}	F40/18 ^{*2}	F10/16 ^{*2}	F16/14 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
	Passage history Ferret number Genetic Group			3	2	2	2	3	3	3	3	3
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	160	80	80	40	40	20	40	10
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	160	80	320	160	80	40	320	10
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	80	20	320	80	80	20	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	80	20	320	160	160	80	320	40
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	40	20	160	80	160	40	160	20
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	640	160	160	320	320	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	40	10	160	80	80	40	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	1280	160	40	80	80	80	80	160	80
TEST VIRUSES												
B/Palestine/2Q2018/2018	3	2018-02-15	MDCK1	1280	40	20	40	40	40	40	40	40
B/Palestine/6R2018/2018	3	2018-03-04	MDCK1	1280	160	80	80	80	80	80	160	80
B/Mauritius/SF39/2018	3	2018-07-04	MDCKx/MDCK1	1280	80	20	40	80	40	40	80	40
B/Mauritius/I-762/2018	3	2018-09-02	MDCK1/MDCK1	1280	80	20	80	80	80	40	80	80
B/Mauritius/I-784/2018	3	2018-09-06	MDCK1/MDCK1	1280	80	20	80	80	80	40	160	80
B/Norway/3367/2018	3	2018-10-19	MDCK1	1280	80	40	40	80	40	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine#

Table 13-8. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-01-23

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection ferret antisera								
					B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
					SH614 ^{*1,3}	F40/18 ^{*1}	F10/16 ^{*2}	F06/17 ^{*1}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*2}
		Genetic Group			3	2	2	2	3	3	3	3	3
REFERENCE VIRUSES													
B/Estonia/55669/2011		2	2011-03-14	MDCK2/MDCK3	640	160	40	<	20	20	20	40	<
B/Massachusetts/02/2012		2	2012-03-13	MDCK1/C2/MDCK3	1280	160	80	320	80	80	40	320	20
B/Massachusetts/02/2012		2	2012-03-13	E3/E4	640	40	20	160	40	80	10	160	<
B/Wisconsin/1/2010		3	2010-02-20	E3/E2	2560	80	20	160	80	160	20	320	40
B/Stockholm/12/2011		3	2011-03-28	E4/E1	1280	20	10	80	40	160	20	160	<
B/Phuket/3073/2013		3	2013-11-21	MDCK2/MDCK3	2560	640	160	160	160	160	320	320	160
B/Phuket/3073/2013		3	2013-11-21	E4/E3	1280	40	40	80	40	80	20	160	20
B/Mauritius/1791/2017		3	2017-09-20	MDCK1/MDCK4	2560	<	40	<	20	40	40	80	40
TEST VIRUSES													
B/Iran/107824/2018		3	2018-01-06	SIAT3/MDCK1	2560	80	160	40	80	80	80	160	80
B/Lyon/CHU-R18.98.84/2018		3	2018-09-26	MDCK2/MDCK1	2560	80	40	<	40	40	80	80	80
B/Iceland/99/2018		3	2018-10-28	MDCK1/MDCK1	5120	640	640	160	320	320	320	320	320
B/Ghana/3978/2018		3	2018-10-30	C2/MDCK1	2560	160	160	<	80	40	160	160	80
B/Oman/5758/2018		3	2018-11-05	MDCK1	2560	<	<	<	20	20	40	80	<
B/Ireland/78367/2018		3	2018-11-06	MDCK1/MDCK1	2560	80	40	<	40	40	80	80	40
B/Ghana/3761/2018		3	2018-11-07	MDCK1	2560	80	40	<	40	40	80	160	80
B/England/609/2018		3	2018-11-15	MDCK1/MDCK1	2560	80	80	<	40	80	80	160	160
B/Oman/6163/2018		3	2018-11-18	MDCK1	1280	<	<	<	20	20	20	40	20
B/Oman/6266/2018		3	2018-11-19	MDCK2	2560	160	80	<	40	80	80	160	80
B/Oman/6339/2018		3	2018-11-25	MDCK2	2560	80	40	<	40	40	80	160	80
B/Iran/146039/2018		3	2018-11-28	SIAT1/MDCK1	1280	<	40	<	40	20	40	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-9. Antigenic analyses of influenza B viruses (Yamagata lineage) 2019-02-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Phuket 3073/13 Egg	B/Estonia 55669/11 MDCK	B/Mass 02/12 MDCK	B/Mass 02/12 Egg	B/Wis 1/10 Egg	B/Stock 12/11 Egg	B/Phuket 3073/13 MDCK	B/Phuket 3073/13 Egg	B/Maur 1791/17 MDCK
				SH614 ^{*1,3}	F40/18 ^{*2}	F10/16 ^{*2}	F06/17 ^{*2}	F36/15 ^{*2}	F05/17 ^{*2}	F27/15 ^{*2}	F25/17 ^{*2}	F04/18 ^{*4}
	Passage history			3	2	2	2	3	3	3	3	
	Ferret number											
	Genetic Group											
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	640	160	40	40	20	40	<	40	<
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	640	160	80	160	40	80	<	160	<
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	640	40	20	160	20	80	<	80	<
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	2560	40	20	80	40	80	<	160	20
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	20	10	40	20	80	<	80	<
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	2560	640	320	320	160	320	160	320	160
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	40	20	80	40	80	<	160	20
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	2560	80	40	40	20	80	10	80	40
TEST VIRUSES												
B/Antananarivo/4039/2018		2018-09-03	MDCK1/MDCK1	1280	80	40	20	20	40	10	80	20
B/Hong Kong/1401/2018	3	2018-11-20	MDCK1/MDCK1	1280	80	20	20	20	40	10	80	40
B/Antsirabe/6255/2018		2018-11-26	MDCK1/MDCK1	1280	40	20	20	20	40	10	80	20
B/Hong Kong/1405/2018		2018-12-01	MDCK1/MDCK1	1280	80	40	20	20	40	10	80	40
B/Hong Kong/1409/2018	3	2018-12-06	MDCK1/MDCK1	1280	80	<	20	20	20	10	40	20
B/Lisboa/39/2018	3	2018-12-12	SIAT1/MDCK1	640	40	<	20	20	20	<	20	<
B/Croatia/187/2019		2019-01-04	MDCK1	1280	80	40	40	20	80	10	80	40
B/Luxembourg/1617/2019		2019-01-10	MDCK1	640	40	<	20	20	20	<	40	20

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

1 < = <40; 2 < = <10; 3 hyperimmune sheep serum; 4 < = <20

B/Yamagata-lineage virus recommended for use in quadravalent vaccines NH 2018-19 and SH 2019

Sequences in phylogenetic trees

Vaccine[#]

Table 13-10. Antigenic analyses of influenza B viruses (Yamagata lineage) with collection dates after 2018-08-31 - Summary

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				B/Estonia 55669/11	B/Mass 02/12	B/Mass 02/12	B/Wis 1/10	B/Stock 12/11	B/Phuket 3073/13	B/Phuket 3073/13	B/Maur 1791/17	
				MDCK	MDCK	Egg	Egg	Egg	MDCK	Egg	MDCK	
				F40/18 ⁺¹	F10/16 ⁺²	F06/17 ⁺¹	F36/15 ⁺²	F05/17 ⁺²	F27/15 ⁺²	F25/17 ⁺²	F04/18 ⁺²	
				2	2	2	3	3	3	3	3	3
REFERENCE VIRUSES												
B/Estonia/55669/2011	2	2011-03-14	MDCK2/MDCK3	160	40	<	20	20	20	40	<	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	160	80	320	80	80	40	320	20	
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	40	20	160	40	80	10	160	<	
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	80	20	160	80	160	20	320	40	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	20	10	80	40	160	20	160	<	
B/Phuket/3073/2013	3	2013-11-21	MDCK2/MDCK3	640	160	160	160	160	320	320	160	
B/Phuket/3073/2013	3	2013-11-21	E4/E3	40	40	80	40	80	20	160	20	
B/Mauritius/1791/2017	3	2017-09-20	MDCK1/MDCK4	<	40	<	20	40	40	80	40	
TEST VIRUSES												
Number of viruses tested*				22	22	22	22	22	22	22	22	
No with titre reduction ≤2-fold				16	13	1	20	10	2	18	20	
%				72.6	59.1	4.5	91.0	45.3	9.0	81.8	91.0	
No with titre reduction =4-fold				3	4	3	2	9	9	3	1	
%				13.7	18.2	13.7	9.0	41.0	41.0	13.7	4.5	
No with titre reduction ≥8-fold				3	5	18		3	11	1	1	
%				13.7	22.7	81.8		13.7	50.0	4.5	4.5	

* All viruses sequenced in time for compilation of this report fell in clade 3

B/Yamagata-lineage virus recommended for use in quadravalent vaccines

Reference virus results are taken from an individual table as an example. Summaries for each antiserum are based on fold-reductions observed on the days that HI assays were performed.

Vaccine[#]

in HI tables



Figure 17. HA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Florida/4/2006	DR	IT	SS	NS	PH	V	K	T	A	T
B/Estonia/55669/2011	DR	IT	SS	NS	PH	V	K	T	A	T
B/Massachusetts/02/2012	DR	IT	SS	NS	PH	V	K	T	A	T
B/Stockholm/12/2011	DR	IT	SS	NS	PH	V	K	T	A	T
B/Wisconsin/1/2010	DR	IT	SS	NS	PH	V	K	T	A	T
B/Phuket/3073/2013	DR	IT	SS	NS	PH	V	K	T	A	T
B/El Salvador/962/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Hawaii/36/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Ghana/3761/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Netherlands/10600/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Yunnan-Yingjiang/316/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Mauritius/1791/2017	DR	IT	SS	NS	PH	V	K	T	A	T
B/South Australia/1003/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/England/609/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Hawaii/35/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Oman/6339/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Michigan/01/2019	DR	IT	SS	NS	PH	V	K	T	A	T
B/Oman/5758/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Hong Kong/1409/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Pennsylvania/129/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Oman/6266/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Mauritius/I-762/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Iceland/99/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Iran/146039/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Yunnan-Zhaoyang/11211/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Idaho/17/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Wisconsin/01/2019	DR	IT	SS	NS	PH	V	K	T	A	T
B/Wyoming/14/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Ireland/78367/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/New Mexico/17/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Sweden/62/2018	DR	IT	SS	NS	PH	V	K	T	A	T
B/Norway/3367/2018	DR	IT	SS	NS	PH	V	K	T	A	T

	110	120	130	140	150	160	170	180	190	200
B/Florida/4/2006	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Estonia/55669/2011	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Massachusetts/02/2012	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Stockholm/12/2011	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Wisconsin/1/2010	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Phuket/3073/2013	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/El Salvador/962/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Hawaii/36/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Ghana/3761/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Netherlands/10600/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Yunnan-Yingjiang/316/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Mauritius/1791/2017	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/South Australia/1003/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/England/609/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Hawaii/35/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Oman/6339/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Michigan/01/2019	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Oman/5758/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Hong Kong/1409/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Pennsylvania/129/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Oman/6266/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Mauritius/I-762/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Iceland/99/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Iran/146039/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Yunnan-Zhaoyang/11211/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Idaho/17/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Wisconsin/01/2019	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Wyoming/14/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Ireland/78367/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/New Mexico/17/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Sweden/62/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV
B/Norway/3367/2018	RT	KI	RL	PN	LL	RG	YEN	IR	LS	QNV

	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Estonia/55669/2011	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Massachusetts/02/2012	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Stockholm/12/2011	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Wisconsin/1/2010	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Phuket/3073/2013	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/El Salvador/962/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Hawaii/36/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Ghana/3761/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Netherlands/10600/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Yunnan-Yingjiang/316/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Mauritius/1791/2017	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/South Australia/1003/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/England/609/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Hawaii/35/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Oman/6339/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Michigan/01/2019	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Oman/5758/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Hong Kong/1409/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Pennsylvania/129/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Oman/6266/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Mauritius/I-762/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Iceland/99/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Iran/146039/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Yunnan-Zhaoyang/11211/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Idaho/17/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Wisconsin/01/2019	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Wyoming/14/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Ireland/78367/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/New Mexico/17/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Sweden/62/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH
B/Norway/3367/2018	KN	LY	GD	SN	PQ	KFT	SS	ANG	V	TH

Figure 18. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

Vaccine virus

Reference viruses

Collection date

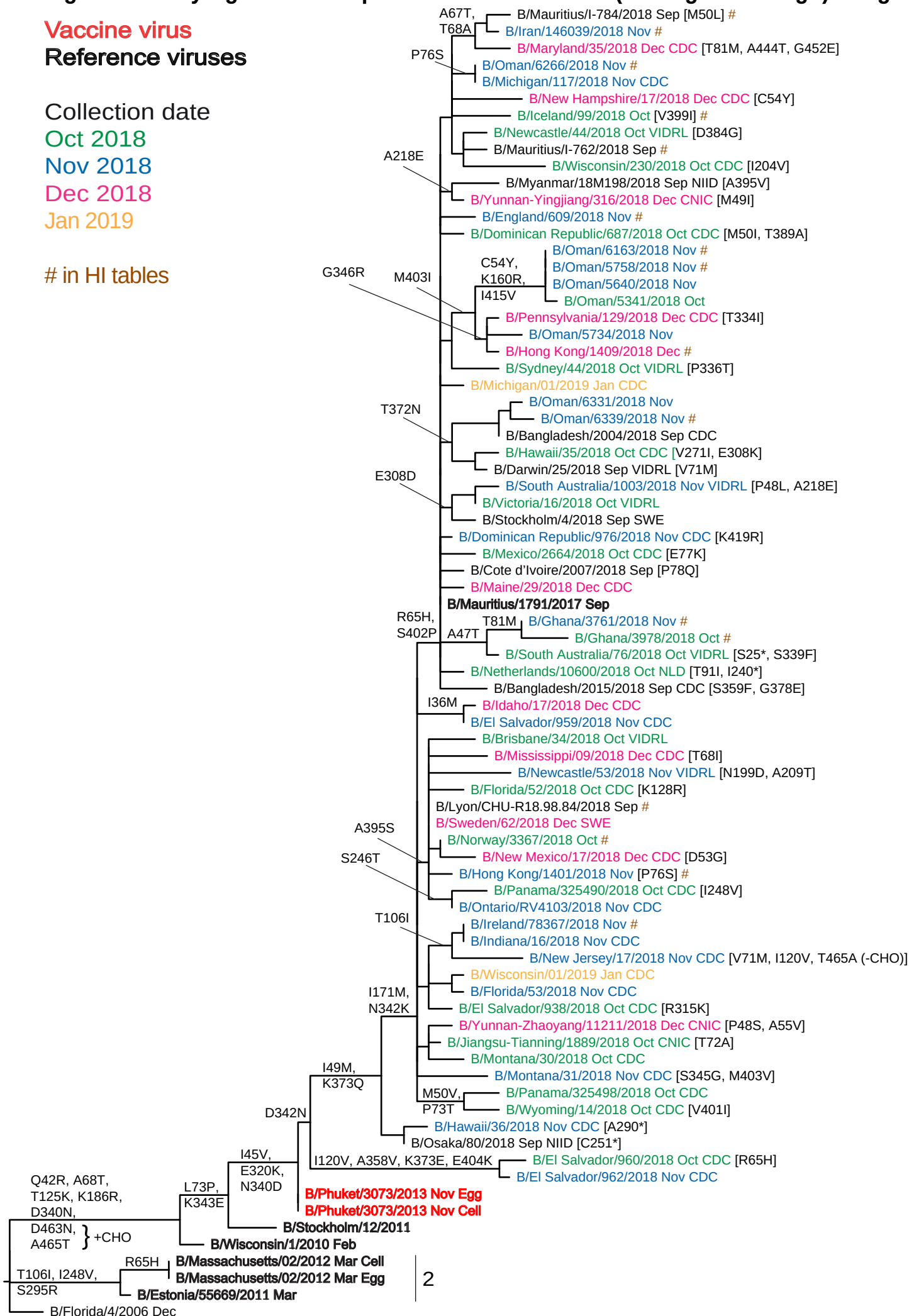
Oct 2018

Nov 2018

Dec 2018

Jan 2019

in HI tables



3

Figure 19. NA protein alignment for B (Yamagata-lineage) viruses

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100
B/Florida/4/2006	MLPSTIQTLLTFLTSGGVLLSLVFSASLSYLLYSDDLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLP	EP	EWY	PRLSCP	STFG	KALLIS	P			
B/Estonia/55669/2011										
B/Massachusetts/02/2012							H			
B/Stockholm/12/2011				R			T	P		
B/Wisconsin/1/2010				R			T			
B/Phuket/3073/2013				R	V		T	P		
B/El Salvador/962/2018				R	V		T	P		
B/Hawaii/36/2018				R	V	M	T	P		
B/Ghana/3761/2018				R	V	T	M			
B/Netherlands/10600/2018				R	V	M			I	
B/Yunnan-Yingjiang/316/2018				R	V		H	T	P	
B/Mauritius/1791/2017				R	V	M	H	T	P	
B/South Australia/1003/2018				R	V	LM	H	T	P	
B/England/609/2018				R	V	M	H	T	P	
B/Hawaii/35/2018				R	V	M	H	T	P	
B/Oman/6339/2018				R	V	M	H	T	P	
B/Michigan/01/2019				R	V	M	H	T	P	
B/Oman/5758/2018				R	V	M	Y	H	T	P
B/Hong Kong/1409/2018				R	V	M	H	T	P	
B/Pennsylvania/129/2018				R	V	M	H	T	P	
B/Oman/6266/2018				R	V	M	H	T	P	S
B/Mauritius/I-762/2018				R	V	M	H	T	P	
B/Iceland/99/2018				R	V	M	H	T	P	
B/Iran/146039/2018				R	V	M	H	T	P	
B/Yunnan-Zhaoyang/11211/2018				R	V	SM	V	H	T	P
B/Idaho/17/2018			M	R	V	M	H	T	P	
B/Wisconsin/01/2019				R	V	M	H	T	P	
B/Wyoming/14/2018				R	V	MV	H	T	P	
B/Ireland/78367/2018				R	V	M	H	T	P	
B/New Mexico/17/2018				R	V	M	G	H	T	P
B/Sweden/62/2018				R	V	M	H	T	P	
B/Norway/3367/2018				R	V	M	H	T	P	
	110	120	130	140	150	160	170	180	190	200
B/Florida/4/2006	HRFGETKGN	SAPLI	IREP	FFIACG	PTCKH	FALTY	AAQ	PGGY	NGT	REDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWYIGVDG
B/Estonia/55669/2011										
B/Massachusetts/02/2012										
B/Stockholm/12/2011				K						R
B/Wisconsin/1/2010				K						R
B/Phuket/3073/2013				K						R
B/El Salvador/962/2018			V	K						R
B/Hawaii/36/2018				K						R
B/Ghana/3761/2018				K						R
B/Netherlands/10600/2018				K				M		R
B/Yunnan-Yingjiang/316/2018				K				M		R
B/Mauritius/1791/2017				K				M		R
B/South Australia/1003/2018				K				M		R
B/England/609/2018				K				M		R
B/Hawaii/35/2018				K				M		R
B/Oman/6339/2018				K				M		R
B/Michigan/01/2019				K				M		R
B/Oman/5758/2018				K			R	M		R
B/Hong Kong/1409/2018				K				M		R
B/Pennsylvania/129/2018				K				M		R
B/Oman/6266/2018				K				M		R
B/Mauritius/I-762/2018				K				M		R
B/Iceland/99/2018				K				M		R
B/Iran/146039/2018				K				M		R
B/Yunnan-Zhaoyang/11211/2018				K				M		R
B/Idaho/17/2018				K				M		R
B/Wisconsin/01/2019				K				M		R
B/Wyoming/14/2018				K				M		R
B/Ireland/78367/2018				I				M		R
B/New Mexico/17/2018				K				M		R
B/Sweden/62/2018				K				M		R
B/Norway/3367/2018				K				M		R
	210	220	230	240	250	260	270	280	290	300
B/Florida/4/2006	LLKIKYGEAY	TDYHSYAK	NILRTQES	ACNCIGGD	CYLMTDGP	PASGIS	ECRFLKIREGRI	IKEIFPTGRVKHTEECTCGFASNT	IEAC	RDNSY
B/Estonia/55669/2011										
B/Massachusetts/02/2012							V			R
B/Stockholm/12/2011							V			R
B/Wisconsin/1/2010										
B/Phuket/3073/2013										
B/El Salvador/962/2018										
B/Hawaii/36/2018										
B/Ghana/3761/2018										X
B/Netherlands/10600/2018										
B/Yunnan-Yingjiang/316/2018										
B/Mauritius/1791/2017										
B/South Australia/1003/2018										
B/England/609/2018										
B/Hawaii/35/2018										
B/Oman/6339/2018										
B/Michigan/01/2019										
B/Oman/5758/2018										
B/Hong Kong/1409/2018										
B/Pennsylvania/129/2018										
B/Oman/6266/2018										
B/Mauritius/I-762/2018										
B/Iceland/99/2018										
B/Iran/146039/2018										
B/Yunnan-Zhaoyang/11211/2018										
B/Idaho/17/2018										
B/Wisconsin/01/2019										
B/Wyoming/14/2018										
B/Ireland/78367/2018										
B/New Mexico/17/2018										
B/Sweden/62/2018										
B/Norway/3367/2018										

	310	320	330	340	350	360	370	380	390	400																				
B/Florida/4/2006	PFVKLN	VETD	TAEIRLM	CTET	YLDTP	RPNDGS	ITGP	CESDG	DGKG	SGG	IKGGF	VHQ	RMA	SKIG	RWYS	RTMS	KT	KRM	GML	YV	KY	GD	PW	TD	SE	AL	AL	SG	VM	1
B/Estonia/55669/2011																														2
B/Massachusetts/02/2012																														2
B/Stockholm/12/2011																														3
B/Wisconsin/1/2010																														3
B/Phuket/3073/2013																														3
B/El Salvador/962/2018																														3
B/Hawaii/36/2018																														3
B/Ghana/3761/2018																														3
B/Netherlands/10600/2018																														3
B/Yunnan-Yingjiang/316/2018																														3
B/Mauritius/1791/2017																														3
B/South Australia/1003/2018																														3
B/England/609/2018																														3
B/Hawaii/35/2018																														3
B/Oman/6339/2018																														3
B/Michigan/01/2019																														3
B/Oman/5758/2018																														3
B/Hong Kong/1409/2018																														3
B/Pennsylvania/129/2018																														3
B/Oman/6266/2018																														3
B/Mauritius/I-762/2018																														3
B/Iceland/99/2018																														3
B/Iran/146039/2018																														3
B/Yunnan-Zhaoyang/11211/2018																														3
B/Idaho/17/2018																														3
B/Wisconsin/01/2019																														3
B/Wyoming/14/2018																														3
B/Ireland/78367/2018																														3
B/New Mexico/17/2018																														3
B/Sweden/62/2018																														3
B/Norway/3367/2018																														3

	410	420	430	440	450	460																										
B/Florida/4/2006	VS	ME	EP	GW	YS	FG	FE	IK	DK	KD	VC	IG	IE	MV	HD	GG	KT	TW	HS	AA	TA	IY	CL	MG	SG	QL	LD	WT	VT	GV	DM	AL
B/Estonia/55669/2011																																
B/Massachusetts/02/2012																																
B/Stockholm/12/2011																																
B/Wisconsin/1/2010																																
B/Phuket/3073/2013																																
B/El Salvador/962/2018																																
B/Hawaii/36/2018																																
B/Ghana/3761/2018																																
B/Netherlands/10600/2018																																
B/Yunnan-Yingjiang/316/2018																																
B/Mauritius/1791/2017																																
B/South Australia/1003/2018																																
B/England/609/2018																																
B/Hawaii/35/2018																																
B/Oman/6339/2018																																
B/Michigan/01/2019																																
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B/Sweden/62/2018																																
B/Norway/3367/2018																																

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 14. Antiviral (NAI) testing of isolates with collection dates after 2018-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2018									
September	8		8		1		4		21
October	13		22		6		3		44
November	45		50		9		9		113
December	81	1	67		11		3		163
2019									
January	70		39				2		111
Totals	217	1 ^{\$}	186	0	27	0	21	0	452

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2019-01-13

	Collection Date	NA substitution	Oseltamivir		Zanamivir		Clinical details
			IC ₅₀	Phenotype	IC ₅₀	Phenotype	
^{\$} A/Antalya/5/2018	2018-12-18	H275Y	792.61	HRI	1.36	NI	Outpatient, unknown drug treatment